Mentor-Protégé Joint Venture of Citrine and Olsson

United States Department of Agriculture

Forest Service

Black Hills National Forest

Black Hills Campground Reconstruction

Comanche Park Campground Specifications

Client Contract No.: 140F0822D0212

Client Order No.: 1282B124F0147

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LIST OF DRAWINGS AND SPECIFICATIONS

PART 1 – GENERAL

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SECTION 010150
GENERAL REQUIREMENTS

PART 1 – SUMMARY OF WORK

1.1 DESCRIPTION

A. The intent of this section is to provide a general outline of the work required under this contract and identify special project conditions and/or requirements. It is not intended to provide detailed specifications for materials, labor, and workmanship. These are covered in other sections of these specifications.

B. The Comanche Park Campground is a seasonal campground operated from approximately mid-May to mid-September annually. It will referred to as “Comanche CG” or “Comanche” throughout the Specifications and Contract documents.

C. Comanche Campground is located 6.5 miles west of Custer on US16.

D. The Campground is located on a Forest Service loop road off US Highway 16 west of Custer. The Forest Service Road, NFSR 618, is approximately 14 ft wide.

E. Comanche G currently has 20 camping spurs inside the Campground gate and 2 sites outside the gate that are primitive and poorly defined. The Campground will be improved to have a total of 34 camping spurs as Base Items ~~and there are 17 additional sites as Option Items.~~

1. The site count does include the Host and Volunteer sites described in detail below.

F. There is currently one Host Site, with power, water, and wastewater vault, located near the center of the campground. The Host Site and associated infrastructure will be relocated to a spot near the beginning of the Loop, for improved service to campers. The existing Host Site will become a “Volunteer Site” with upgraded water and electric service and continued use of the sewage vault; ~~there is an Option Item to remove and replace the sewage vault with a new concrete vault.~~

G. There are currently five stick-built toilet buildings dating from 1965 at Comanche CG. These toilet buildings will be demolished. 4 new pre-cast concrete vault toilets will be installed as Base items at new locations throughout the redesigned Campground. ~~An additional toilet may be installed as an Option Item with the Optional Campground Loop. An additional toilet may be installed as an Option Item at the Optional Trailhead site.~~

H. The water system at Comanche CG consists of a well with a submersible pump in a wooden pumphouse, a 1300-gallon polyethylene above ground water storage tank, and distribution system with 5 points of use. This system is classified as a Transient Non-Community Public water system. The water system is in poor condition. The well needs pump controls, treatment system, and pump house are not functioning properly, are not compliant with codes and require replacement. The distribution system has numerous leaks, is in poor condition and requires replacement. The existing hydrants are not accessible and require upgrades. The pumphouse building is in poor condition with insufficient flooring, limited space and inadequate lighting.

1. Water system work for this project consists of decommissioning the existing pumphouse, installing a new pre-cast concrete treatment building in a new location, with upgraded plumbing and controls, removing the existing aboveground water storage tank and installing a new below ground fiberglass water storage tank, replacing all of the distribution piping and installing new hydrants.

I. The road will be modified with curve widening and shaping for drainage and new surfacing. Generally the road width will not be increased and width will be limited by the CCC-era stone drainage structures that are to be protected.

II. Camp sites in front of gate

III. Host and volunteer sites

~~IV. Optional Item – Trailhead with toilet~~

V. Optional Loop with optional toilet and water system.

1.2 LOCATION

A. This project is located at the Comanche Park Campground, off US Highway 16, approximately 6.5 miles west of Custer, Custer County, South Dakota.

1.3 SPECIFICATIONS

A. The specifications establish the construction procedures and materials to be employed throughout.

1.4 GENERAL SITE CONDITIONS AND WEATHER

A. The elevation of the site is approximately 5,500 feet above mean sea level. The site generally experiences moderate snowfalls beginning in October and November and ending in March or April. The average temperature is 70 °F in July and 25° F in February. The site is generally accessible May through September.

1.5 SITE ACCESS

A. To access Grizzly CG from downtown Custer, South Dakota, travel west on US Highway 16 for approximately 6.5 miles. Comanche Park Campground is on the left (south) side of the Highway.

1.6 USE OF FACILITIES AND UTILITIES

A. Water will not be available for contractor use at the site during the project.

B. There is no cell phone coverage in this area and there is no land-line phone at the site.

C. Vault toilets will not be available for contractor use at the site during the project.

D. Electrical services are available at this site. Contractor will be responsible for all communication with the utility provider and all charges associated with the use of in place and newly installed electricity.

1.7 STAGING AREA

A. An area at or near the project site will be made available for use as a staging area. Staging areas shall be approved by the COR prior to use. No camping will be allowed at the site. Security and clean-up of this area shall be the responsibility of the Contractor.

1.8 SCHEDULE

A. The Contractor shall submit, at the pre-work conference, a construction schedule showing the proposed scheduling of all work items. The contractor may plan on working 7 days per week during daylight hours.

B. Work shall start approximately September 15, 2025, and shall be completed before November 13, 2026. The water system shall be operational by November 13, 2026.

1.9 WORK RESTRICTIONS

A. Nonsmoking Site: Prohibit smoking by all persons within Project work and staging areas.

1.10 PROTECTION OF PUBLIC

A. Install and maintain suitable barriers to ensure the protection of the public, employees, and facilities.
   1. Fence, barricade, or otherwise block off the immediate work area to prevent unauthorized entry to the work area.
   2. Erect and maintain barricades and warning signs in accordance with ANSID6.1.
   3. Material may be new or used but shall be suitable for intended purpose.
   4. Fences and barriers shall be structurally adequate and neat in appearance.

B. Protect persons, motor vehicles, surrounding buildings, plants, and surrounding surfaces of buildings from harm resulting from Contractor's work.

   1. Provide temporary barricades, barriers and directional signage to exclude the public from areas where Work is being performed.
   2. Provide shoring, bracing, and supports as necessary. Do not overload structural elements.
   3. Do not attach temporary protection to existing surfaces except as approved by CO.

1.11 FIRE PROTECTION

A. Comply with NFPA 241 requirements unless otherwise indicated.

B. Fire-Prevention Plan: Prepare a written plan for preventing fires during the Work, including placement of fire extinguishers, fire blankets, rag buckets, and other fire-control devices. Include fire-watch personnel's training, duties, and authority to enforce fire safety.

C. Remove and keep area free of combustibles, including rubbish, paper, wastes, and chemicals, unless necessary for the immediate work.
   1. If combustible materials cannot be removed, provide fire blankets to cover such materials.

D. Comply with all Red Flag Warnings issued by the National Weather Service.

E. Heat-Generating Equipment, Combustible Materials, and Open Flames: Comply with the following procedures while performing work with heat-generating equipment, combustible materials, or open flames, including welding, torch-cutting, soldering, brazing, removing paint with heat, or other operations where open flames or implements using heat or combustible solvents and chemicals are anticipated.
   1. Obtain CO's approval for operations involving use of open-flame or welding or other high-heat equipment. Notify CO at least 2 workdays before planned use, indicating location and purpose of such work.
      a. Confirm with CO continued approval of such work on day of planned work as weather and fire conditions may change rapidly at project site.
   2. Do not perform work with heat-generating equipment, combustible materials, or open flames where flammable liquids or explosive vapors are present or thought to be present. Use a combustible gas indicator to ensure that the area is safe.

F. Internal combustion equipment and machinery must be equipped with approved spark arresters.
   1. Spark arrestors shall meet the requirements of the USDA Forest Service "Standard for Spark Arresters for Internal Combustion Engins" (Standard 5100-1).
   2. Spark arresters must be properly installed and maintained in effective working order. Spark arresters shall be inspected and maintained properly, including but not limited to regular cleaning to removed accumulated carbon particles.
   3. Mufflers and catalytic converters are not acceptable substitutes for spark arresters as they may pose their own fire hazards.

G. Fire Watch: Before working with heat-generating equipment, combustible materials, or open flames, station personnel to serve as a fire watch at each location where such work is performed. Fire-watch personnel shall have the authority to enforce fire safety. Station fire watch according to NFPA 51B, NFPA 241, and as follows:
   1. Train each fire watch in the proper operation of fire-control equipment and alarms.
   2. Prohibit fire-watch personnel from other work that would be a distraction from fire-watch duties.
   3. Cease work with heat-generating equipment, combustible materials or open flames whenever fire-watch personnel are not present.

4. Have fire-watch personnel perform final fire-safety inspection each day beginning no sooner than 30 minutes after the conclusion of work in each area to detect hidden or smoldering fire and to ensure that proper fire prevention is maintained.
5. Maintain fire-watch personnel at each area of Project site where heat-generating equipment, combustible materials, or open flames have been used until 60 minutes after conclusion of daily work.

H. Fire-Control Devices: Provide and maintain fire extinguishers, fire blankets, and rag buckets for disposal of rags with combustible materials. Maintain each as suitable for the type of fire risk in each work area.
1. Ensure that personnel are trained in fire extinguisher and fire blanket use.

1.12 PROTECTION DURING APPLICATION OF CHEMICALS

A. Protect buildings, sites, vegetation vehicles, infrastructure, and waterways from harm or spillage resulting from applications of chemicals and adhesives.

B. Use protective materials that are proven to resist chemicals and adhesives selected for Project. Use protective materials that are waterproof and UV resistant and that will not stain or leave residue on surfaces to which they are applied.
1. Apply protective materials according to manufacturer's written instructions.
2. Do not apply liquid masking agents or adhesives to painted or porous surfaces.
3. When no longer required, promptly remove protective materials and dispose of in manner that prevents release of chemicals.

C. Do not apply chemicals during winds of sufficient force to spread them to unprotected surfaces.

D. Collect and dispose of runoff from chemical operations by legal means and in a manner that prevents soil contamination, soil erosion, undermining of paving and foundations, or damage to landscaping,

1.13 CONTRACTOR'S RESPONSIBILITIES

A. Confine storage of materials to areas as approved by the CO.

B. Provide adequate signing and barricades and take necessary safety measures to protect the public during all construction operations. Minimize disturbance of all undisturbed areas.

C. The Contractor, at the Contractor's expense. shall repair Existing Forest Service roads and parking areas disturbed by construction activities. Repair includes any work needed to return the roads and parking areas to their existing condition before construction began and shall include replacing road aggregate if necessary.

D. All areas where the existing grass vegetation is removed or damaged, or destroyed due to construction activities, shall be leveled and seeded. The Contractor shall furnish and apply all seed in accordance with specifications.

E. The Contractor shall be responsible for all damage that results from the Contractor's action during completion of the project work. Any damage that occurs due to action of the Contractor shall be repaired at the Contractor's expense.

F. Protect trees and vegetation within project area from damage.

1.14 EXISTING POWER LINES

A. Existing buried utilities shown on the drawings are located as accurately as existing records permit. The Contractor shall exercise caution when excavating in areas where buried utilities are anticipated. It will be the Contractor's responsibility to contact South Dakota One Call at 811 or 800-781-7474 for utility locates. Any damage to such lines shall be repaired at the Contractor's expense. If a power line is damaged, the Contractor shall stop all work in the immediate vicinity of the damage and immediately notify the Contracting Officer. Work shall not be permitted near a damaged line until such damage is repaired.

1.15 UTILITY LOCATIONS

A. Contractor shall be responsible for hiring a locating service to locate all utility lines including but not limited to electrical, water, and phone. This includes the Forest Service lines.

1.16 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

A. Provide waste-collection containers in sizes adequate to handle waste from construction operations. Containerize and clearly label hazardous, dangerous, or unsanitary waste materials separately from other waste.

B. remove waste materials from Project site and legally dispose of them in a landfill acceptable to authorities having jurisdiction.

C. Do not burn waste materials.

D. Waste Management in Historic Zones or Areas: Transportation equipment and other materials shall be of sizes that clear surfaces within historic spaces, areas, rooms, and openings, by 12 inches (300 mm) or more.

1.17 SITE CLEANUP AND DISPOSAL AREA

A. The Contractor shall cleanup construction debris at the completion of each day’s work. Disposal of construction debris shall be at an established Sanitary Landfill, with all permits and related costs of disposal being the responsibility of the Contractor. The Contractor and the CO shall mutually agree to the frequency of disposal. The Contractor shall not use existing on-site trash receptacles for disposal or interim storage of construction debris. Burning construction debris shall not be allowed on lands administered by the Forest Service.

1.18 FIELD VERIFICATION

A. Field verify all new and existing dimensions affecting the work of this contract before ordering products.

1.19 ELECTRICAL WORK

A. All electrical work shall be in accordance with applicable state and local codes, regulations, and ordinances, and the latest applicable requirements of the National Electrical Code of the NFPA. State Electrical Inspection and approval is required on all electrical work. The Contractor shall be responsible for obtaining inspection and approval, and any associated costs.

1.20 PERMITS

A. Obtain permits for electrical work or any other work required by County, State, or Federal laws or regulations.

PART 2 – ACCIDENT PREVENTION

2.1 DESCRIPTION

This section consists of establishing an effective accident prevention plan and providing a safe environment for personnel and visitors.

2.2 SUBMITTALS

A. Accident Prevention Plan: Before on-site work begins, submit a company approved accident prevention plan. This plan will be posted in the contract file. Design the plan to address Federal, State, and Local Occupational Safety and Health requirements that apply to this project. No progress payments will be processed until the plan is received. As a minimum the plan shall include:

1. Name, position title and contact information of company executive responsible for approving the Accident Prevention Plan.
2. Name and contact information of supervisor responsible for carrying out the plan.
3. Outline of each phase of the work, the hazards associated with each major phase, and the methods proposed to ensure property protection and safety of the public, government personnel, and the Contractor's employees. Identify the work included under each phase by reference to specification section or division numbers.
4. Contingency plans for emergency situations such as medical, fire, hazard material spills and other contract assessed hazard prevention and abatement requirement needs that apply to this project.

B. Certificates: Provide certificates from a mechanic that all mechanical equipment has been inspected and meets OSHA requirements.

C. Submit a copy of test reports, as required by OSHA, for personnel working with hazardous materials.

D. Submit a brief report of safety meetings and of inspections.

E. Upon request, submit proof of employees' qualifications to perform assigned duties in a safe manner.

2.3 QUALITY ASSURANCE

A. Clauses entitled "Accident Prevention" and "Permits and Responsibilities" of the General Provisions." In case of conflicts between Federal, state, and local safety and health requirements, the most stringent shall apply. Equipment or tools not meeting OSHA requirements will not be allowed on the project sites. Failure to comply with the requirements of this section and related sections may result in suspension of work.

B. Qualifications of Employees: Ensure employees are physically qualified to perform assigned duties in a safe manner.

2.4 ACCIDENT REPORTING

A. Reportable Accidents: A reportable accident is defined as death, occupational disease, traumatic injury to employees or the public, property damage by accident in excess of $100, and fires. Within 7 calendar days of a reportable accident, fill out and forward to the CO a CA-1 form, which may be obtained from CO.

B. All Other Accidents: Report all other accidents to the CO as soon as possible and assist the CO and other officials as required in the investigation of the accident.

2.5 FIRST AID FACILITIES

A. Provide adequate facilities for the number of employees and the type of construction at the site.

2.6 PERSONNEL PROTECTIVE EQUIPMENT

A. Meet requirements of NIOSH and MSHA, and ANSI where applicable.

B. Inspect personal protective equipment daily and maintain in a serviceable condition. Clean, sanitize, and repair, as appropriate, personal items before issuing them to another individual.

C. Inspect and maintain other protective equipment and devices before use and on a periodic basis to ensure safe operation.

2.7 EMERGENCY INSTRUCTIONS

A. Post telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and police in conspicuous locations at the work site.

2.8 SAFETY MEETINGS

A. As a minimum, conduct weekly 15-minute "toolbox" safety meetings. These meetings shall be conducted by a foreman and attended by all construction personnel at the worksite.

2.9 HARD HATS AND PROTECTIVE EQUIPMENT AREAS

A. The CO will designate a hard hat area. Post the hard hat area in a manner satisfactory to the CO.

B. It is the Contractor's responsibility to require all those working on or visiting the site to wear hard hats and other necessary protective equipment at all times. As a minimum, provide two hard hats for use by visitors. Change liners before reissuing hats.

2.10 TRAINING

A. First Aid: Provide adequate training to ensure prompt and efficient first aid.
B. Hazardous Material: Train and instruct each employee exposed to hazardous material in safe and approved methods of handling and storage. Hazardous materials are defined as explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substances that could cause death or injury.

PART 3 – SUBMITTAL PROCEDURES

3.1 DESCRIPTION

A. This section includes administrative and procedural requirements for submittals.

B. Submittals may be rejected for not complying with requirements.

3.2 DEFINITIONS

A. Action Submittals: Written and graphic information that requires Contracting Officer’s (CO's) responsive action.

B. Informational Submittals: Written information that does not require CO's approval.

3.3 PROCEDURES

A. Submittal Register: Contractor shall provide submittal register which identifies all required submittals listed in specifications for review by CO prior to submission of any other submittals.

B. Processing Time: Allow enough time for submittal review, including time for re-submittals, as follows. Time for review shall commence on CO's receipt of submittal.

1. Initial Review: Allow 10 working days for initial review of each submittal. Allow additional time if processing must be delayed to permit coordination with subsequent

submittals. CO will advise Contractor when a submittal being processed must be delayed for coordination.

2. If intermediate submittal is necessary, process it in same manner as initial submittal.
3. Allow 10 working days for processing each re-submittal.
4. No extension of the Contract Time will be authorized because of failure to transmit submittals enough in advance of the Work to permit processing.

C. Identification: Place a permanent label or title block on each submittal for identification.

1. Indicate name of firm or entity that prepared each submittal on label or title block.
2. Provide a space approximately 2 by 3 inches on label or beside title block to record Contractor's review and approval markings and action taken by CO.
3. Include the following information on label for processing and recording action taken:
   a. Project name.
   b. Date.
   c. Name and address of Contractor.
   d. Name of manufacturer.
   e. Unique identifier, including revision number.
   f. Number and title of appropriate Specification Section.
   g. Drawing number and detail references, as appropriate.
   h. Other necessary identification.

D. Deviations: Highlight, encircle, or otherwise identify deviations from the Contract Documents on submittals.

E. Additional Copies: Unless additional copies are required for final submittal, and unless CO observes noncompliance with provisions of the Contract Documents, initial submittal may serve as final submittal.

F. Use for Construction: Use only final submittals with mark indicating action taken by CO in connection with construction.

G. Review each submittal and check for compliance with the Contract Documents. Note corrections and field dimensions. Mark with approval stamp before submitting to CO.

H. Approval Stamp: Stamp each submittal with a uniform, approval stamp. Include Project name and location, submittal number, Specification Section title and number, name of reviewer, date of Contractor's approval, and statement certifying that submittal has been reviewed, checked, and approved for compliance with the Contract Documents.

I. CO will not review submittals that do not bear Contractor's approval stamp and will return them without action.

J. Submittals not required by the Contract Documents will not be reviewed and may be discarded.

3.4 ACTION SUBMITTALS

A. General: Prepare and submit Action Submittals required by individual Specification Sections.

1. Number of Copies: Submit four copies of each submittal, unless otherwise indicated. CO will return two copies. ark up and retain one returned copy as a Project Record Document.

B. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.

1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
2. Mark each copy of each submittal to show which products and options are applicable.
3. Include the following information, as applicable:
   a. Manufacturer's written recommendations.
   b. Manufacturer's product specifications.
   c. Manufacturer's installation instructions.
   d. Manufacturer's catalog cuts.
   e. Wiring diagrams showing factory-installed wiring.
   f. Compliance with recognized trade association standards.
   g. Compliance with recognized testing agency standards.

3.5 INFORMATIONAL SUBMITTALS

A. General: Prepare and submit Informational Submittals required by other Specification Sections.

1. Number of Copies: Submit two copies of each submittal, unless otherwise indicated. CO will not return copies.
2. Certificates and Certifications: Provide a notarized statement that includes signature of entity responsible for preparing certification. Certificates and certifications shall be signed by an officer or other individual authorized to sign documents on behalf of that entity.

B. Contractor's Construction Schedule: Provide a construction schedule that incorporates the full contract period. Construction Schedule shall include the following, at a minimum:

1. Activity Duration: treat each separate work area or work item as a separate numbered activity for each main element of the Work.
2. Procurement Activities: Include procurement process activities for long lead items and major items as separate activities in schedule.
3. Submittal Review Time: Include review and resubmittal times in schedule.
4. Startup and Testing: Include no fewer than 5 days for startup and testing.
5. Substantial Completion: Include completion in advance of date established for Substantial Completion and allow time for CO's administrative procedures necessary for certification of Substantial Completion.

6. Punch List and Final Completion: Include not more than 30 days for completion of punch list items and final completion.

C. Upcoming Work Summary: Prepare summary report indicating activities scheduled to occur or commence prior to submittal of next schedule update. Summarize the following issues:
   1. Unresolved issues.
   2. Unanswered Requests for Information.
   3. Rejected or returned submittals.
   4. Notations on returned submittals.
   5. Pending Modifications affecting the Work and the Contract Time.

D. Construction Schedule Updating Reports: Submit with Applications for Payment.
   1. At monthly intervals, update schedule to reflect actual construction progress and activities. Issues schedule one week before each regularly scheduled progress meeting.
      a. Revise schedule after each meeting or other activity where revisions have been recognized or mad. Issue updated schedule concurrently with the report of each such meeting.
      b. As the Work progresses, indicate final completion percentage for each activity.

E. Daily Construction Reports: Submit at monthly intervals.

   1. Prepare a daily construction report recording the following information concerning events at Project site:
      a. List of subcontractors at Project site.
      b. Count of personnel at Project site.
      c. Equipment at Project site.
      d. Material deliveries.
      e. High and low temperatures and general weather conditions, including the presence of rain or snow.
      f. Testing and inspection.
      g. Accidents and Emergency procedures.
      h. Meetings and significant decisions.
      i. Stoppages, delays, shortages, and losses.
      j. Orders and requests of authorities having jurisdiction.
      k. Change Orders received and implemented.
      l. Services connected and disconnected.
      m. Equipment or system tests and startups.
      n. Substantial Completions authorized.

F. Site Condition Reports: Submit at time of discovery of differing conditions.

G. Cost Breakdown: Following award provide a cost breakdown schedule listing main types of work with associated costs for each bid item in CSI format.

PART 4 – EXECUTION REQUIREMENTS

4.1 DESCRIPTION

A. This section consists of general procedural requirements governing execution of the Work including, but not limited to, the following:

1. Construction Layout
2. Installation.
3. Progress cleaning.
4. Starting and Adjusting
5. Protection of installed construction.
6. Correction of the Work.

4.2 QUALITY ASSURANCE

A. Cutting and patching: Comply with requirements for and limitations on cutting and patching of construction elements.

1. Structural Elements: When cutting and patching structural elements, or when encountering the need for cutting and patching of elements whose structural function is not known, notify Contracting Officer of locations and details of cutting and await directions from CO before proceeding. Shore, brace, and support structural elements during cutting and patching. Do not cut and patch structural elements in a manner that could change their load-carrying capacity or increase deflection.
2. Operational Elements: Do not cut and patch operating elements and related components in a manner that results in reducing their capacity to perform as intended or that result in reducing their capacity to perform as intended or that results in increased maintenance or decreased operational life or safety.
3. Other Construction Elements: Do not cut and patch other construction elements or components in a manner that could change their load-carrying capacity, that results in reducing their capacity to perform as intended, or that results in increased maintenance or decreased operational life or safety.
4. Visual Elements: Do not cut and patch construction in a manner that results in visual evidence of cutting and patching. Do not cut and patch exposed construction in a manner that would, in CO's opinion, reduce the building's or site's aesthetic qualities. Remove and replace construction that has been cut and patched in a visually unsatisfactory manner.
5. Use materials for patching identical to in-place materials. For exposed surfaces, use materials that visually match in-place adjacent surfaces to the fullest extent possible.
   a. If identical materials are unavailable or cannot be used, use materials that, when installed, will provide a match acceptable to CO for the visual and functional performance of in-place materials.
   b. Use materials that are not considered hazardous.

B. Manufacturer’s Installation Instructions: Obtain and maintain on-site manufacturer's written recommendations and instructions for installation of specified products and equipment.

4.3 EXAMINATION

A. Existing Conditions: The existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Before beginning sitework, investigate and verify the existence and location of underground utilities, mechanical and electrical systems, and other construction affecting the Work.
   1. Before construction, verify the location and invert elevation at points of connection of water and wastewater and electrical utilities.

B. Examination and Acceptance of Conditions: Before proceeding with each component of the Work, examine substrates, areas, and conditions for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.

C. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surface and conditions.

4.4 PREPARATION

A. Field Measurements: Take field measurements as required to fit the Work properly. Recheck measurements before installing each product. Where portions of the Work are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

B. Existing Utility Information: Furnish information to CO that is necessary to adjust, move, or relocate existing utility structures, utility poles, lines, services, or other utility appurtenances located in or affected by construction. Coordinate with authorities having jurisdiction.

C. Space Requirements; Verify space requirements and dimensions of items shown diagrammatically on drawings.

D. Review of Contract Documents and Field Conditions: Immediately on discovery of the need for clarification of the Contract Documents, submit a request for information to CO. Include a detailed description of problem encountered, together with recommendations for changing the Contract Documents.

4.5 CONSTRUCTION LAYOUT

A. Verification: Before proceeding to lay out the Work, verify layout information shown on Drawings, in relation to the site survey and existing benchmarks and existing conditions. If discrepancies are discovered, notify CO promptly.

B. Site Improvements: Locate and lay out site improvements, including roads, pavements, grading, fill placement, utility slopes, and rim and invert elevations.

C. Building Lines and Levels: Locate and lay out control lines and levels for structures and foundations, including that required for mechanical and electrical work.

4.6 INSTALLATION

A. General: Locate the Work and components of the Work accurately, in correct alignment and elevation, as indicated.
   1. Make vertical work plumb and make horizontal work level.
   2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.

B. Comply with manufacturer's written instructions and recommendations for installing products.

C. Install products at the time and under conditions that will ensure satisfactory results as judged by CO. Maintain conditions required for product performance until Substantial Completion.

D. Conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal operations of occupancy of type expected for Project.

E. Sequence the Work and allow adequate clearances to accommodate movement of construction items on-site and placement in permanent locations.

F. Joints: Make joints of uniform width. Where joint locations in exposed Work are not indicated, arrange joints for the best visual effect, as judged by the CO. Fit exposed connections. Together to form hairline joints.

G. Cutting and Patching: Employ skilled workers to perform cutting and patching. Cut in-place construction to provide for installation of other components or performance of other construction and subsequently patch as required to restore surfaces to their original condition.
   1. Provide temporary support of Work to be cut.
   2. Protect in-place construction during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portions of Project that might be exposed during cutting and patching operations.

H. Cutting: Cut in-place construction by sawing, drilling, breaking, chipping, grinding, and similar operations, including excavation, using methods least likely to damage elements retained or adjoining construction.
   1. In general, use hand or small power tools designed for sawing and grinding, not hammering or chopping. Cut holes and slots neatly to minimum size required, and

with minimum disturbance to adjacent surfaces. Temporarily cover openings when not in use.

2. Finished surfaces: Cut or drill from the exposed or finished side into concealed surfaces.
3. Concrete and Masonry: Cut using a cutting machine such as an abrasive saw or a diamond-core drill.
4. Mechanical and Electrical Services: Cut off pipe or conduit to be removed. Cap, valve, or plug and seal remaining portion of pipe or conduit to prevent entrance of moisture or other foreign matter after cutting.

I. Patching: Patch construction by filling, repairing, refinishing, closing up, and similar operations following performance of other Work. Patch with durable seams that are as invisible as practicable, as judged by CO.

J. Existing Utility Services and Mechanical/Electrical Systems: Where existing services/ systems are required to be removed, relocated, or abandoned, bypass such services/systems before cutting to minimize interruption to occupied areas.

K. Hazardous Materials: Use products, cleaners, and installation materials that are not considered hazardous.

4.7 PROGRESS CLEANING

A. General: Clean Project site and work areas daily, including common areas. Dispose of materials lawfully.

1. Comply with requirements in NFPA 241 for removal of combustible waste materials and debris.
2. Do not hold materials more than 7 days during normal weather or 3 days if the temperature is expected to rise above 80 deg F (27 deg C).
3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, according to regulations.
4. Use containers intended for holding waste materials of type to be stored.

B. Site: Maintain Project site free of waste materials and debris

C. Work Areas: Clean areas where work is in progress to the level of cleanliness necessary for proper execution of the Work.

D. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.

E. Exposed Surfaces: Clean exposed surfaces and protect as necessary to ensure freedom from damage and deterioration.

F. Cleaning Agents: Use cleaning materials and agents recommended by manufacturer or fabricator of the surface to be cleaned.

1. Do not use cleaning agents that are potentially hazardous to health or property or that might damage finished surfaces.
2. Do not use cleaning agents that are hazardous to water, water sources, wildlife, or plant life.

G. Waste Disposal: Do not bury or burn waste materials on site. Do not wash waste materials down sewers or into waterways.

H. During handling and installation, clean and protection construction in progress and adjoining materials already in place. Apply protective covering where required to ensure protection from damage or deterioration.

I. Limiting Exposures: Supervise construction operations to ensure that no part of the construction, completed or in progress, is subject to harmful, dangerous, damaging or otherwise deleterious exposure during the construction period.

4.8 STARTING AND ADJUSTING

A. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.

B. Adjust equipment for proper operation. Adjust operating components for proper operation without binding.

C. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

4.9 PROTECTION OF INSTALLED CONSTRUCTION

A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.

B. Protection of Existing Items: Provide protection and ensure that existing items to remain undisturbed by construction are maintained in condition that existed at commencement of the Work.

C. Comply with manufacturer's written instructions for temperature and relative humidity.

4.10 CORRECTION OF THE WORK

A. Repair or remove and replace defective construction. Restore damaged substrates and finishes.

1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.

B. Repair work previously completed and subsequently damaged during construction period. Repair to like-new condition.

C. Restore permanent facilities used during construction to their specified condition.

D. Remove and replace damaged surfaces that are exposed to view if surfaces cannot be repaired without visible evidence or repair.

E. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.

F. Remove and replace chipped, scratched, and broken glass or reflective surfaces.

PART 5 – CLOSEOUT PROCEDURES

5.1 DESCRIPTION

A. This section consists of administrative and procedural requirements for contract closeout, including, but not limited to, the following:

1. Inspection procedures.
2. Project Record Documents.
3. Final cleaning.

5.2 SUBSTANTIAL COMPLETION

A. Definition of Substantial Completion: The Date certified by the Contracting Officer when construction is sufficiently complete, in accordance with the Contract Documents, so the Government can occupy or utilize the Work or designated portion thereof for the use for which it is intended, as expressed in the Contract Documents.

B. Preliminary Procedures: Before requesting inspection for determining date of Substantial Completion, complete the following. List items below that are incomplete in request.

1. Prepare a list of items to be completed and corrected (punch list), the value of items on the list, and reasons why the Work is not complete.
2. Advise Contracting Officer (CO) of pending insurance changeover requirements.
3. Submit specific warranties, workmanship bonds, final certifications, and similar documents.
4. Obtain and submit releases permitting Government unrestricted use of the Work and access to services and utilities. Include occupancy permits, operating certificates, and similar releases.
5. Prepare and submit Project Record Documents, operation and maintenance manuals, Final Completion construction photographs and photographic negatives, and similar final record information.
6. Deliver tools, spare parts, extra materials, and similar items to location designated by Government. Label with manufacturer's name and model number where applicable.
7. Complete startup testing of systems.

8. Submit changeover information related to Government occupancy, use, operation, and maintenance.
9. Complete final cleaning requirements, including touchup painting.
10. Touch up and otherwise repair and restore marred exposed finishes to eliminate visual defects.

C. Inspection: Submit a written request for inspection for Substantial Completion at least 24 hours before inspection is needed. On receipt of request, CO will either proceed with inspection or notify Contractor of unfulfilled requirements. O will prepare the Certificate of Substantial Completion after inspection or will notify Contractor of items, either on Contractor's list or additional items identified by CO, that must be completed or corrected before certificate will be issued.

1. Reinspection: Request reinspection when the Work identified in previous inspections as incomplete is completed or corrected.
2. Results of completed inspection will form the basis of requirements for Final Completion.

5.3 FINAL COMPLETION

A. Preliminary Procedures: Before requesting final inspection for determining date of Final Completion, complete the following:

1. Submit certified copy of CO’s Substantial Completion inspection list of items to be completed or corrected (punch list), endorsed and dated by CO. The certified copy of the list shall state that each item has been completed or otherwise resolved for acceptance.
2. Submit evidence of final, continuing insurance coverage complying with insurance requirements.
3. Instruct Government personnel in operation, adjustment, and maintenance of products, equipment, and systems.

B. Inspection: Submit a written request for final inspection for acceptance at least 24 hours before inspection is needed. On receipt of request, CO will either proceed with inspection or notify Contractor of unfulfilled requirements. CO will prepare a final Certificate for Payment after inspection or will notify Contractor of construction that must be completed or corrected before certificate will be issued.

1. Reinspection: Request reinspection when the Work identified in previous inspections as incomplete is completed or corrected.

5.4 LIST OF INCOMPLETE ITEMS (PUNCH LIST)

A. Preparation: Submit three copies of list. Include name and identification of each space and area affected by construction operations for incomplete items and items needing correction including, if necessary, areas disturbed by Contractor that are outside the limits of construction.

5.5 PROJECT RECORD DOCUMENTS

A. General: Do not use Project Record Documents for construction purposes. Protect Project Record Documents from deterioration and loss. Provide access to Project Record Documents for CO's reference during normal working hours.

B. Closeout Submittals:

1. Record Drawings: Submit one (1) paper copy and one (1) electronic PDF copy (of marked-up record prints as “As-Built” drawings.
2. Record Specifications: Submit annotated PDF electronic files and one (1) paper copy of Project’s Specifications, including addenda and Contract Modifications.
3. Record Product Data: Submit annotated PDF Electronic files and directories and one (1) paper of each submittal.
   a. Where Record Product Data are required as part of Operations and Maintenance Manuals, submit duplicate marked-up Product Data as a component of Manual.

C. Record Drawings: Maintain one set of marked-up paper copies of Contract Drawings and Shop Drawings incorporating new and revised drawings as modifications are issued.

1. Preparation: Mark Record Prints to show the actual installation, where installation varies from that shown originally. Require individual or entity who obtained record data, whether individual or entity is Installer, subcontractor, or similar entity, to provide information for preparation of corresponding marked-up Record Prints.

   a. Give particular attention to information on concealed elements that cannot be readily identified and recorded later.
   b. Accurately record information in an acceptable drawing technique,
   c. Record data as soon as possible after obtaining it.
   d. Record and check the markup before enclosing concealed installations.
2. Content: Types of items requiring marking include, but are not limited to, the following:
   a. Dimensional changes to Drawings.
   b. Revisions to details shown on Drawings.
   c. Depths of foundations.
   d. Locations and depths of underground facilities.
   e. Revisions to routing of piping and conduit.
   f. Revisions to electrical circuitry.
   g. Actual equipment locations.
   h. Locations of concealed internal utilities.
   i. Changes made by Change Order.
   j. Details not on the original Contract Drawings.
   k. Field records for variable or concealed conditions.
   l. Record information on the Work that is shown only schematically.
3. Mark the Contract Drawings and Shop Drawings completely and accurately.

4. Mark record prints with erasable, red-colored pencil. Use other colors to distinguish between changes for different categories of the Work at the same location.
5. Mark important additional information that was either shown schematically or omitted from original Drawings.
6. Note Construction Change Directive numbers, Change Order numbers, alternate numbers, and similar identification where applicable.

D. Record Digital Data Files: Immediately before inspection for Certificate of Substantial Completion, review marked-up record prints with Contracting Officer. When authorized, prepare a full set of corrected digital data files of the Contract Drawings, as follows:
1. Format: Annotated PDF electronic file with Comment function enabled.
2. Incorporate changes and additional information previously marked on record prints. Delete, redraw, and add details and notations where applicable.
3. Refer instances of uncertainty to CO for resolution.
4. CO will furnish Contractor with one set of digital data files of the Contract Drawings for use in recording information.

E. Format: Identify and date each Record Drawing; include the designation "PROJECT RECORD DRAWING" in a prominent location.
1. Record Prints: Organize into manageable sets; bind each set with durable paper cover sheets. Include identification on cover sheets.
2. Format: Annotated DF electronic files with Comment function enabled.
3. Record Digital Data files: Organize digital data information into separate electronic files that correspond to each sheet of the Contract Drawings. Name each file with the sheet identification. Include identification in each digital data file.
4. Indication: As follows:
   a. Project Name.
   b. Date.
   c. Designation “PROJECT RECORD DRAWINGS”
   d. Name of Contractor.

5.6 RECORD SPECIFICATIONS

A. Preparation: Mark Specifications to indicate the actual product installation, where installation varies from that indicated in Specifications, addenda, and Contract modifications.
1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
2. For each principal product, indicate whether Record Product Data has been submitted in operation and maintenance manuals instead of submitted as Record Product Data.
3. Note related Change Orders, and Record Drawings where applicable.

B. Format: Submit Record Specifications as annotated PDF electronic file.

5.7 RECORD PRODUCT DATA

A. Recording: Maintain one (1) copy of each submittal during the construction period for Project Record Document purposes. Post changes and revisions to Project Record Documents as they occur. Do not wait until end of Project.

B. Preparation: Mark Product Data to indicate the actual product installation where installation varies substantially from that indicated in Product Data submittal.
   1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
   2. Include significant changes in the product delivered to Project site and changes in manufacturer's written instructions for installation.
   3. Note related Change Orders and Record Drawings where applicable.

C. Format: Submit Record Product Data as annotated PDF electronic file.
   1. Include Record Product Data directory organized by Specification Section number and title, electronically linked to each item of Record Product Data.

5.8 OPERATION AND MAINTENANCE MANUAL

A. Before scheduling a final inspection, furnish two complete sets of operation and maintenance (O&M) data to the Contracting Officer for review. Should CO find the manual to be substantially incomplete, the final inspection will be delayed.

B. Within 30 days following receipt of review comments, deliver three complete sets of the O&M Manual.

C. O&M format shall be as follows:

   1. Bound in white, commercial quality, hard back, three-ring, with clear window pockets on front and side and inside pockets on cover and back.
   2. Index sheet with mylar reinforced edges at binder holes and tabbed divider sheets with mylar reinforced edges and pre-printed numbered tabs aligned with numbers and title lines on index sheet or with labeled tabs.
   3. Cover Sheet: Identify the project title, location, forest, contract number, prime contractor's name and address, date of substantial completion, and binder volume number. Insert cover sheet into clear plastic view pocket on front of binder. Insert sheet with project title and "Operation and Maintenance" into side clear plastic view pocket.
   4. Index System: Organize data into sections by common subjects and subsystems. Place a consecutively numbered tabbed divider sheet in front of each section. Place index sheet at the beginning of each binder, listing sections by subject name. If multiple binders are used, place a table of contents of all data provided behind the index sheet in each binder.
   5. Data: Fill binders to no more than 75 percent of capacity. Punch holes shall not obscure any data. Normal sheet size shall be 8-1/2 inches by 11 inches. Fold oversize sheets and insert them in 8-1/2 by 11-inch clear pocket sheet protectors placed in binders. When the contents of a single tabbed section covers more than one item, provide colored paper sheets to separate the data for each item.

a. Manufacturers' Data: Provide originals for color or copyrighted data. Black and white data may be originals or clean, good quality reproductions. Where originals are printed on both sides of the page, reproductions shall also be printed on both sides of the page. Copies produced by facsimile transmission and sheets with stamps, such as submittal approval stamps, will not be acceptable. Include only sheets that apply to items installed; cross out inapplicable data.
b. Vendor Furnished As-Built Drawings: Maximum 24-inch by 36-inch sheets with minimum character or lettering size of 1/8 inch. Reduced-size reproductions may be provided instead of full-size drawings if the reproductions are clear and legible. If reduced-size drawings are used, identify as "REDUCED SIZE" and provide graphic scales, if applicable.
c. Custom Written Data: Typewritten text, supplemented by drawings and schematics necessary to describe systems adequately.
d. Equipment Data Sheet: Typewritten data, using form at the end of this section.
e. Schedules: Clean, typewritten schedules reflecting final, as-installed conditions. Hand-written mark-ups of schedules submitted earlier are not acceptable.
f. Data that is poorly reproduced or in any way illegible will be rejected.

D. O&M Content shall be as follows:

1. Manufacturers' Published Data: Provide all available data, including installation and operating instructions, parts lists, electrical and mechanical schematics, control circuit documentation, performance data, safety instructions, cleaning and care instructions, and illustrations and instructions for maintenance, including lubrication, disassembly and repair, cleaning, and service. Indicate catalog numbers, sizes, colors, options, and other information pertaining to the products furnished which would be required when ordering replacements. For equipment assemblies, provide data for each separate item of equipment furnished as part of the assembly.
2. Custom Written Data: For data not in manufacturer's standard literature, provide text, drawings, and schematics specifically applicable to installed systems. Include step-by-step descriptions of operating procedures; identification of individual components and their functions; descriptions of how system components relate to one another and operate together to accomplish a common process or function; and sequence of operation for system control circuits.
3. For seasonally operated systems, provide clear, step by step start-up and shutdown instructions written so that nonprofessionals can do the job.
4. Equipment Data Sheets: For each item of equipment included in the operation and maintenance data, provide an Equipment Data Sheet using the form at the end of this section. For equipment consisting of both a driven machine and a driver (for example, a pump and a motor), the equipment data shall cover both the driven machine and the driver. For similar type equipment (for example, multiple exhaust

fans of the same model and type), provide a single equipment data sheet with an attached schedule listing the individual equipment items.

5. Vendor Furnished As-Built Drawings: Provide for each electrical and each mechanical control system.
   a. For each control system, provide control circuit schematic drawings, if control system is not Government furnished. Identify each wire and terminal block number. Show terminal numbers on all control devices. Show control wires and devices remote from the control panel.
   b. For each control panel, provide a general arrangement drawing showing location of each control component and terminal block on the panel front and interior. Include a materials list of all panel-mounted control components as well as field-installed control components remote from the panel, identifying components, manufacturer, model number, and initial set points or sensing ranges of devices where applicable.
   c. For packaged equipment systems, provide general arrangement drawings showing interrelationships of the various items of equipment and components.
   d. In addition to the control wiring schematic, provide a power wiring schematic drawing showing the power flow to each motor. Identify each power conductor. Show all overcurrent protection and motor starting devices.
6. Warranties: Place a copy of each manufacturer, supplier, and installer warranty extending for a period greater than one year in a single separately identified tabbed section of the manual.
7. Test Results: Include in the operation and maintenance data copies of test results for mechanical and electrical equipment and systems as listed in the individual specification sections.
8. Subcontractor and Supplier List: List all subcontractors and major suppliers who worked on the project. Include each subcontractor's or supplier's address and telephone number and identify work performed.

E. O&M Training:
   1. Contractor shall provide half-day training session on system operations for government personnel.

5.9 FINAL CLEANING

A. General: Provide final cleaning. Conduct cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.

B. Cleaning Agents: Use cleaning materials and agents recommended by manufacturer or fabricator of the surface to be cleaned. Do not use cleaning agents that are potentially hazardous to health or property or that might damage finished surfaces.

C. Complete the following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a portion of Project:

1. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
2. Remove tools, construction equipment, machinery, and surplus material from Project site.
3. Touch up and otherwise repair and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.

D. Comply with safety standards for cleaning. Do not burn waste materials. Do not bury debris or excess materials on Government property. Do not discharge volatile, harmful, or dangerous materials into drainage systems. Remove waste materials from Project site and dispose of lawfully.

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SECTION 010250
DEFINITION OF CONTRACT ITEMS
AND
MEASUREMENT AND PAYMENT

PART 1 – GENERAL

1.1 SUMMARY

A. The intent of this section is to explain, in general, what is and what is not included in a contract item; the limit or cut-off points where one item ends and another begins: and method of measurements and basis of payment for work items listed in the Schedule of Items.

B. Work: Furnishing all labor, materials, equipment, and other incidentals necessary to successfully complete the project or any portion of it and carrying out all duties and obligations imposed by the contract on the Contractor.

C. Payment: For each individual item listed here and in the Schedule of Items, payment shall be full compensation for all work related to the particular item in accordance with these specifications and as shown on the Drawings.

D. Measurement and payment for contract work shall be made only for and under those pay items included in the Schedule of Items. All other work and materials shall be considered incidental or as include in the payment for items shown.

1.2 DETERMINATION OF QUANTITIES

A. The Contractor shall perform, or cause to be performed, all measurement of quantities of materials incorporated into the work processes that are to be measured under the provisions of the contract.

B. Quantity Measurements
   1. The Contractor shall make all measurements for computation of quantities for all work items except those specified for payment by Actual Quantity (AQ), Designed Quantity) or Lump Sum Quantity (LSQ)
   2. The Contractor shall compute the quantities for periodic progress payments. The CO shall compute the quantities for the final payment based on measurements taken by the Contractor.
   3. All Contractor measurements are subject to verification.
   4. The Contractor shall submit all field notes, calculation sheets, and other data used to determine quantities.
   5. The Contractor shall certify in writing as to the accuracy of the measurement and computations submitted.

1.3 UNITS OF MEASUREMENT

A. Payment shall be by units defined and determined according to U.S. Standard measure and by the following:

1. Cubic Yard (CY): A measurement computed by one of the following methods:
   a. Excavation, Embankment, or Borrow. The measurement computed by the average end area method from measurements made longitudinally along a centerline or reference line.
   b. Material in Place or Stockpile. The measurement computed using the dimensions of the in-place material.
   c. Material in the Delivery Vehicle. The measurement computed using measurements of material in the hauling vehicles at the point of delivery. Vehicles shall be loaded to at least their water level capacity. Leveling of the loads may be required when vehicles arrive at the delivery point.
2. Each (EA): One complete unit, which may consist of one or more parts.
3. Linear Feet (LF): Linear feet measured horizontally.
4. Lump Sum Quantities (LSQ): Do not measure directly. The bid amount is complete payment for all work described in the contract and necessary to complete the work for that item.
5. Square Feet: Square Feet measured horizontally.
6. Square Yard: Square Yard (3 ft x 3 ft) measured horizontally.

PART 2 – METHOD OF MEASUREMENT

2.1 GENERAL

A. One of the following methods of measurement for determining final payment is designated on the Schedule of Items for each pay item.

B. ACTUAL QUANTITIES (AQ)
   1. These estimated quantities which are determined from actual measurements of completed work.

C. DESIGNED QUANTITIES (DQ)
   1. These quantities denote the final number of units to be paid for under the terms of the contract. They are based upon the original design data available prior to advertising the project. Original design data include the preliminary survey information, design assumptions, calculations, drawings, and the presentation in the contract. Changes in the number of units shown in the Schedule of Items may be authorized under any of the following conditions:
      a. As a result of changes in the work authorized by the CO.
      b. As a result of the CO determining that errors exist in the original design that cause a pay item quantity to change by 15 percent or more.
      c. As a result of the Contractor submitting to the CO a written request showing evidence of errors in the original design that cause a pay item quantity to change by 15 percent or more. The evidence must be verifiable and consist of

calculations, drawings, or other data that show how the designed quantity is believed to be in error.

D. LUMMP SUM QUANTITIES (LSQ)
   1. These quantities denote one complete unit of work as required by or described in the contract, including necessary material, equipment, and labor to complete the job. They shall be measured complete and in-place.

PART 3 – DEFINITION OF CONTRACT ITEMS

3.1 SCHEDULE OF ITEMS

A. The project Schedule of Items includes several sections to organize the contract items into Base Items and Optional Items. When a pay item is described as a Base Item, the description will be the same when that item or a similar item is included as an Optional Item.

B. Provide item pricing for all items listed in the Schedule of Items. Contract award will be made for all Base Items. Those items listed as Optional Items will be awarded based on available project funding. The Optional Items may be awarded in any order and in any combination.

3.2 DEFINITION OF BASE ITEMS (BI)

A. Base Item No. 1 – General Requirements/Mobilization
   1. Description of Work: This Item includes, but is not limited to, all preparatory work and operations necessary for the movement of personnel, equipment, supplies, bonding and incidentals to the project site; all quality control testing and quality control plan; all construction signing; any other work and operations that must be incurred prior to the beginning of work; and demobilization at the completion of work. This item also includes accident prevention and safety plan, protection of the public, submittals, construction site exit/drive off mat, concrete washout, erosion and sediment control, seeding, construction surveying, clearing and grubbing, O&M manuals and as-built drawings.
   2. Measurement: This item is measured as a Lump Sum Quantity.
   3. Progress payments shall be made as follows:
      a. When 5 percent or more of the original contract amount is earned from other base pay items (except Base Pay Item No. 1), 50 percent of the amount of this base pay item or 2.5 percent of the original contract amount will be paid, whichever is less.
      b. When 10 percent or more of the original contract amount is earned from other base pay items (except Base Pay Item No. 1), 100 percent of the amount of this base pay item or 5 percent of the original contract amount will be paid, whichever is less.
      c. Upon completion of all work on the project any unpaid amount for this item will be paid.

d. The total of all payments, including bonding, shall not exceed the original contract amount for this item

B. Base Bid Item No. 2 – Toilet Buildings Demolition
1. Description of Work: This Base Item consists of demolition, removal and disposal of five existing toilet buildings as shown on the Drawings and described in the Specifications. Demolition of the existing toilet buildings includes but is not limited to demolition of the building and associated concrete; transport and legal off-site disposal of demolition wastes. The toilet concrete vaults will be broken so as to not hold water, sides will be broken, and vault can be buried in place and covered with compacted fill. Government will have toilets vaults pumped prior to Contractor's work.
2. Measurement: This Item is measured as Each, based on demolition of five toilet buildings.
3. Payment: The contract unit price of Each shown in the Schedule of Items includes all materials, equipment, labor, and incidentals required for Demolition, removal, disposal and backfill of these buildings.

C. Base Bid Item No. 3 – Pumphouse Building Demolition
1. Description of Work: This Base Item consists of demolition, removal and disposal of existing pumphouse as shown on the Drawings and described in the Specifications. Contractor shall protect the existing well which is located inside the pumphouse and retain well for continued use. Demolition of the Pumphouse includes disconnect and removal of all electric services to pumphouse; disconnect and removal of all water system components including but not limited to water shutoff valves, water treatment chemical tanks and meters.
2. Measurement: This Item is measured as Each, based on demolition of one pumphouse.
3. Payment: The contract unit price of Each shown in the Schedule of Items includes all materials, equipment, labor, and incidentals required for Demolition, removal, disposal of these buildings.

D. Base Item No. 4 – Water System Demolition
1. Description of Work: This Base Item consists of demolition of existing water system components as shown on the Drawings and described in the Specifications. Work also includes salvaging existing site components as shown on the drawings and described in the specifications. Removal and disposal of the existing polyethylene above ground water tank, water hydrants and their concrete pads, and existing above ground valves, drains, and other incidentals associated with the existing water system. Abandoned water lines may be abandoned in place, however any part of abandoned water system lines that are an above ground safety risk or hazard shall be removed and site regraded to eliminate such hazards. Disposal to an approved landfill is included in this pay item.
2. Measurement: This item is measured as a Lump Sum Quantity.

3. Payment: The contract lump sum price shown in the Schedule of Items includes all materials, equipment, labor, and incidentals required for demolition and disposal of the existing water system.

E. Base Item No. 5 – General Demolition, Campground Furnishings and Amenities
   1. Description of Work: This Base Item includes all deliverables, materials, equipment, labor and incidental to remove and dispose of campground components as shown on the Drawings and describe in the Specifications. Items for Demolition in this Base Item include tables, fire rings, information boards, kiosks, fee tubes, gates, barriers, signs and posts, markers, and similar furnishings, amenities and components. Work also includes grading and revegetation of disturbed areas.
   2. Measurement: This item is measured as a Lump Sum Quantity.
   3. Payment: The contract lump sum prices as shown in the Schedule of Items includes all materials, equipment, labor, and incidentals required for demolition and disposal of the campground furnishings and amenities.

F. Base Item No. 6 – Existing Campground Road Improvements
   1. Description of Work: This Base Item includes all materials, equipment, labor and incidentals to reconstruct and improve the existing campground roads as shown on the Drawings and described in the Specifications. Work includes all earthwork, grading, aggregate and revegetation of disturbed areas. Drainage items including, but not limited to, culverts and associated bedding materials, dips, and ditches are included. Item also includes providing, placing, and compacting road surface aggregate materials and all associated incidental watering needed for meeting compaction requirements. Work includes construction of dumpster pads and pull-offs and similar road enhancements as shown on the Drawings.
   2. Work does not include the construction of any camping spurs or paths.
   3. Measurement: This item is measured as a Lump Sum Quantity.
   4. This contract Lump Sum Price as shown in the Schedule of Items includes all materials, equipment, labor, and incidentals required for complete and in-place Improvements to Existing Campground Roads.

G. Base Item No. 7 – Site Work for Camp Sites
   1. Description of Work: This Base Item includes all materials, equipment, labor and incidentals to install and construct camp sites, including parking spurs and living areas. This includes but is not limited to clearing and grubbing; soil removal; tree stump removal; cut and fill earthmoving; providing, hauling, and placing fill and removing unsuitable material by hauling off-site; providing, placing, grading, and compacting aggregate surfacing; providing and placing parking barriers and wheel stops and rock barriers; and grading and revegetation of disturbed areas.
   2. Site (Unit) Markers are NOT included in this item. They are included in Signage Base Item.
   3. Measurement: This item is measured as a Lump Sum Quantity.

4. Payment: This contract Lump Sum Price as shown in the Schedule of Items includes all materials, equipment, labor, and incidentals required for complete and in-place development and construction of Camp Sites.

H. Base Item No. 8 – New Pumphouse Building

1. Description of Work: This Base Item includes all materials, personnel, equipment, labor, supplies and incidentals for providing and installing pre-fabricated concrete pumphouse building to be fully constructed and operational. This Base Item includes site preparation, and furnishing and installing all components including excavation, bedding, backfill, final site grading, seeding, mulching, exterior concrete flatwork and subgrade, all interior plumbing and equipment, and all water lines outside the building up to 5 feet from the building.
2. Measurement: This item is measured as a Lump Sum Quantity.
3. Payment: The contract lump sum price shown in the Schedule of Items includes all materials, equipment, labor, and incidentals required for the New Pumphouse Building.

I. Base Item No. 9– Water Storage Tank

1. Description of Work: This Base Item consists of furnishing and installing a new underground water tank and appurtenances including, ladder, overflow/drain, perimeter drain, vent and liquid level sensors including all material and piping to within 10 feet horizontally from the edge of the tank. Line from tank to daylight drain for perimeter drain and overflow/tank drain are included in this item. Control panel for submersible well pump and float controls are included in this item. Also included in this item is tank bedding, excavation backfill, grading, seeding and mulching. Valves outside the tank are included in a separate pay item.
2. Measurement: This item is measured as a Lump Sum Quantity.
3. Payment: The contract lump sum price shown in the Schedule of Items includes all materials, equipment, labor, and incidentals required for new underground water storage tank.

J. Base Item No. 10 – Water System

Description of Work: This Base Item consists of all materials, personnel, equipment, labor, supplies, and incidentals for providing and constructing new water and wastewater systems as shown on Drawings and described in Specifications. This Item includes improvements to the well - new well cap, and positive drainage away from the well;; installation of new supply, return, and distribution lines from well to storage tank and to hydrants and drain lines, and construction of new hydrants on concrete pads in new locations. This item also includes new water treatment and storage equipment including but not limited to water meter, chemical pumps, chemical storage tank, and meters. This item also includes all valves, curb stops, drains, pipes, tees, connections, elbows, unions, thrust blocks, sensors, and other incidentals. Excavation, bedding, backfill, and compaction for water lines and water system components are also included in this

Base Item. This item also includes grading, stabilization, and revegetation of all disturbed areas in the direct vicinity of the water system and its components.

1. Measurement: This Item is measured as a Lump Sum Quantity
2. Payment: The contract lump sum price shown in the Schedule of Items includes all materials, equipment, labor and incidentals required for construction of the new water system complete and in-place. .

K. Base Item No.11– Host /Volunteer Site Construction,
   1. Description of Work: This Base Item includes all materials, personnel, equipment, labor, supplies and incidentals to complete all site work to install and construct a host site including earthwork, grading, slope stabilization, aggregate and surfacing, and revegetation of disturbed areas.
   2. Work does not include electrical work at the host sites or water or water connections or wastewater tanks or wastewater connections. These are in separate bid items.
   3. Measurement: This item is measured as Each Host Site.
   4. Payment: The Contract Unit Price of Each includes all materials, equipment, labor, materials, supplies, and incidentals for Each Host and Volunteer Site. Payment shall be made per each Host Site constructed and installed.

L. Base Item No. 12 – Host Site Wastewater Connections and Tanks
   1. Description of Work: This Base Item includes all materials, personnel, equipment, labor, supplies and incidentals to install wastewater holding tanks (vaults) and associated plumbing connections at each host site as shown on the Drawings and described in Specifications. Trenching and backfilling of connection piping is also included in this item. Grading, seeding and mulching of all disturbed areas associated with this work is included in this item.
   2. Measurement: This item is measured as a Lump Sum Quantity.
   3. Payment: The contract lump sum price as shown in the Schedule of Items includes all deliverables, materials, equipment, labor, and incidentals to install complete wastewater holding tanks and connections at each host site.

M. Base Item No. 13 –Electrical
   1. Description of Work: This Base Item includes all materials, equipment, labor and incidentals associated with electrical systems at the Campground including but not limited to the pumphouse, well, water tank, host site, and volunteer sites. This includes but is not limited to: power poles, transformers, service and distribution lines, breakers, switches, trenching and backfilling where electrical is routed independent of other utilities. Item also includes all interior electrical including all materials and labor for installation of switches, lighting fixtures, outlets, heaters and control panels. RV pedestals and associated conductors, conduit, and associated trenching and backfilling for providing full electric service at Host and Volunteer sites is also included. Work also includes providing and installing surge suppressors, breakers grounding equipment, utility meters and connections to meters, electrical panels, and wiring and site electrical. Additionally, this Base Item

includes trenching and backfilling where electrical is routed independent of other utilities. Decommissioning existing electric system components as shown on Drawings and reconstructing and improving electrical system components as shown on Drawings and described in Specifications, EXCEPT removal of Electrical System at the existing pumphouse is included in the Bid Item for Pumphouse Demolition.

2. Measurement: This item is measured as a Lump Sum Quantity.
3. Payment: The contract Lump Sum Price shown in the Schedule of Items includes all materials, equipment, labor, and incidentals required for site work.

N. Base Item No. 14 – New Toilet Buildings

1. Description of Work: -This Base Item includes materials, personnel, equipment, labor, supplies and incidentals for providing and installing pre-fabricated double riser concrete vault toilet building. Work of this Base Item includes site preparty work, clearing, grubbing, excavation, bedding, backfill, grading, compaction, and construction of approach paths and barriers. This work includes providing and placing, and ensuring full functionality of a pre-fabricate concrete, double riser, concrete vault toilet incorporating Sweet Smelling Technology (SST) design. This item also includes all toilet room amenities detailed in the Drawings and specifications, including but not limited to, vent stacks with bird screens, toilet paper dispensers, grab bars, and coat hooks as shown on the Drawings and described in the Specifications. Grading and revegetation of disturbed areas is included in this Item
2. Measurements: This item is measured as Each, based upon 4 new toilet buildings.
3. Payment: The contract unit price of Each shown in the Schedule of Items includes all materials, equipment, labor, and incidentals required for providing and installing new toilet buildings, complete and in-places as shown on the Drawings and described in the Specifications.

O. Base Item No. 15 – Gates

1. Description of Work: This Base Item includes all materials, personnel, equipment, labor, supplies and incidentals to provide and construct new campground road closure gate assemblies as shown on the Drawings and described in Specifications. Work includes associated excavation, earthwork, concrete, grading, and revegetation of disturbed areas in the direct area of the gates. Gates signage is included in separate Bid Item, Signage.
2. Measurement: This item is measured as Each.
3. The contract unit price of Each as shown in the Schedule of items includes all materials, personnel, equipment and supplies to install Each complete new gate assembly.

P. Base Item No. 16 – Campground Entrance Sign

1. Description of Work: This Base Bid Item includes all materials, personnel, equipment, labor, supplies and incidentals to install and construct a new campground entrance sign as shown on the Drawings and described in the

Specifications. Work includes all associated earthwork, excavation, concrete, fill and backfill, grading compaction, and revegetation of disturbed areas in the direct vicinity of the new sign.

2. Measurement: This item is measured as Each.
3. Payment: The contract unit price of Each as shown in the Schedule of items includes all materials, personnel, equipment and supplies to install complete new campground entrance sign.

Q. Base Item No. 17 – Signage

1. Description of Work: This Base Item includes all materials, personnel, equipment, labor, supplies and incidentals to provide and all signs throughout the campground as shown on the Sign Plan and Drawings and described in the Specifications. Work includes providing and installing signs and posts, clearing, excavation and footings as necessary, and revegetation of disturbed areas in the direct area of the signs.
2. This Base Item requires Contractor to provide all signs, posts and hardware shown on the Sign Plan, i~~ncluding those that are for Optional Items. If Optional Items are not awarded and signs related thereto are not installed, Contractor shall deliver all signs, posts and hardware that have not been installed to Government. No additional compensation to Contractor shall be due Contractor for not installing these signs pertaining to Option Items.~~
3. Measurement: This item is measured as a Lump Sum Quantity.
4. Payment: The contract lump sum price as shown in the Schedule of Items includes all materials, personnel, equipment and supplies to install all signs as identified on the Sign Plan, Drawings and Specifications.

END OF SECTION 010250

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SECTION 014000
QUALITY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes administrative and procedural requirements for quality assurance and quality control.

B. Testing and inspection services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with the Contract Document requirements.

1. Specified tests, inspections, and related actions do not limit Contractor's other quality-assurance and quality-control procedures that facilitate compliance with the Contract Document requirements.
2. Requirements for Contractor to provide quality-assurance and quality-control services required by Contracting Officer (CO) or authorities having jurisdiction are not limited by provisions of this Section.

1.2 DEFINITIONS

A. Field Quality-Control Tests and Inspections: Tests and inspections that are performed on-site for installation of the Work and for completed Work.

B. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, Subcontractor, or Sub-subcontractor, to perform a particular construction operation, including installation, erection, application, assembly, and similar operations.

1. Use of trade-specific terminology in referring to a Work result does not require that certain construction activities specified apply exclusively to specific trade(s).
2. constructed on-site in their actual final location as part of permanent construction.

C. Preconstruction Testing: Tests and inspections performed specifically for Project before products and materials are incorporated into the Work, to verify performance or compliance with specified criteria. Unless otherwise indicated, copies of reports of tests or inspections performed for other than the Project do not meet this definition.

D. Product Tests: Tests and inspections that are performed by a nationally recognized testing laboratory (NRTL) according to 29 CFR 1910.7, by a testing agency accredited according to NIST's National Voluntary Laboratory Accreditation Program (NVLAP), or by a testing agency qualified to conduct product testing and acceptable to authorities having jurisdiction, to establish product performance and compliance with specified requirements.

E. Source Quality-Control Tests and Inspections: Tests and inspections that are performed at the source; for example, plant, mill, factory, or shop.

F. Testing Agency: An entity engaged to perform specific tests, inspections, or both. The term "testing laboratory" has the same meaning as the term "testing agency."

G. Quality Assurance Services: Activities, actions, and procedures performed before and during execution of the Work to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.

H. Quality Control Services: Tests, inspections, procedures, and related actions during and after execution of the Work to evaluate that actual products incorporated into the Work and completed construction comply with requirements. Contractor's quality-control services do not include contract administration activities performed by CO.

1.3 CONFLICTING REQUIREMENTS

A. Conflicting Standards and Other Requirements: If compliance with two or more standards or requirements is specified and the standards or requirements establish different or conflicting requirements for minimum quantities or quality levels, inform the CO regarding the conflict and obtain clarification prior to proceeding with the Work. Refer conflicting requirements that are different, but apparently equal, to CO for clarification before proceeding.

B. Minimum Quantity or Quality Levels: The quantity or quality level shown or specified is the minimum provided or performed. The actual installation may comply exactly with the minimum quantity or quality specified, or it may exceed the minimum within reasonable limits. To comply with these requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to CO for a decision before proceeding.

1.4 INFORMATIONAL SUBMITTALS

A. Testing Agency Qualifications: For testing agencies specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include proof of qualifications in the form of a recent report on the inspection of the testing agency by a recognized authority.

B. Permits, Licenses, and Certificates: For Government's record, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents established for compliance with standards and regulations bearing on performance of the Work.

1.5 REPORTS AND DOCUMENTS

A. Test and Inspection Reports: Prepare and submit certified written reports specified in other Sections. Include the following:

1. Date of issue.
2. Project title and number.
3. Name, address, telephone number, and email address of testing agency.
4. Dates and locations of samples and tests or inspections.
5. Names of individuals making tests and inspections.
6. Description of the Work and test and inspection method.
7. Identification of product and Specification Section.
8. Complete test or inspection data.

9. Test and inspection results and an interpretation of test results.
10. Record of temperature and weather conditions at time of sample taking and testing and inspection.
11. Comments or professional opinion on whether tested or inspected Work complies with the Contract Document requirements.
12. Name and signature of laboratory inspector.
13. Recommendations on retesting and reinspecting.

B. Manufacturer's Technical Representative's Field Reports: Prepare written information documenting manufacturer's technical representative's tests and inspections specified in other Sections. Include the following:

1. Statement on condition of substrates and their acceptability for installation of product.
2. Statement that products at Project site comply with requirements.
3. Summary of installation procedures being followed, whether they comply with requirements and, if not, what corrective action was taken.
4. Results of operational and other tests and a statement of whether observed performance complies with requirements.
5. Other required items indicated in individual Specification Sections.

C. Factory-Authorized Service Representative's Reports: Prepare written information documenting manufacturer's factory-authorized service representative's tests and inspections specified in other Sections. Include the following:

1. Statement that equipment complies with requirements.
2. Results of operational and other tests and a statement of whether observed performance complies with requirements.
3. Other required items indicated in individual Specification Sections.

1.6 QUALITY ASSURANCE

A. Qualifications paragraphs in this article establish the minimum qualification levels required; individual Specification Sections specify additional requirements.

B. Manufacturer Qualifications: A firm experienced in manufacturing products or systems similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units. As applicable, procure products from manufacturers able to meet qualification requirements, warranty requirements, and technical or factory-authorized service representative requirements.

C. Fabricator Qualifications: A firm experienced in producing products similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

D. Installer Qualifications: A firm or individual experienced in installing, erecting, applying, or assembling work similar in material, design, and extent to that indicated for this Project, whose work has resulted in construction with a record of successful in-service performance.

E. Testing and Inspecting Agency Qualifications: An NRTL, an NVLAP, or an independent agency with the experience and capability to conduct testing and inspection indicated, as documented according to ASTM E329; and with additional qualifications specified in individual Sections; and, where required by authorities having jurisdiction, that is acceptable to authorities.

F. Manufacturer's Technical Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to observe and inspect installation of manufacturer's products that are similar in material, design, and extent to those indicated for this Project.

G. Factory-Authorized Service Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to inspect, demonstrate, repair, and perform service on installations of manufacturer's products that are similar in material, design, and extent to those indicated for this Project.

1.7 QUALITY CONTROL

A. Contractor Responsibilities: Tests and inspections not explicitly assigned to Government are Contractor's responsibility. Perform additional quality-control activities, whether specified or not, to verify and document that the Work complies with requirements.

1. Engage a qualified testing agency to perform quality-control services.
   a. Contractor will not employ same entity engaged by Government, unless agreed to in writing by CO.
2. Notify testing agencies at least 24 hours in advance of time when Work that requires testing or inspection will be performed.
3. Where quality-control services are indicated as Contractor's responsibility, submit a certified written report, in duplicate, of each quality-control service.
4. Testing and inspection requested by Contractor and not required by the Contract Documents are Contractor's responsibility.
5. Submit additional copies of each written report directly to authorities having jurisdiction, when they so direct.

B. Retesting/Reinspecting: Regardless of whether original tests or inspections were Contractor's responsibility, provide quality-control services, including retesting and reinspecting, for construction that replaced Work that failed to comply with the Contract Documents.

C. Testing Agency Responsibilities: Cooperate with CO and Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.

1. Notify CO and Contractor promptly of irregularities or deficiencies observed in the Work during the performance of its services.
2. Determine the locations from which test samples will be taken and in which in-situ tests are conducted.
3. Conduct and interpret tests and inspections and state in each report whether tested and inspected Work complies with or deviates from requirements.
4. Submit a certified written report, in duplicate, of each test, inspection, and similar quality-control service through Contractor.
5. Do not release, revoke, alter, or increase the Contract Document requirements or approve or accept any portion of the Work.
6. Do not perform duties of Contractor.

D. Manufacturer's Field Services: Where indicated, engage a factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing as specified in Section 013300 "Submittal Procedures."

E. Manufacturer's Technical Services: Where indicated, engage a manufacturer's technical representative to observe and inspect the Work. Manufacturer's technical representative's services include participation in pre-installation conferences, examination of substrates and conditions, verification of materials, observation of Installer activities, inspection of completed portions of the Work, and submittal of written reports.

F. Contractor's Associated Requirements and Services: Cooperate with agencies and representatives performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide the following:

1. Access to the Work.
2. Incidental labor and facilities necessary to facilitate tests and inspections.
3. Adequate quantities of representative samples of materials that require testing and inspection. Assist agency in obtaining samples.
4. Facilities for storage and field curing of test samples.
5. Preliminary design mix proposed for use for material mixes that require control by testing agency.
6. Security and protection for samples and for testing and inspection equipment at Project site.

G. Coordination: Coordinate sequence of activities to accommodate required quality-assurance and quality-control services with a minimum of delay and to avoid necessity of removing and replacing construction to accommodate testing and inspection.

1. Schedule times for tests, inspections, obtaining samples, and similar activities.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 TEST AND INSPECTION LOG

A. Test and Inspection Log: Prepare a record of tests and inspections. Include the following:

1. Date test or inspection was conducted.
2. Description of the Work tested or inspected.
3. Date test or inspection results were transmitted to CO.
4. Identification of testing agency or special inspector conducting test or inspection.

B. Maintain log at Project site. Post changes and revisions as they occur. Provide access to test and inspection log for CO reference during normal working hours.

1. Submit log at Project closeout as part of Project Record Documents.

3.2 REPAIR AND PROTECTION

A. General: On completion of testing, inspection, sample taking, and similar services, repair damaged construction and restore substrates and finishes.

1. Provide materials and comply with installation requirements specified in other Specification Sections or matching existing substrates and finishes. Restore patched areas and extend restoration into adjoining areas with durable seams that are as invisible as possible. Comply with the Contract Document requirements for cutting and patching in Section 017300 "Execution."

B. Protect construction exposed by or for quality-control service activities.

C. Repair and protection are the Contractor’s responsibility, regardless of the assignment of
D. responsibility for quality-control services.

END OF SECTION 014000

SECTION 015000
TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes requirements for temporary utilities, support facilities, and security and protection facilities.

1.2 USE CHARGES

A. Installation, removal, and use charges for temporary facilities shall be included in the Contract Sum unless otherwise indicated. Allow other entities engaged in the Project to use temporary services and facilities without cost, including, but not limited to, Contracting Officer, testing agencies, and authorities having jurisdiction.

B. Water and Sewer Service from Existing System: Water from Owner's existing water system will not be available for use.

C. Electric Power Service from Existing System: Electrical services are available at this site. Contractor will be responsible for all communication with the utility provider and all charges associated with the use of in place and newly installed electricity.

1.3 INFORMATIONAL SUBMITTALS

A. Site Utilization Plan: Show temporary facilities, temporary utility lines and connections, staging areas, construction site entrances, vehicle circulation, and parking areas for construction personnel.

B. Fire-Safety Program: Show compliance with requirements of NFPA 241 and authorities having jurisdiction. Indicate Contractor personnel responsible for management of fire-prevention program.

1.4 QUALITY ASSURANCE

A. Electric Service: Comply with NECA, NEMA, and UL standards and regulations for temporary electric service. Install service to comply with NFPA 70.

B. Tests and Inspections: Arrange for authorities having jurisdiction to test and inspect each temporary utility before use. Obtain required certifications and permits.

1.5 PROJECT CONDITIONS

A. Temporary Use of Permanent Facilities: Engage Installer of each permanent service to assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before Owner's acceptance, regardless of previously assigned responsibilities.

1.6 TEMPORARY TREE AND PLANT PROTECTION

A. Plant Protection Zone: Area surrounding individual trees or groups of trees to be protected during construction and with a minimum radius of 96 inches unless otherwise indicated.

PART 2 - PRODUCT

2.1 TEMPORARY FACILITIES

A. Field Offices: Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.

2.2 EQUIPMENT

A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.

B. HVAC Equipment: Unless Owner authorizes use of permanent HVAC system, provide vented, self-contained, liquid-propane-gas or fuel-oil heaters with individual space thermostatic control.

1. Use of gasoline-burning space heaters, open-flame heaters, or salamander-type heating units is prohibited.

C. Plant Protection Zone Fencing: Fencing fixed in position and meeting the follogin requirements:

1. Plastic Protection-Zone Fencing: Plastic construction fencing constructed of high-density extruded and stretched polyethylene fabric with 2-inch (50-mm) maximum opening in pattern and supported by tubular or T-shape galvanized-steel posts spaced not more than 96 inches (2400 mm) apart. Height not less than 48 inches (1200 mm). High-visibility orange color.

D. Plant Protection Zone Signage: Shop-fabricated, rigid plastic or metal sheet with attachment holes prepunched and reinforced; legibly printed with nonfading lettering.

E.

PART 3 - EXECUTION

3.1 TEMPORARY FACILITIES, GENERAL

A. Conservation: Coordinate construction and use of temporary facilities with consideration given to conservation of energy, water, and materials. Coordinate use of temporary utilities to minimize waste.

3.2 INSTALLATION, GENERAL

A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.

B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.3 TEMPORARY UTILITY INSTALLATION

A. General: Install temporary service or connect to existing service.

1. Arrange with utility company, Owner, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services.

B. Water Service: Install water service and distribution piping in sizes and pressures adequate for construction.

C. Sanitary Facilities: Provide temporary toilets, wash facilitiesand drinking water for use of construction personnel. Provide safety shower and eyewash facilities for construction personnel in accordance with OSHA requirements and risk assessment and management. Comply with requirements of authorities having jurisdiction for type, number, location, operation, and maintenance of fixtures and facilities.

D. Temporary Heating and Cooling: Provide temporary heating and cooling required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of low temperatures or high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed.

E. Electric Power Service: Provide electric power service and distribution system of sufficient size, capacity, and power characteristics required for construction operations.

1. Install electric power service overhead unless otherwise indicated.

F. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, inspections, and traffic conditions.

1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.

3.4 SUPPORT FACILITIES INSTALLATION

A. Comply with the following:

1. Request approval from Contracting Officer (CO) for location of temporary field office and support facilities.
2. Utilize area designated by CO for temporary field office.
3. Maintain support facilities for the minimum duration necessary. Remove before Substantial Completion..

B. Temporary Roads and Paved Areas:

1. Contractor shall not be permitted to construct new temporary roads.
2. Limit construction activities to areas indicated on Drawings.

C. Traffic Controls: Comply with requirements of Contracting Officer and authorities having jurisdiction.

1. Protect existing site improvements to remain including curbs, pavement, and utilities.
2. Maintain access for fire-fighting equipment.
3. The Gate at the road accessing the point shall be closed and locked at all times except when construction traffic is entering/exiting the site.

D. Parking: Contracting parking shall be located within the project area.

1. Do not park on highway.
2. Do not block access bridge.
3. Use designated areas of existing site areas for parking for construction personnel.

E. Storage and Staging: An area at or near the project site will be made available for use as a staging area. Staging areas shall be approved by the CO prior to use.

F. Camping: Contractor will be allowed to camp at the site, with approval from the Contracting Officer

G. Contractor may use existing electric service at the site.

1. The Campground will be closed and all services will be shut off. Contractor will be responsible for contacting electric utility to have power turned on. Contractor shall pay all fees for power connection and usage.

H. Security and clean-up of the site shall be the responsibility of the Contractor.

1. Contractor shall keep the site gated and locked at all times..

I. Dewatering Facilities and Drains: Comply with requirements of authorities having jurisdiction. Maintain Project site, excavations, and construction free of water.

1. Dispose of rainwater in a lawful manner that will not result in flooding Project or adjoining properties or endanger permanent Work or temporary facilities.
2. Remove snow and ice as required to minimize accumulations.

J. Contractor Identification Project signs are not permitted.

K. Temporary Signs: Provide other signs as indicated and as required to inform public and individuals seeking entrance to Project.

1. Maintain and touch up signs so they are legible at all times.

L. Waste Disposal Facilities: Provide waste-collection containers in sizes adequate to handle waste from construction operations. Comply with requirements in Section 010150 “General Requirements” for Waste Management and Disposal.

3.5 SECURITY AND PROTECTION FACILITIES INSTALLATION

A. Protection of Existing Facilities: Protect existing vegetation, equipment, structures, utilities, and other improvements at Project site and on adjacent properties, except those indicated to be removed or altered. Repair damage to existing facilities.

   1. Where access to adjacent properties is required in order to affect protection of existing facilities, obtain written permission from adjacent property owner to access property for that purpose.

B. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction as required to comply with environmental regulations and that minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.

C. Temporary Erosion and Sedimentation Control: Provide measures to prevent soil erosion and discharge of soil-bearing water runoff and airborne dust to undisturbed areas and to adjacent properties and walkways, according to requirements of EPA Construction General Permit or authorities having jurisdiction, whichever is more stringent.

   1. Verify that flows of water redirected from construction areas or generated by construction activity do not enter or cross Plant Protection Zones.
   2. Inspect, repair, and maintain erosion- and sedimentation-control measures during construction until permanent vegetation has been established.
   3. Clean, repair, and restore adjoining properties and roads affected by erosion and sedimentation from Project site during the course of Project.
   4. Remove erosion and sedimentation controls and restore and stabilize areas disturbed during removal.

D. Stormwater Control: Comply with requirements of authorities having jurisdiction. Provide barriers in and around excavations and subgrade construction to prevent flooding by runoff of stormwater from heavy rains.

E. Tree and Plant Protection:

   1. Notify Contracting Officer of trees of any size that interfere with or are affected by construction. Do not cut or remove trees without approval of CO.
   2. Notify CO of scope and extent of pruning of existing trees to meain that interfere with or are affected by construction. Do not trim or prune vegetation without approval of CO.
   3. Do not direct vehicle or equipment exhaust toward plant protection zones.
   4. Prohibit heat sources, flames, ignition sources, and smoking within or near Protection Zones and organic mulch.
   5. Protect tree root systems from damage caused by runoff or spillage of noxious materials while mixing, placing, or storing construction materials. Protect root systems from ponding, eroding, or excessive wetting caused by dewatering operations.
   6. : Install protection-zone fencing along edges of protection zones in a manner that will prevent people from easily entering protected areas
   7. Install temporary fencing located as indicated or outside the drip line of trees to protect vegetation from damage from construction operations. Protect tree root systems from damage, flooding, and erosion.
   8. Install protection-zone signage in visibly prominent locations in a manner approved by CO.
   9. Maintain Protection Zones free of weeds and trash.

10. Maintain Protection Zones fencing and signage in good condition as acceptable to CO and remove when construction operations are complete and equipment has been removed from the site.

F. Security Enclosure and Lockup: Install temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security. Lock entrances at end of each workday.

G. Barricades, Warning Signs, and Lights: Comply with requirements of authorities having jurisdiction for erecting structurally adequate barricades, including warning signs and lighting.

H. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.

I. Temporary Fire Protection: Install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241; manage fire-prevention program.

1. Prohibit smoking in construction areas. Comply with additional limits on smoking specified in other Sections.
2. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition according to requirements of authorities having jurisdiction.
3. Develop and supervise an overall fire-prevention and -protection program for personnel at Project site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
4. Provide temporary standpipes and hoses for fire protection. Hang hoses with a warning sign stating that hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.

3.6 MOISTURE AND MOLD CONTROL

A. Exposed Construction Period: Before installation of weather barriers, when materials are subject to wetting and exposure and to airborne mold spores, protect as follows:

1. Protect porous materials from water damage.
2. Protect stored and installed material from flowing or standing water.

B. Partially Enclosed Construction Period: After installation of weather barriers but before full enclosure and conditioning of building, when installed materials are still subject to infiltration of moisture and ambient mold spores, protect as follows:

1. Do not load or install drywall or other porous materials or components, or items with high organic content, into partially enclosed building.
2. Keep interior spaces reasonably clean and protected from water damage.
3. Periodically collect and remove waste containing cellulose or other organic matter.
4. Discard or replace water-damaged material.
5. Do not install material that is wet.
6. Discard and replace stored or installed material that begins to grow mold.
7. Perform work in a sequence that allows wet materials adequate time to dry before enclosing the material in gypsum board or other interior finishes.

C. Controlled Construction Period: After completing and sealing of the building enclosure but prior to the full operation, control moisture and humidity inside building by maintaining effective dry-in conditions.

3.7 OPERATION, TERMINATION, AND REMOVAL

A. Supervision: Enforce strict discipline in use of temporary facilities. To minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.

B. Maintenance: Maintain facilities in good operating condition until removal.

1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on a 24-hour basis where required to achieve indicated results and to avoid possibility of damage.

C. Termination and Removal: Remove each temporary facility when need for its service has ended, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.

1. Materials and facilities that constitute temporary facilities are property of Contractor. Owner reserves right to take possession of Project identification signs.
2. At Substantial Completion, repair, renovate, and clean permanent facilities used during construction period. Comply with final cleaning requirements specified in Section 010150 “Genera Requirements, subpart "Closeout Procedures."

END OF SECTION 015000

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SECTION 016440
GOVERNMENT FURNISHED MATERIALS

PART 1 – SUMMARY OF WORK

1.1 DESCRIPTION

A. The Government’s intent is that there will be no Government Furnished Materials (GFMs) that Contractor will Install. There are Materials that Government intends to provide and self-install.

B. In the event Government conditions and plans change, and Government needs to issue a Contract Modification for Contractor to install these materials, this Section will apply. In this event, Contractor would be responsible for labor, equipment, and incidentals necessary to acquire these materials and for placement in accordance with the plans and specifications.

1.2 INSPECTION

A. The Contractor shall notify the Contracting Officer (CO) at least 5 calendar days prior to picking up or returning Government-furnished materials.

1. Government-furnished materials are in storage at the Black Hills National Fores Supervisor’s Office at 1019 N. 5th Street, Custer, SD.
2. Time for material pick-up will be coordinated between Contractor and CO.
3. Material may be picked up between the hours of 8:00 AM and 4:00 PM, Monday through Friday, as agreed upon with the Contractor and CO.

B. The Contractor and the CO shall jointly inspect all materials during the loading operation at the storage area and at the project site.

1.3 LIABILITY

A. The Contractor's liability for GFMs shall start at the time he accepts the materials at the storage area and shall terminate upon completion and final acceptance of the work.

1. It shall be implied that the Contractor has accepted GFMs prior to the materials being loaded and ready for transportation from the storage area to the worksite.
2. The Contractor shall provide all labor, equipment, and incidentals necessary to load, unload, and transport Government furnished materials between the worksite and the specified storage area.

1.4 REPLACEMENT AND/OR SUPPLEMENTAL MATERIALS

A. Any defective materials discovered prior to the Contractor's acceptance of the materials will be replaced by the Government.

B. If the quantity of GFMs accepted by the Contractor is insufficient to complete the specified work, supplemental materials complying with these specifications shall be provided by the Government at no expense to the Contractor.

C. The Government shall not replace any materials that, in the opinion of the CO, are damaged or lost due to the Contractor's negligence after the Contractor accepts the materials. Such materials shall be replaced at the Contractor's expense.

1.5 SURPLUS MATERIAL

A. After completion and final acceptance of the work, surplus GFMs shall be returned to the specified storage area in the same mechanical and physical condition as originally accepted.

B. Damaged surplus materials shall be replaced at the Contractor's expense.

C. The Contractor’s liability for surplus materials shall terminate upon unloading at the storage area.

PART 2 – PRODUCTS

2.1 GOVERNMENT-FURNISHED MATERIALS

A. Installation and testing of GFMs shall comply with the applicable contract specifications covering work requiring the use of such materials.

B. It is the Government's intent that there will be no Government-Furnished Materials installed by the Contractor for this project.

C. The Government will be providing the following Materials with the intent of having them installed by the Government. Contractor is not required to install these items. Contractor is not required to transport these materials from Government storage site to project site.

1. Picnic Tables
2. Fire Rings
3. Kiosk/Information Boards
4. Fee Tubes

PART 3 – EXECUTION

3.1 GENERAL

A. It is the Government's intent that there will be no Government-Furnished Materials installed by the Contractor for this project.

END OF SECTION 016440

Section 002222 –
Excavation, Backfilling and Compaction

GENERAL

1.01 Scope: Includes excavation, general backfill, gravel bedding, special bedding and compaction around and under concrete flatwork.

1.02 Protection:

A. Protect excavations from accumulation of water. If soil becomes saturated, compact after it has dried as specified herein.

B. Protect excavations from freezing. If frost action occurs, compact ground after thawing as specified herein.

C. Provide shoring and bracing as necessary to protect structures employees, and the public. Open excavations shall comply with the requirements and restrictions of OSHA.

D. All footer excavations shall be in reasonably close conformity with the dimensions, lines, and grades shown on the drawings. Any shifting or change from the drawings must receive prior approval of the CO.

1.03 Classification: All excavation shall be unclassified.

1.04 References: Standard Moisture Density Test AASHTO T-99, Method C or D.

1.05 Approvals Required:

A. Excavation: The Contractor shall notify the CO, a minimum of 2 working days, when excavation is completed and prior to placing any bedding material or concrete.

B. Backfill: The Contractor shall not place any backfill until the CO has approved the work.

PRODUCTS

2.01 Sources of Backfill and Fills: Excess excavated material from work required by this contract and/or borrow material from the following:

A. No borrow source available

2.02 Materials: Use the excavated material as specified below.

A. Remove rocks over 4 inches in maximum dimension, ice or frozen earth, muck, debris, and earth with high void content.

B. Materials removed in stripping shall not be used for backfill.

C. Backfill shall have moisture content within +1 to -2% of optimum.

2.03 Gravel Fills: Clean pit-run or crushed gravel uniformly graded with less than 5% minus #200 material, ¾ inch maximum size, free of deleterious materials.

EXECUTION

3.01 Stripping: Strip organic material in accordance with Section 2215.

3.02 Excavation: Perform to the lines, grades, and elevations indicated. Extend excavation a sufficient distance for safety, placing and removal of forms, installation of services, and for inspection. Provide proper depth with allowance made for gravel fills, formed voids and concrete flatwork.

A. Bottoms for slabs and pathways shall be level, clean, true to size, and clear of loose materials.

B. If rock is encountered, remove it for a minimum depth of 6 inches below the excavation level.

C. Roots projecting into the excavation shall be cut flush with the face of the excavation.

D. Surplus excavated material shall be used as needed at other site locations in this contract. Large rock and other unacceptable material that cannot be incorporated into the work shall be removed from the site and disposed of by the Contractor.

3.03 Backfilling: Place and compact fills and backfills adjacent to structures in such a manner as to prevent wedging action or eccentric lodging upon or against the concrete.

A. Remove forms, and clean the excavation of trash and debris prior to backfilling.

B. Do not place backfill on frozen or wet material.

C. Do not place backfill against any concrete without prior permission of the CO and in no case less than 7 days after placement.

D. Place backfill in horizontal layers not more than 6 inches thick with the proper moisture content for the required degree of compaction. Flooding or puddling not allowed. Compact each layer as specified. Backfill layer thickness under concrete flatwork limited to 6 inches maximum.

E. Provide for anticipated settlement and shrinkage of the backfill and for the finished grades required, including topsoil grading.

F. Prepare sub grades for gravel fill areas level and free from soft spots. Place and compact earth fill where required to raise sub grade. Place gravel to the lines and grades shown. Compact the gravel sub-base as specified until a smooth, uniform use surface is obtained.

G. For concrete flatwork, a minimum of 4" gravel bedding shall be required.

3.04 <u>Compaction</u>: Compact each layer to following percentages of maximum density determined by AASHTO T-99, Method C or D. It is the Contractor's responsibility to hire a third party testing firm to perform the needed compaction tests.

A. Everywhere: 95%.

3.05 <u>General Borrow</u>: If additional material is required to bring subsequent backfill to the finished grade specified, the Contractor shall be required to supply the backfill at no additional cost to the Government. All materials used for backfill shall contain no rocks larger than 4" in diameter and with moisture content within +1% to -2% of optimum.

END OF SECTION

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Section 002240 – Crushed Aggregate Base or Surface Course

GENERAL

1.01 Scope: Includes furnishing, hauling, and placing of one or more courses of aggregate bases or surface courses, on an approved subgrade or base.

1.02 References:

AASHTO T11: Materials Finer Than 75-pm (no. 200) Sieve in Mineral Aggregates by Washing Unit Weight and Voids of Aggregate
AASHTO T27: Sieve Analysis of Fine and Course Aggregates
AASHTO T89: Determining the Liquid Limit of Soils
AASHTO T90: Determining the Plastic Limit and Plasticity Index of Soils
AASHTO T96: Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
AASHTO T99: The Moisture-Density Relations of Soils Using a 2.5kg (5.5-lb) Rammer and a 305-mm (12-in) Drop
AASHTO T176: Plastic Fines in Graded Aggregates and Soils by Use of the Sand Equivalent Test
AASHTO T210: Aggregate Durability Index

PRODUCTS

2.01 Materials shall be obtained from a commercial source. Crushed aggregate for surface courses shall be Minnekhata limestone. The gradations and materials shall be approved by the CO prior to placing.

2.02 Gradation – The crushed aggregate gradation shall meet the following requirements for the grading shown on the drawing or listed in the Schedule of Items. Aggregate shall be well graded from coarse to fine within the gradation band.

Sieve Size (inch)	% by Weight Passing Designated Sieve (AASHTO T27 and T11) — Grading Designation: 3/4" Minus	1/4" Minus Crusher Fines
3/4 inch	100	
1/2 inch		
1/4 inch		100
No. 4	46-67	
No. 8	29-53	53-83
No. 16		33-53
No. 40	10-28	20-40
No. 50		
No. 100		
No. 200	3-12	8-20

A. Aggregate Certification: The commercial source shall provide a statement from a certified testing facility stating that aggregate from this source meets the requirements of these specifications and the designated gradation. The CO may at any time sample the deliverance and make compliance tests.

B. Variance from the Specified Requirements: Variances from the crushed aggregate quality and gradation requirements shall be approved in writing by the CO prior to use of the material on the project.

EXECUTION

3.01 Preparation of the Subgrade:

A. The subgrade shall conform to the grades and lines shown on the drawings. Rock requiring blasting, rippable rock, and boulders shall be excavated to a minimum depth of 6 inches below subgrade unless otherwise shown on the drawings and replaces with a suitable cushion material obtained from within the excavation area or from sources approved by the CO. Rocks larger than 4 inches that do not protrude above the subgrade more than one third of the depth of the base or surface course, or 3 inches whichever is less, may be left in place.

B. Suitable excavated material shall be utilized in the preparation of the subgrade. Some suitable material may be needed at other locations within the site. Unsuitable excavated material shall be disposed of as directed by the CO. Embankments shall be placed in compacted layers not exceeding 8 inches (loose measure).

C. Subgrade shall be compacted by three passes of a mechanical type tamper or other approved compaction machine or until visual displacement ceases.

D. The subgrade shall be completed and approved by the CO before placing base or surface course.

3.02 Mixing and Placing:

A. The moisture content of the aggregate shall be suitable for compaction and uniform throughout. The aggregate shall be spread in a uniform layer, with no segregation of size, and to a loose depth that shall have the required thickness when compacted. Hauling equipment shall be dispersed over the surface of the previously constructed layer to minimize rutting or uneven compaction.

B. Water shall be added to the crusher fines during placement if necessary to enhance the compaction process.

3.03 Compaction:

A. The aggregate shall be compacted by Compaction B.

1. Compaction B: Aggregate shall be moistened or dried to uniform moisture content suitable for compaction. A mechanical vibrating tamper or other approved compaction machine shall be operated over the full width of each layer until visual displacement ceases, but not fewer than three complete passes.

B. Thickness Requirements: The thickness of the compacted aggregate shall not vary more than ½ inch for aggregates with a nominal maximum particle size of 1 inch or less from the thickness shown on the drawings. The compacted thickness shall not be consistently above or below the specified thickness.

END OF SECTION

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SECTION 033000
CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 SUMMARY

A. This Section specifies cast-in-place concrete, including reinforcement, concrete materials, mix design, placement procedures, and finishes.

1.2 SUBMITTALS

A. Product Data: For each type of manufactured material and product indicated.

B. Design Mixes: For concrete mix.

1.3 QUALITY ASSURANCE

A. Source Limitations: Obtain each type of cement of the same brand from the same manufacturer's plant, each aggregate from one source, and each admixture from the same manufacturer.

B. Comply with ACI 301, "Specification for Structural Concrete," including the following, unless modified by the requirements of the Contract Documents.

1. General requirements, including submittals, quality assurance, acceptance of structure, and protection of in-place concrete.
2. Formwork and form accessories.
3. Steel reinforcement and supports.
4. Concrete mixtures.
5. Handling, placing, and constructing concrete.

PART 2 – PRODUCTS

2.1 FORMWORK

A. Furnish formwork and form accessories according to ACI 301.

2.2 STEEL REINFORCEMENT

A. Reinforcing Bars: ASTM A 615/A 615M, Grade 60 (Grade 420), deformed.

2.3 CONCRETE MATERIALS

A. Portland Cement: ASTM C 150, Types I or II.

B. Normal-Weight Aggregate: ASTM C 33, uniformly graded, not exceeding 1-inch (25-mm) nominal size.

C. Water: Potable and complying with ASTM C 94.

2.4 ADMIXTURES

A. General: Admixtures certified by manufacturer to contain not more than 0.1 percent water-soluble chloride ions by mass of cement and to be compatible with other admixtures. Do not use admixtures containing calcium chloride.

B. Air-Entraining Admixture: ASTM C 260.

C. Water-Reducing Admixture: ASTM C 494, Type A.

2.5 CURING MATERIALS

A. Evaporation Retarder: Waterborne, monomolecular film forming, manufactured for application to fresh concrete.

B. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. (305 g/sq. m) dry.

C. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.

D. Water: Potable.

2.6 CONCRETE MIXES

A. Comply with ACI 301 requirements for concrete mixtures.

B. Prepare design mixes, proportioned according to ACI 301, for normal-weight concrete determined by either laboratory trial mix or field test data bases, as follows:

1. Compressive Strength (28 Days): 4000 psi (20.7 MPa).
2. Slump: 4 inches (100 mm).

C. Add air-entraining admixture at manufacturer's prescribed rate to result in concrete at point of placement having an air content of 6.0 percent within a tolerance of plus 1.0 or minus 1.5 percent.

2.7 CONCRETE MIXING

A. Ready-Mixed Concrete: Comply with ASTM C 94.

1. When air temperature is between 85 and 90 deg F (30 and 32 deg C), reduce mixing and delivery time from 1-1/2 hours to 75 minutes; when air temperature is above 90 deg F (32 deg C), reduce mixing and delivery time to 60 minutes.

2. Do not place concrete when air temperature is below 50 degrees F (10 degrees C).

B. Project-Site Mixing: Measure, batch, and mix concrete materials and concrete according to ASTM C 94. Mix concrete materials in appropriate drum-type batch machine mixer.

1. For mixer capacity of 1 cu. yd. (0.76 cu. m) or smaller, continue mixing at least one and one-half minutes, but not more than five minutes after ingredients are in mixer, before any part of batch is released.
2. For mixer capacity larger than 1 cu. yd. (0.76 cu. m), increase mixing time by 15 seconds for each additional 1 cu. yd. (0.76 cu. m).
3. Provide batch ticket for each batch discharged and used in the Work, indicating Project identification name and number, date, mix type, mix time, quantity, and amount of water added. Record approximate location of final deposit in structure.

PART 3 – EXECUTION

3.1 FORMWORK

A. Design, construct, erect, shore, brace, and maintain formwork according to ACI 301.

3.2 STEEL REINFORCEMENT

A. Comply with CRSI's "Manual of Standard Practice" for fabricating, placing, and supporting reinforcement.

3.3 JOINTS

A. General: Construct joints true to line with faces perpendicular to surface plane of concrete.

B. Construction Joints: Locate and install so as not to impair strength or appearance of concrete, at locations indicated or as approved by Contracting Officer.

C. Contraction (Control) Joints in Slabs-on-Grade: Form weakened-plane contraction joints, sectioning concrete into areas as indicated. Construct contraction joints for a depth equal to at least one-fourth of the concrete thickness, as follows:

1. Grooved Joints: Form contraction joints after initial floating by grooving and finishing each edge of joint with groover tool to a radius of 1/8 inch (3 mm). Repeat grooving of contraction joints after applying surface finishes. Eliminate groover marks on concrete surfaces.
2. Sawed Joints: Form contraction joints with power saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut 1/8-inch- (3-mm-) wide joints into concrete when cutting action will not tear, abrade, or otherwise damage surface and before concrete develops random contraction cracks.

3.4 CONCRETE PLACEMENT

A. Comply with recommendations in ACI 304R for measuring, mixing, transporting, and placing concrete.

B. Do not add water to concrete during delivery, at Project site, or during placement.

3.5 FINISHING FORMED SURFACES

A. Rough-Formed Finish: As-cast concrete texture imparted by form-facing material with tie holes and defective areas repaired and patched, and fins and other projections exceeding 1/4 inch (6 mm) in height rubbed down or chipped off.

1. Apply to concrete surfaces not exposed to public view.

3.6 FINISHING UNFORMED SURFACES

A. General: Comply with ACI 302.1R for screeding, restraightening, and finishing operations for concrete surfaces. Do not wet concrete surfaces.

B. Screed surfaces with a straightedge and strike off. Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane before excess moisture or bleedwater appears on the surface.

1. Do not further disturb surfaces before starting finishing operations.

C. Nonslip Broom Finish: Apply a nonslip broom finish to surfaces indicated and to exterior concrete platforms, steps, and ramps. Immediately after float finishing, slightly roughen trafficked surface by brooming with fiber-bristle broom perpendicular to main traffic route.

3.7 TOLERANCES

A. Comply with ACI 117, "Specifications for Tolerances for Concrete Construction and Materials."

3.8 CONCRETE PROTECTION AND CURING

A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection, and follow recommendations in ACI 305R for hot-weather protection during curing.

B. Evaporation Retarder: Apply evaporation retarder to concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 lb/sq. ft. x h (1 kg/sq. m x h) before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.

C. Begin curing after finishing concrete, but not before free water has disappeared from concrete surface.

D. Curing Methods: Cure formed and unformed concrete for at least seven days by moisture curing, moisture-retaining-cover curing, curing compound, or a combination of these as follows:

1. Moisture Curing: Keep surfaces continuously moist for not less than seven days with the following materials:
   a. Water.
   b. Continuous water-fog spray.
   c. Absorptive cover, water saturated and kept continuously wet. Cover concrete surfaces and edges with 12-inch (300-mm) lap over adjacent absorptive covers.
2. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 inches (300 mm), and sealed by waterproof tape or adhesive. Immediately repair any holes or tears during curing period using cover material and waterproof tape.
3. Curing Compound: Apply uniformly in continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period.

3.9 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified independent testing and inspecting agency to sample materials, perform tests, and submit test reports during concrete placement according to requirements specified in this Article. Perform tests according to ACI 301.

B. Testing Frequency:

1. Slump of Concrete – one for each truck delivery

2. Air Content of Concrete – one for each truck delivery

3. Obtain one composite sample for each day's pour of each concrete mix exceeding 5 cu. yd. (4 cu. m), but less than 25 cu. yd. (19 cu. m), plus one set for each additional 50 cu. yd. (38 cu. m) or fraction thereof.
4. Obtain at least one composite sample for each 100 cu. yd. (76 cu. m) or fraction thereof of each concrete mix placed each day.

3.10 REPAIRS

A. Remove and replace concrete that does not comply with requirements in this Section.

END OF SECTION

GENERAL

1.01 Description: Perform all work necessary and required to waterproof the exterior foundation walls below finish grade and to seal the interior floor, ceiling, and walls of the concrete vault.

1.02 Environmental Conditions: Waterproofing and sealing shall not be applied when the temperature is below 40 degrees F and falling or when the vault is wet or any standing water in the vault.

PRODUCTS

2.01 Materials: Materials shall conform to the following requirements and shall be delivered in sealed containers bearing the manufacturer's original labels:

A. Interior of Vault and Exterior of Vault: Rainguard BG-500 as manufactured by Rainguard Products Company, 3334 E. Coast Hwy #201, Corona del Mar, CA 90302. Phone 949-675-2811 or an approved equal such as Koppers Foundation Coating.

EXECUTION

3.01 Preparation of Surfaces: Surfaces to receive waterproofing and sealing shall be smooth, clean, and dry. Holes, joints, and cracks shall be pointed flush with mortar and high spots ground level with surrounding surface. Before waterproofing and sealing, all surfaces shall be swept clean of all foreign matter, any latent defects shall be removed and the surfaces shall be inspected and approved by the CO. The concrete is to be cured according to Item Specification 03300 and the vault is to be tested for water tightness and all standing water removed and vault dried for at least 1 1/2 days before applying waterproofing and sealing the concrete vault.

3.02 Application: The Contractor shall coordinate waterproofing and sealing operations with other phases of the work to prevent staining or damaging finished work. Two coats of BG-500 shall be applied to the exterior of foundation walls and shall extend from the bottom of the vault to within four inches of the finished grade. The interior concrete vault floor, ceiling, and walls are to be covered with two coats of BG-500, color black, are to be applied at the rates and in accordance with manufactures recommendations. All safety precautions shall be adhered to and all necessary safety equipment is to be used as recommended by the manufacturer. Care is to be taken during backfilling procedures to protect the coating from any damage. Rocks larger than 2 inches shall not be placed against the exterior concrete wall.

3.03 Repair: The Contractor shall repair or replace damaged finished work to its original condition without additional cost to the Government.

3.04 Testing: The contractor shall place water (in the vault(s) as directed by the CO. Before backfilling concrete vaults, see water tightness test requirements under Section 3400 Precast Concrete. For cast in place concrete fill with clean water to the top of the tank, notify the CO, and let stand for 3-hour period. Observe the level of the water and check for leaks. If leaks occur, mark the areas, drain the tank, and allow to dry thoroughly. Grind these areas down and repair with bonding agent and grout complying with Section 03000 or 3300. Recoat area with appropriate surface coatings and repeat water test. All tanks shall be watertight prior to backfilling.

END OF SECTION

Section 007849 - PVC Vent Stack

GENERAL

1.01 Description - Perform all work necessary and required for the proper installation of tubular polyvinyl chloride vent stacks as detailed on the 2-UNIT toilet drawings and specified herein.

PRODUCTS

2.01 Polyvinyl Chloride Pipe - Vent stacks shall be 12-inch PVC pipe with a .360 minimum wall thickness. The vent stack shall be mounted and secured as detailed on the drawings.

2.02 Polyvinyl Chloride Coupler - The coupler shall be PVC and the proper size for the 12-inch PVC pipe. The coupler shall be installed as shown on the drawings. When solvent welds are called for on the drawings, use only solvents and cleaners that are approved for use on PVC pipes and as recommended by the manufacturer.

2.03 Metal Cap and Storm Collar - The metal cap shall be fabricated from 24 gauge galvanized steel sheeting. A bolt on storm collar will be installed on the vent pipes approximately 1" above the 2" lip on the metal cap.

2.04 Painting - The exposed portion of the vent stack, both inside and outside, and to a point 6 inches beyond the exposed portion, is to be painted Tudor Brown. See Specification 09900.

END OF SECTION

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Section 007850 – Manhole Cover

GENERAL

1.01 Scope: Applies to the manhole and cover of the holding tank and installation as shown on drawings.

1.02 Related Work Specified Elsewhere

A. Painting: Section 09900.

PRODUCTS

2.01 Manhole Cover: Pressure-type aluminum manhole cover sealed with rubber gasket and fastened to the frame with cam lock. Size as shown on the drawings. Similar to Model # IMH-85-24 AL as manufactured by Castings Inc. of Grand Junction, CO. Phone Number 970-243-2032. Manhole shall have a gasket, shall have a minimum clear opening of 22".

EXECUTION

3.01 Set the manhole frame in place and to the required elevation prior to pouring the concrete slab. The base is to be sealed to the concrete slab with waterproof mastic so as to be water and airtight.

A. If bolts require a special wrench, the contractor shall provide a minimum of 1 such special wrench.

END OF SECTION

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PART 1 - GENERAL

1.1 SUMMARY

A. This section includes all materials and labor required in the placement of Contractor supplied pre-engineered concrete vault comfort stations (toilets) and Contractor supplied pre-engineered concrete pump house building. The work includes excavation, placement of aggregate base, backfill, concrete entrance construction, and final site grading. Work also includes coordination with the CO for delivery times and schedules.

1.2 LOCATIONS

A. Install the comfort stations and pump house building at the locations shown on the drawings and as approved by the Contracting Officer (CO).

1.3 CODE AND COMPLIANCE STANDARDS

A. All work and materials shall comply with the latest industry building codes and regulations, including but not limited to the following:

1. Uniform Building Code
2. National Electrical Code
3. Uniform Plumbing Code

1.4 DELIVERY AND STORAGE

A. Pre-engineered units, accessories and other manufactured items shall be delivered to the site. Materials shall be handled, transported, and erected to prevent damage and/or deformation of materials.

B. Contractor supplied material stored at the project site shall be protected from weather.

C. Delivery to the site shall be made on normal highway trucks and trailers. Ensure that delivery conditions and access at the site is not hazardous or unsuitable for truck and equipment due to weather, physical constraints, roadway width or grade to provide a safe and quality installation.

1.5 BUILDING DESCRIPTION

A. Double Unit Vault Toilets: Provide Double Unit Rocky Mountain as produced by CXT Incorporated, Precast Concrete Products, Spokane, WA (800) 696-5766. Or approve equal.
   1. Floor plan and building elevations are shown on the drawings.

B. Pump House Building: Provide utility building as produced by CXT Incorporated, Precast Concrete Products, Spokane, WA (800) 696-5766. Or approved equal.
   1. Floor plan and building elevations are shown on the drawings.
   2. Note: building to have double hollow metal ((2) 3'-0" x 7'-0") door and frame.

PART 2 – PRODUCTS

2.1 EXCAVATION, BEDDING MATERIALS AND BACKFILL

A. See Section 002222 and 002240 for excavation, bedding and backfill materials.

2.2 CONCRETE WORK

A. See Section 033000.

PART 3 - EXECUTION

3.1 GENERAL

A. Coordinate with the CO for exact delivery times and schedules.

B. The building supplier will use a crane to set the building on prepared aggregate base.

C. Obtain manufacturers supplied materials necessary to complete construction as listed in the installation manuals and as modified in these specifications.

D. Excavate hole for double unit toilet building to accommodate the toilet vaults and a leveling bed for the vaults large enough to be workable and allow the vaults to fit on the leveling bed when placed. Backfill, in 1-foot lifts, and compact to rough grade after vaults have been placed.

E. Excavate hole for pump house building to accommodate the fill for the base large enough to be workable and allow the floor of the building to fit on the fill for the base when placed.

F. Compact the bottom of the area for the buildings after it has been excavated to a minimum bearing of 1,500 psf.

G. Construct pump house building aggregate base to the dimensions shown on the drawings. Slope aggregate base from back to front to drain water out of the pump house building and off the concrete pad.

H. Complete adjacent site work as detailed on the drawings. Grade site to obtain positive drainage away from structure.

I. At the entrance to all prefabricated buildings, construct reinforced concrete entrance slab to the dimensions shown on the drawings.

J. Thoroughly clean all surfaces, interior and exterior, making facilities ready for use.

END OF SECTION

SECTION 220500
COMMON WORK RESULTS FOR PLUMBING

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes the following basic work results for plumbing to compliment other 22 and 33 Sections.

1. Piping materials and installation instructions common to most piping systems.
2. Installation requirements common to equipment specification sections.

B. Pipe and pipe fitting materials are specified in Division 22 and 33 piping Sections.

1.2 QUALITY ASSURANCE

A. Comply with ASME A13.1 for lettering size, length of color field, colors, and viewing angles of identification devices.

B. Equipment Selection: Equipment of higher electrical characteristics, physical dimensions, capacities, and ratings may be furnished provided such proposed equipment is approved by Contracting Officer (CO) and connecting mechanical and electrical services, circuit breakers, conduit, motors, bases, and equipment spaces are increased. If minimum energy ratings or efficiencies of equipment are specified, equipment must meet design and commissioning requirements.

1.3 DELIVERY, STORAGE, AND HANDLING

A. Deliver pipes and tubes with factory-applied end caps. Maintain end caps through shipping, storage, and handling to prevent pipe end damage and prevent entrance of dirt, debris, and moisture.

B. Protect stored pipes and tubes from moisture and dirt. Elevate above grade. Do not exceed structural capacity of floor, if stored inside.

C. Protect flanges, fittings, and piping specialties from moisture and dirt.

D. Store plastic pipes protected from direct sunlight. Support to prevent sagging and bending.

1.4 SEQUENCING AND SCHEDULING

A. Coordinate connection of mechanical systems with exterior underground services. Comply with requirements of governing regulations.

PART 2 - PRODUCTS

2.1 PIPE AND PIPE FITTINGS

A. Refer to individual Division 33 piping Sections for pipe and fitting materials and joining methods.

B. Pipe Threads: ASME B1.20.1 for factory-threaded pipe and pipe fittings.

2.2 JOINING MATERIALS

A. Refer to individual Division 33 piping Sections for special joining materials not listed below.

B. Solvent Cements: Manufacturer's standard solvent cements for the following:

1. PVC Piping: ASTM D 2564. Include primer according to ASTM F 656.

2.3 PIPE SEALS THROUGH CONCRETE SLAB

A. Foam for Annular Space

1. High density, low expansion insulate foam, Dap Tex as manufactured by Dap, Inc., Baltimore, MD, (888) 327-8477 or equal.

PART 3 - EXECUTION

3.1 PIPING SYSTEMS - COMMON REQUIREMENTS

A. General: Install piping as described below, unless piping Sections specify otherwise. Individual Division 15 piping Sections specify unique piping installation requirements.

B. General Locations and Arrangements: Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated, unless deviations to layout are approved by COR.

C. Install piping to drain.

D. Install components with pressure rating equal to or greater than system operating pressure.

E. Install piping free of sags and bends.

F. Install exposed interior and exterior piping at right angles or parallel to building walls. Diagonal runs are prohibited, unless otherwise indicated.

G. Install piping tight to slabs, beams, joists, columns, walls, and other building elements. Allow sufficient space above removable ceiling panels to allow for ceiling panel removal.

H. Install piping to allow 1-inch clearance around pipe.

I. Locate groups of pipes parallel to each other, spaced to permit valve servicing.

J. Install fittings for changes in direction and branch connections.

K. Install couplings according to manufacturer's written instructions.

L. Aboveground, Exterior-Wall, Pipe Penetrations: Seal penetrations using sleeves and mechanical sleeve seals. Size sleeve for 1-inch annular clear space between pipe and sleeve for installing mechanical sleeve seals.

1. Install steel pipe for sleeves smaller than 6 inches in diameter.
2. Assemble and install mechanical sleeve seals according to manufacturer's written instructions. Tighten bolts that cause rubber sealing elements to expand and make watertight seal.

M. Pipe penetrations through concrete slab:

1. Holes through slab of treatment building shall be performed by manufacturer. Contractor shall make entry water tight using foam insulation. Contractor shall fill annular space with styrofoam and seal with foam.

N. Piping Joint Construction: Join pipe and fittings as follows and as specifically required in individual piping specification Sections:

1. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
2. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.
3. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
   a. Note internal length of threads in fittings or valve ends, and proximity of internal seat or wall, to determine how far pipe should be threaded into joint.
   b. Apply appropriate tape or thread compound to external pipe threads, unless dry seal threading is specified.
   c. Align threads at point of assembly.
   d. Tighten joint with wrench. Apply wrench to valve end into which pipe is being threaded.

e. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds.

4. Plastic Piping Solvent-Cement Joints: Clean and dry joining surfaces by wiping with clean cloth or paper towels. Join pipe and fittings according to the following:
   a. Comply with ASTM F 402 for safe-handling practice of cleaners, primers, and solvent cements.
   b. PVC Pressure Piping: ASTM D 2672.

O. Piping Connections: Make connections according to the following, unless otherwise indicated:

1. Install unions, in piping 2-inch NPS and smaller, adjacent to each valve and at final connection to each piece of equipment with 2-inch NPS or smaller threaded pipe connection.

3.2 EQUIPMENT INSTALLATION - COMMON REQUIREMENTS

A. Install equipment to provide maximum possible headroom, if mounting heights are not indicated.

B. Install equipment according to approved submittal data. Portions of the Work are shown only in diagrammatic form. Refer conflicts to CO.

C. Install equipment level and plumb, parallel and perpendicular to other building systems and components in exposed interior spaces, unless otherwise indicated.

D. Install mechanical equipment to facilitate service, maintenance, and repair or replacement of components. Connect equipment for ease of disconnecting, with minimum interference to other installations. Extend grease fittings to accessible locations.

E. Install equipment giving right of way to piping installed at required slope.

END OF SECTION

SECTION 223000
PLUMBING EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes furnishing and installing all treatment building plumbing and appurtenances within the building and beneath it to a line 5 feet outside the walls.

B. Pre-Cast Concrete Treatment Building

1. See Section 133128.

1.2 QUALITY STANDARDS

A. Work shall comply with the Uniform Plumbing Code (UPC), Latest Edition.

B. Plumbing work shall be performed by a licensed plumber.

1.3 SUBMITTALS

A. Submit manufacturer's data for the following:

1. Pipe
2. Fittings
3. Valves
4. Water Meter
5. Pressure Gauge
6. Y Strainer

PART 2 –PRODUCTS

2.1 PIPES AND PIPE FITTINGS

A. Potable Water Piping, Above Ground:

1. Galvanized Steel Pipe:

a. As specified in Section 331116.

2. Plastic Pipe and Pipe Fittings:

a. Polyvinyl Chloride Pipe (PVC): ASTM D 1785, Schedule 80.
b. PVC Fittings:
1) Schedule 80 socket: ASTM D 2467.
2) Schedule 80 threaded: ASTM D 2464.

3) Solvent Cement: ASTM D 2564.
4) Thread Sealing Tape: Teflon.

3. Plastic Tubing and Tube Fittings:

a. High Density Polyethylene Pipe: As specified in Section 331116.

B. Potable Water Piping, Below Ground:

1. Galvanized Steel Pipe:

a. As specified in Section 331116.

2. High Density Polyethylene Pipe:

a. As specified in Section 331116.

2.2 SUPPORTS, HANGERS AND SHIELDS

A. Provide factory-fabricated horizontal-piping hanger and supports complying with MSS SP-58 selected by installer to suit horizontal-piping system, in accordance with MSS SP-69.

B. Use only one type by one manufacturer for each piping service.

C. Select size of hanger and support to exactly fit pipe size for bare piping, and to exactly fit around piping insulation with saddle or shield for insulated piping.

D. Grouped Horizontal Piping and Piping on Walls

1. Cold formed, lipped channels, not less than 1 1/2 inch No. 12 gauge with mounting holes. Hangers shall support five times the weight or thrust applied without failure.

2.3 PIPE PENETRATIONS OF WALLS

A. As specified in Section 220500.

2.4 COUPLINGS FOR DISSIMILAR PIPE

A. As specified in Section 220500 and 331116.

2.5 VALVES

A. General:

1. Provide factory-fabricated valve recommended by manufacturer for use in service indicated.

a. Provide valves of types and pressure ratings indicated; provide proper selection as determined by installer to comply with installation requirements.
b. Provide end connections that properly mate with pipe, tube, and equipment connections.

2. Sizes:

a. Unless otherwise indicated, provide valves of same size as upstream pipe size.

B. Ball Valve

1. PVC, single entry, thermoplatic ball valve, economical industrial grade, quarter-turn, polypropylene handle with union nuts. Pressure rated at 235 psi. Valves shall be PVC Single Entry Ball Valves as manufactured by Spears Manufacturing Company, Sylmar, CA (818) 364-1611 or equal.
2. 2.5 gpm flow controller shall be installed on each ball valve used as a hose bib or sampling tap.

C. Check Valve

1. Check valve with an unleaded body, unleaded bronze poppet, Buna-N seal and stainless steel spring. Drilled and tapped connections on both the inlet and outlet sides.
2. Check Valve model no. 80BE as manufactured by Flomatic, Glens Falls, NY, (518) 761-9797, and supplied by CPS Distributors, Inc, Denver, CO (303) 394-6040 or equal.

2.6 VACUUM BREAKER

A. As specified in Section 331219.

2.7 Y STRAINER

A. PVC Y Strainer, 1”, with 30 mesh stainless steel screen to protect piping system components from dirt and debris. Strainer shall have a removable strainer basket with 0-ring sealed drain plug and shall be pressure rated to 150 psi. Remove nut and add ball valve, IPS/hose bibb nipple and vacuum breaker.

2.8 WATER METER

A. The meter shall be a nutating disc type, in line meter, with bronze body, 3/4" threaded ends.

B. The meter shall conform to the requirements of NSF Standard 61.

C. Meter shall be rated for a maximum pressure of 150 psi.

D. The meter shall be rated for an accuracy of plus or minus 1 percent of actual flow for the flow range of 2 gpm to 3 gpm.

E. The register shall be direct reading and be of rolled seal construction.

F. The meter shall be a Neptune T-10, size 3/4”, as manufactured by Schlumberger Water, Tallassee, AL, (800) 645-1892 or equal, and supplied by National Water Works, Denver, CO (303) 292-1112.

2.9 DISINFECTANT FEED EQUIPMENT

A. As specified in Section 332600.

PART 3 - EXECUTION

3.1 INSPECTION

A. Examine roughing-in work of potable water and waste piping systems to verify actual locations of piping connections prior to installing fixtures.

B. Also examine floors and substrates, and conditions under which fixture work is to be accomplished. Correct any incorrect locations of piping, and other unsatisfactory conditions for installation of plumbing fixtures.

C. Do not proceed with work until unsatisfactory conditions have been corrected in manner acceptable to Contracting Officer (CO).

3.2 INSTALLATION OF ABOVE GROUND WATER PIPING

A. As specified in Section 220500.

3.3 INSTALLATION OF EXTERIOR WATER PIPING

A. As specified in Section 331116.

B. Extend water service piping of size and in location indicated on drawings.

3.4 VALVES

A. General:

1. Install valves where required for proper operation of piping and equipment, including valves in branch lines where necessary to isolate sections of piping.
2. Locate valves so as to be accessible and so that separate support can be provided when necessary.
3. Install valves with stems pointed up, in vertical position where possible, but in no case with stems pointed downward from horizontal plane unless unavoidable.

B. Shutoff Valves:

1. Provide shutoff valve at water service entry inside building.
2. Install at inlet of each plumbing equipment item and elsewhere as indicated.

C. Drain Valves:

1. Install in each plumbing equipment item located to completely drain equipment for service or repair.
2. Install at base of each rise or drop in piping system, and elsewhere where indicated or required to completely drain potable water system.

D. Hose Bibbs:

1. IPS/Hose bibb bushing on ball valve. Install on exposed piping where indicated, with vacuum breaker.

3.5 INSTALLATION OF WATER METER

A. The water meter shall be installed with standard pipe unions at each end. The Contractor shall make an idler, with the female half of a standard pipe union at each end, to maintain water service as well as proper spacing and alignment of piping in the absence of the water meter.

B. The Contractor in the presence of the CO, shall remove the meter, install and remove the idler, and reinstall the meter. The idler shall be taped to the meter inlet piping.

C. The water meter shall be installed with 10 pipe diameters of straight pipe upstream and 5 pipe diameters of straight pipe downstream to minimize turbulence and ensure accuracy of the meter readings.

3.6 SUPPORTS, HANGERS AND SHIELDS

A. Install hanger, supports and shields to support piping properly from building structure; comply with MSS SP-69.

B. Install hangers and support to allow controlled movement of piping system.

3.7 FIELD TESTS, INSPECTION, AND DISINFECTION

A. General:

1. The Contractor shall perform all field tests and provide all labor, equipment and incidentals required for the tests.
2. The CO will witness all field tests and conduct all field inspections.
3. The Contractor shall give the CO ample notice of the dates and times scheduled for tests.

4. Any deficiencies found shall be rectified and work affected by such deficiencies shall be completely retested at no additional cost to the Government.

B. Inspection:

1. Inspection shall continue during installation and testing.
2. Perform a final inspection of the equipment prior to installation to determine conformity to the type, class, grade, size, capacity, and other characteristics specified herein or indicated.
3. Correct or replace all equipment rejected prior to installation.

C. Water Distribution Piping Test:

1. As Specified in Section 331116.

D. Disinfection:

1. As specified in Section 331116.

3.8 FIELD QUALITY CONTROL

A. Upon completion of installation of plumbing appurtenances and equipment and after system is pressurized, test appurtenances and equipment to demonstrate capability and compliance with requirements.

B. When possible, correct malfunctioning units at site, then retest to demonstrate compliance; otherwise, remove and replace with new units and proceed with retesting.

C. Remove cracked or dented units and replace with new units.

END OF SECTION

SECTION 260100
GENERAL ELECTRICAL PROVISIONS

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes basic requirements of a common or administrative nature that pertain to all electrical work.

1.2 WORK INCLUDED

A. Furnish, install, test, and place in satisfactory and successful operation all equipment, materials, devices, and necessary appurtenances to provide a complete and operable electrical system. The work also includes the completion of those details of electrical work not mentioned or shown which are necessary for the successful operation of all electrical systems.

1.3 QUALITY ASSURANCE

A. Workers Qualifications: All electrical work shall be performed by licensed electricians or under the direct supervision of a licensed electrician.

B. Codes and Regulations:

1. All work shall meet requirements of governing codes and regulations, NFPA 70-2014 (NEC), and NESC.
2. Comply with the State of South Dakota requirements for electrical installations.
3. Advise the Contracting Officer of conflicting codes or conflicts between codes drawings, and specifications.
4. When the requirements of specifications or drawings are more stringent than the codes, regulations, or standards, the specifications or drawings shall prevail.
5. The electrical installation shall meet the requirements of NECA Standard of Installation, except where otherwise specified.

C. UL Listing:

1. All electrical materials and equipment shall meet requirements of the applicable standards of UL if UL standards exist for such materials and equipment.
2. The UL authorized listing mark is acceptable as evidence that the materials meet this requirement.
3. In lieu of UL authorized listing mark, the Contractor may submit independent proof satisfactory to the Contracting Officer that the materials meet the standards.
4. Install materials and equipment only for their intended operational purpose.

D. Standard Products: Provide only new electrical equipment of current standard design.

1.4 INTENT OF CONSTRUCTION DRAWINGS

A. Electrical drawings do not attempt to show complete details of construction that affect installation. Diagrams are schematic only and do not necessarily show physical arrangement of equipment. Refer to drawings of other trades for additional details which affect work.

B. Conduit, conductor and ground connections are shown diagrammatically only. Layout does not necessarily show total number of conduits or conductors for circuits required and should not be used for obtaining quantities for linear runs of conduits or wires. Locations of indicated runs are not intended to show actual routing of conduits. Provide additional conduits and wire wherever needed to complete installation of specific equipment furnished.

C. Locations of devices on drawings are approximate and may be distorted for clarity in representation.

D. Make changes such as offsetting conduit runs, moving outlets, or other minor changes necessary to facilitate installation at no additional expense to the Government.

PART 2 – PRODUCTS NOT USED

PART 3 – EXECUTION

3.1 INSPECTION

A. Demonstrate that electrical work operates satisfactorily and in accordance with the requirements of the drawings and specifications.

3.2 TESTING

A. General:

1. Make all specified tests in the presence of the Contracting Officer.
2. Furnish all instruments and provide qualified personnel to perform all tests in accordance with the drawings and specifications.
3. Perform all tests at no additional expense to the Government.
4. Operate all electrical equipment within the ranges specified by manufacturer.
5. Correct defects revealed by the tests.

B. Conductors: Prior to energization of circuitry, test installed wires and cables to ensure that insulation resistance levels are adequate, test wires and cables for electrical continuity and for short-circuits.

C. Wiring devices: Ensure proper polarity of connections. Prior to system energization, test wiring devices in accordance with manufacturer's recommendations.

END OF SECTION

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SECTION 260500
BASIC ELECTRICAL MATERIALS AND METHODS

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes the following for a complete and operable electric system:

1. Wire and connectors.
2. Raceways.
3. Supporting devices.
4. Grounding system.
5. Surge protection devices (SPD).
6. Receptacles.
7. Switches.
8. Outlet, pull, and junction boxes.
9. Device plates and covers.
10. Panelboards and circuit breakers.
11. Electric heaters.
12. Holding tank alarm.
13. RV pedestals
14. Luminaires.

1.2 COORDINATION

A. See Division 26 Section "Electrical Service" for coordination with utility companies for connection to their systems.

B. See Division 31 Section "Trenching, Bedding, and Backfill for Utilities" for buried utilities.

1.3 SUBMITTALS

A. Product Data: As specified in Division 1, "Submittal Procedures." Submit manufacturer's literature for items listed below:

1. Wire and cable.
2. Raceways.
3. Ground rods and grounding electrode conductor.
4. Surge protection device (SPD).
5. Receptacles.
6. Switches.
7. Outlet, pull, and junction boxes.
8. Panelboards and circuit breakers.
9. Electric heaters.

10. Holding tank alarm.
11. RV pedestals.
12. Luminaires.

B. Closeout Submittals: As specified in Division 1, “Operation and Maintenance Manuals” and “Closeout Submittals.” Submit manufacturer’s warranties for luminaires and surge protection devices.

PART 2 – PRODUCTS

2.1 WIRES, CABLES, AND CONNECTIONS

A. Feeders and Branch Circuits: Copper, unless otherwise indicated, 600 volt insulation; solid for No. 10 AWG and smaller, stranded for No. 8 AWG and larger.
Type THHN/THWN, XHHW, USE, or UF depending on application.

B. Wire Connectors and Splices: Units of size, ampacity rating, material, type, and class suitable for service indicated.

C. Color Coding: Color code secondary conductors for the electrical system.

1. Phase A: Black.
2. Phase B: Red.
3. Neutral: White.
4. Ground: Green.

2.2 RACEWAYS

A. Metal Conduit:

1. RMC: Rigid metal conduit; galvanized rigid steel; ANSI C80.1.
2. Fittings and Conduit Bodies: ANSI/NEMA FB 1; material to match conduit.

B. EMT: Electrical metallic tubing; ANSI C80.3, zinc-coated steel.

1. Fittings and Conduit Bodies: ANSI/NEMA FB 1; steel compression type.

C. RNC: Rigid nonmetallic conduit; NEMA TC 2, Schedule 80 PVC.

1. Fittings and Conduit Bodies: NEMA TC 3.

2.3 SUPPORTING DEVICES

A. Material: Cold-formed steel, with corrosion-resistant coating.

B. Metal Items for Use Outdoors or in Damp Locations: Hot-dip galvanized steel.

2.4 GROUNDING SYSTEM

A. Electrode Conductors: Copper, without splice throughout length, sized according to NEC; bare or insulated wire.

B. Ground Rods: One-piece, copper-clad steel rods, minimum 3/4 inches by 120 inches.

C. Equipment Grounding Conductors: Copper, size in accordance with NEC. If insulated wire is use the same type as other wires in the circuit with green-colored insulation.

D. Connectors: Bronze, copper, or stainless steel listed according to UL 467.

2.5 SURGE PROTECTION DEVICES (SPD)

A. UL Listed to 1449 3rd edition for Type 1 surge protection device. ANSI/IEEE C-62.41, C62.45 and NEC Article 100/285. 120/240 Vac, single phase, NEMA 4X enclosure with LED lights indicating operational status.

1. Maximum continuing operating voltage (MCOV): 275 V.
2. Short circuit current rating (SCCR): 200kA
3. Nominal Discharge Rating (I_N): 20kA.
4. Surge current per mode: 50kA.
5. Surge current per phase: 50kA.
6. Voltage protection Rating (VPR):

    a. L-G: 700V.
    b. L-L: 1200V.

B. LEA model SP50-2403SP or equal.

2.6 RECEPTACLES

A. Ground Fault Circuit Interrupter (GFCI) Receptacles: UL 943, Class A, Specification grade, straight blade, duplex 20-ampere with NEMA 5-20R configuration. Provide with built-in surge and load noise suppression, feed-through capability, with test and reset buttons. Indicator light glows only when tripped. Ivory color.

2.7 SWITCHES

A. Wall Switches: Specification grade, 20-ampere, 120/240 volts, ac rated, quiet type, NEMA WD heavy duty grade, with ivory handle.

2.8 OUTLET, PULL, AND JUNCTION BOXES

A. General: Provide one for each outlet, switch, receptacle, or combination, at each junction point.

1. Metal Raceway Systems in Interior Dry Locations: Galvanized sheet metal.
2. Outdoor and Wet Locations: Weatherproof, cast metal, threaded hub.
3. Buried or Subject to Constant Moisture: Watertight.

B. Exterior Pull or Splice Boxes: NEMA 4X, fiberglass or composolite enclosure. Provide base and extension to extend depth to allow level entry of conduit.

1. Cover: Nonskid design, composolite cover with lifting eye and recessed stainless steel penta-head fasteners. Cast in label with "Electric" for voltages of 600 volts or below, and "Electric-High Voltage" above 600 volts.

2.9 DEVICE PLATES AND COVERS

A. Single and combination types to match corresponding wiring devices.

1. Material for Wet Locations: Listed and labeled for use in "wet locations," gasketed and weatherproof, with spring loaded neoprene gasketed doors for receptacles, lever operators for switches and cast aluminum cover plates for junction boxes.

2.10 PANELBOARDS

A. Enclosures: Surface mounted cabinets suitable for environmental conditions at installed location.

B. Front: Hinged door-in-door construction with flush lock. Secured to box with concealed trim clamps. For surface-mounted fronts, match box dimensions.

C. Directory Card: With transparent protective cover, mounted in metal frame, inside panelboard door.

D. Main Bus: Copper bus bar sized in accordance with UL standards.

E. Equipment Ground Bus: Adequate for feeder and branch-circuit equipment ground conductors; bonded to box.

F. Neutral Bus: Full sized, 100% rated in service entrance panels. Arrange panel bus bar taps with single-pole branches for sequence phasing of the branch circuit devices. Provide neutral bushing lug for each outgoing feeder requiring a neutral connection.

G. Panelboard Short-Circuit Rating: Short-circuit ratings as shown on drawings but not less than 22,000 amperes RMS symmetrical.

H. Panelboards with Main Service Disconnect: Listed for use as service equipment.

I. Spaces for Future Devices: Mounting brackets, bus connections, and necessary appurtenances required for future installation of devices.

2.11 CIRCUIT BREAKERS

A. Molded-Case Circuit Breaker: NEMA AB 1, with interrupting capacity to meet available fault currents. Inverse time-current element for low-level overloads, and instantaneous magnetic trip element for short circuits. Heavy-duty, bolt-on or snap-in, quick make, quick break type. Minimum interrupting rating of 10,000 amperes symmetrical at 240 volts, unless otherwise noted on drawings.

1. Multiple Pole Circuit Breakers: Thermal-magnetic type with common type handle. Minimum 100-ampere frame and through 100-ampere trip sizes shall take up same pole spacing.
2. Application Listing: Appropriate for application.

2.12 ELECTRIC HEATERS

A. Unit Heater: UL listed, steel sheathed type heating elements with aluminum fins. Qmark model MUH-03-21 or equal.

1. Housing: Steel.
2. Fan: Dynamically balanced, 350 CFM.
3. Motor: Totally enclosed, permanently lubricated with automatic-reset thermal protection.
4. Louvers: Built-in.
5. Thermostat: Built-in, SPST with 40-85ºF temperature range.
6. Safeties: Automatic reset, thermal cut-out, fan delay.
7. Mounting: Provide with wall mount bracket.
8. Electrical Requirements: 3,000 Watts, 240 volt, 60 Hz, 12.5 amps.

2.13 HOLDING TANK ALARM

A. UL Listed, outdoor alarm to monitor high liquid levels. Model A-Pak II 10-0623 by Zoeller Pump Company or equal.

1. Enclosure: Type 4X water-tight.
2. Float: Corrosion resistant PVC housing for use in sewage. Mechanical activated switch not containing mercury. Provide control switch with 15 feet of cable and mounting clamp.
3. Alarm:

   a. Visual: Read beacon light mounted on top of enclosure.
   b. Audible: 85 decibels at 10 feet.

2.14 RV PEDESTALS

A. Model U075CP6031 manufactured by Midwest Electric Products, Inc., P.O. Box 910, Mankato, MN 56002-0910, or equal, having following essential characteristics.

1. General: UL-listed, galvanized zinc coated, steel pedestal consisting of post and power outlets with welded flange. NEMA 3R construction, suitable for underground services. Terminals rated for copper or aluminum wire.
2. Post: Factory wired from loop-feed lugs and twin 2-250 MCM terminals per phase, suitable for Cu/Al. Provide with stabilizer base.
3. Outlets: 50/30/20 amp pedestal with 20 amp GFCI receptacle and TV/Telephone.

2.15 LUMINAIRES

A. Ceiling Mounted Luminaires, (Type A): Surface-mounted enclosed linear LED fixture for rough service applications, UL listed for damp locations.

1. Housing: Heavy-duty, 16-gauge steel, one-piece housing
2. Lamps: Integrated LED module providing a minimum of 4000 lumens at 4000K color temperature with a CRI of 80 or higher.
3. Optical: Clear prismatic, UV-stabilized, polycarbonate lens.
4. Driver: High-efficiency electronic driver, universal voltage (120-277V), with 0-10V dimming capability.
5. Manufacturer: Lithonia VTL LED 48IN MVOLT 40K 80CRI or equal.

B. Wall Mounted Luminaires, (Type B): Wall pack, UL listed for wet locations.

1. Construction: Cast aluminum housing with hinged door and one-piece gasket. -
2. Driver: MVOLT.
3. Lens: Prismatic glass.
4. Electrical: High-output LEDs integrated on a circuit board. Integral 6kV surge protection. L87 at 100,000 hours.
5. Lumens: 2,161 at 4,000K color temperature and 75 CRI.
6. Finish: Bronze powder paint.
7. Control: Provide photo control.
8. Warranty: 5 year limited warranty.
9. Manufacturer: Lithonia TWR1 LED 1 40K MVOLT PE or equal.

PART 3 – EXECUTION

3.1 EXAMINATION

A. Field verify circuiting arrangements and all existing conditions prior to beginning work. Beginning of removal means installer accepts existing conditions.

B. Verify that abandoned wiring and equipment serve only abandoned facilities. Maintain electrical continuity to remaining wiring and equipment.

C. Removal information is based on existing record documents. Report discrepancies to Contracting Officer before disturbing existing installation.

3.2 REMOVAL AND EXTENSION OF EXISTING ELECTRICAL WORK

A. Remove, relocate, abandon, and extend existing installations to accommodate new construction.

B. Connect new work to existing work in a manner that will assure proper operation and grounding throughout in conformance with the National Electrical Code.

C. Removed conduits, wire, devices, etc. shall become the property of the Contractor unless otherwise noted.

D. Coordinate installation of meter can and final connection with Utility Company.

3.3 RACEWAY APPLICATION

A. Outdoor Locations, Above Grade: RMC.

B. Underground Installations:

1. More than 5-feet from Foundation Wall: PVC Schedule 80 RNC except as otherwise noted.
2. In or under Slab on Grade: PVC Schedule 80 RNC.

C. Wet and Damp Locations: RMC.

3.4 WIRING METHODS

A. Underground Feeders and Branch Circuits: Insulated single conductors, direct burial.

B. Conductor Size:

1. Meet requirements of NFPA 70 or the sizes shown on the drawings if larger. Use no wire smaller than No. 12 AWG for power and lighting circuits.
2. Unless otherwise specified, use No. 10 AWG conductor for 20 ampere, 120 volt branch circuit home runs longer than 75 feet.

C. Wire and Cable Application: Use wiring and cabling methods specified below to the extent permitted by applicable codes.

1. Concealed Interior Feeders and Branch Circuits: AC, MC, or MI cable.
2. Wet or Damp Interior Locations: Building wire in raceways.
3. Exterior Locations: Building wire in raceways.
4. Underground: Building wire in raceways or direct buried cable as indicated.
5. Below Slabs or Slabs: Building wire in raceways.

D. Color Coding: Factory color-code wires No. 8 AWG and smaller. Color code wire No. 6 AWG and larger with color tape at least 6-inches at exposed ends and at every accessible junction box. For single phase voltages, black and red for phases, white for neutral, and green for ground.

3.5 WIRING INSTALLATION

A. Make splices and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.

B. Conductor Size: Meet requirements of NFPA 70 or the sizes shown on the drawings if larger.

C. Underground Conductors and Conduit:

1. Perform trenching and backfilling in accordance with Division 31 Section "Trenching, Bedding, and Backfill for Utilities."
2. Install minimum depth to top of underground conduit in accordance with NEC unless otherwise indicated.
3. Install underground conduit with watertight joints.

3.6 GROUNDING

A. Ground the electrical service system neutral at service entrance equipment to supplementary grounding electrodes.

B. Provide a separate, insulated equipment grounding conductor in feeder and branch circuits. Terminate each ground conductor to the bushing and ground lug.

C. Route grounding connections, conductors to ground, and grounding conductors to protective devices in the shortest and straightest path possible to minimize transient voltage rises.

D. Provide a corrosion-resistant finish to field connections, buried metallic bonding products, and where factory applied protective coatings have been destroyed.

E. Equipment Grounding Conductors: Comply with NEC for types, sizes, and quantities of equipment grounding conductors, unless specific types, larger sizes, or more conductors than required by NFPA 70 are indicated.

F. Connections: Make connections so galvanic action or electrolysis possibility is minimized. Select connectors, connection hardware, conductors, and connection methods so metals in direct contact will be galvanically compatible.

1. Equipment Grounding Conductor Terminations: For No. 8 AWG and larger, use pressure-type grounding lugs.

2. Moisture Protection: If insulated grounding conductors are connected to ground rods or grounding buses, insulate entire area of connection and seal against moisture penetration of insulation and cable.

G. Field Welding:

1. Make exothermic-welded connections where grounding conductors connect to underground grounding conductors and to underground grounding electrodes, and for bonding to steel.
2. Exothermic weld all underground connections.

3.7 PANELBOARDS

A. Field verify available voltage and phase prior to ordering panelboards.

B. Install panelboards and accessories according to NEMA PB 1.1.

C. Mounting Heights: Top of trim 72 inches above finished floor.

D. Mounting: Plumb and rigid without distortion of box.

E. Install filler plates in unused protective device spaces.

3.8 EXTERIOR PULL BOXES

A. Set pullboxes on 6 inches of level, 90 percent compacted crushed rock, 3/4 inch to 1 inch size, extending 12 inches on each side.

B. Set pullbox sections in grout, making watertight joints.

C. Install top of pullboxes flush with adjacent soil or paved areas.

3.9 HOLDING TANK ALARM

A. Mount alarm enclosure on treated wood post with enclosure visible from the host site.

B. Extend control cable underground in conduit to holding tank.

C. Install float control switches in holding tank per manufacturer's instructions. Attach cable with mounting clamp to the inlet pipe. Set alarm on level as specified. Field verify setting height.

END OF SECTION

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SECTION 262210
ELECTRICAL SERVICE

PART 1 - GENERAL

1.1 SUMMARY

A. The work of this section consists of furnishing and installing service entrance equipment, raceways, and conductors.

1.2 COORDINATION

A. Permanent electric power of the phasing, voltage, and characteristics shown will be supplied by the power company.

B. Coordinate the work of this section with the Utility Company for the associated campground.
1. Comanche Campground – Black Hills Electric Cooperative, (605) 673-4461
2. Grizzly Bear Campground – Black Hills Energy, (888) 890-5554
3. Pactola Loop B Campground – Black Hills Energy, (888) 890-5554

C. Where work by the power company is required in conjunction with construction, such as installation of cable in common trench with other utilities, the Contractor is responsible for coordinating the work and ensuring that all required work is completed.

D. Payment to Power Company:

1. Where work is to be performed by the power company that requires payment to the power company, payment will be made by the Contractor.
2. Where the Contractor identifies the need for work by the power company that requires payment and no known contractual arrangement by the Government has been made, the Contractor shall immediately notify the Contracting Officer.

1.3 SUBMITTALS

A. As specified in Division 1 Section “Submittal Procedures.”

1. Product Data: Submit manufacturer’s literature for the meter socket.

1.4 PERMITS

A. Applications for utility connection permits will be completed by the Government. Copies will be provided to the Contractor.

PART 2 – PRODUCTS

2.1 SERVICE CONDUCTORS

A. From Transformer to Service Entrance: Provided and installed by utility company.

2.2 METER SOCKET

A. Weather proof type approved by Utility Company. Provide with temporary blank cover.

2.3 METERS

A. Furnished by Utility Company.

PART 3 – EXECUTION

3.1 METERS AND SERVICE CONDUCTORS

A. Utility company will install meter and service conductors. Coordinate installation requirements with Utility Company.

3.2 INSPECTION

A. Comply with utility company requirements for inspections before making final connections.

END OF SECTION 262210

PART 1 - GENERAL

1.1 SUMMARY

A. This work consists of clearing, grubbing, trimming, removing, and disposing of, or treatment of, live and dead timber, construction slash, and debris within the areas shown on the drawings or designated on the ground. Areas may include but are not limited to: water, and electrical systems; structures, buildings, and related appurtenances; roads, spurs, trails, camp unit use pad spaces, tent pads and other areas to be graded. This work shall include the removal and disposal of designated trees outside the clearing limits. Also included is the protection from injury or defacement of vegetation and other objects designated to remain and treatment or removal of damaged trees.

PART 2 – PRODUCTS (NOT APPLICABLE)

PART 3 - EXECUTION

3.1 CLEARING

A. Clearing Limits:

1. Spurs and Paths: as shown on the drawings.
2. Utility Trenches: Width of clearing shall not exceed 15 feet unless otherwise agreed to by the Contracting Officer or as designated on the drawings.
3. Structures/Buildings: 10 feet outside the extreme limits of the structure or building.
4. Areas to be graded: 2 feet past the intersection of the existing ground with the cut slope and to the toe of fill as applicable.

B. Remove all trees, down timber, snags, underbrush, rubbish, and other objectionable materials within the clearing limits.

C. Individual marked trees, within the clearing limits, shall be left standing and uninjured.

D. Protect existing trees and other vegetation indicated to remain in place, against unnecessary cutting, breaking, or skinning of roots, skinning and bruising of bark, smothering of trees by stockpiling construction materials or excavated materials within drip line, excess foot or vehicular traffic, or parking of vehicles within drip line.

E. Provide temporary guards to protect trees and vegetation to be left standing.

F. Where any trees are damaged, or where limbs are required to be trimmed or removed because of operations under this contract, or because of interference with new construction, any cutting or removal of limbs or branches shall be done only with prior approval of Contracting Officer.

G. Solvents, oils, and other material which may be harmful to plant life shall be disposed of in containers and removed from the site. At the completion of the work, any contaminated soil shall be removed and replaced with topsoil.

H. Trees shall be felled into the area being cleared when ground conditions, tree lean, and shape of clearing permit. Controlled felling shall be used that will ensure the direction of fall to prevent damage to property, structures, trees designated to remain, or traffic.

I. Sound logs that exceed 8 feet in length and 4 inches in diameter at the small end, that are capable of being skidded by chokers or tongs, shall be decked at the location designated by the Contracting Officer.

J. Material smaller than described above shall be treated as slash and disposed of according to 3.4.

K. Felling, cutting, and trimming methods shall not cause bark damage to standing timber. Trees with major roots exposed by construction that are rendered unstable shall be felled and disposed of in accordance with the above paragraphs.

3.2 GRUBBING

A. All debris, stumps, roots and other protruding vegetative material within the clearing limits, not designated to remain, shall be grubbed, removed and disposed of, except the following:

1. System Mains, Service Laterals, and Absorption Fields: Inside the clearing limits, cut stumps and projecting roots flush with the natural ground.
2. Trails: On all trails, stumps within the trailbed shall be removed. Stumps located between the edge of the trailbed and the edge of the trailway that cannot be cut flush with the finished slope, or are not tightly rooted, shall be removed.
3. Roads: Undisturbed stumps outside the roadway or in embankment areas, provided they can be cut flush above the original ground (measured from the uphill side) nor closer than 2 feet to the finished subgrade or 1 foot to any slope surface or as otherwise shown on the drawings and they do not interfere with the placement or compaction of embankments.
4. All roots over 3 inches in diameter within the roadbed area shall be grubbed to a minimum depth of 6 inches below subgrade. Roots over 1 inches in diameter protruding from the excavated slope shall be cut flush with the excavated slope surface.
5. Fire-dangerous dead trees or unstable live trees, designated by the Contracting Officer within 200 feet slope distance of the centerline of roads and capable of falling and reaching the roadway shall be cut off not more than 12 inches above the uphill ground line and treated in accordance with Subsections 3.1 and 3.3.

6. Branches on remaining trees or shrubs shall be trimmed to give a clear height of 14 feet above the roadbed unless otherwise shown on the drawings. Tree limbs shall be trimmed as near flush with the trunk as practicable.
7. When grubbing for structures, buildings, and areas to be landscape graded; grub and remove all stumps inside the clearing limits, Grub and remove all roots to a depth of at least 6 inches below the natural ground when fill is required for site grading. When excavation is required for site grading, grub and remove all roots to a depth of at least 6 inches below finished grade.
8. Stumps, roots, and material developed in grubbing operations shall be removed from National Forest lands.

3.3 UTILIZATION OF TIMBER

A. Merchantable timber is timber that meets Utilization Standards. Utilization Standards shall be trees larger than 8-inches in diameter measured 12-inches above grade.

B. Utilization and Removal of Timber: Trees that meet "Merchantable Timber" standards shall be treated as follows.

1. Removal from Government Land. Merchantable timber, designated for removal, shall become the property of the contractor without charge and removed from Government land. This timber shall not be exported from the United States nor used as substitution (as defined in 36 CFR 223 Subpart D) for timber from private lands exported by the contractor or an affiliate directly or indirectly.
2. Timber commercially sold will be purchased from the Forest Service at the current base rates. Base rates to be determined at the time of harvest. Total cost for merchantable timber harvested is estimated a $1,130.00.

3.4 SLASH TREATMENT

A. Treatment of construction slash shall be accomplished by one or more of the following methods:

1. Chipping
2. Removal

B. All Methods. No construction slash shall be deposited in lakes, meadows, streams, or streambeds. Construction slash that interferes with drainage structures shall be removed immediately.

C. Specific Methods

1. Chipping. Construction slash up to at least 4 inches in diameter shall be processed through a chipping machine. Chips shall be deposited on embankment slopes or outside the roadway to a loose depth not exceeding 6 inches. Minor amounts of chips may be permitted within the roadway if they are thoroughly mixed with soil and do not form a layer.

2. Removal. Construction slash shall be hauled to locations designated on the ground or removed from National Forest Lands at an approved landfill.

END OF SECTION

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes earthwork and excavation products and requirements for a variety of field conditions that may be encountered during the project.

1.2 DEFINITIONS

A. Backfill: Soil materials used to fill an excavation.

1. Initial Backfill: Backfill placed below, beside and over pipe in a trench, including haunches to support sides of pipe.
2. Subsequent Backfill: Backfill placed over initial backfill to fill a trench.

B. Aggregate Base Course: Layer placed between the subgrade and concrete slab, concrete walk, or structure.

C. Borrow: Satisfactory soil imported from off-site for fill or backfill.

D. Excavation: Removal of material encountered above subgrade elevations and to lines and dimensions indicated.

1. Additional Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions as directed by CO. Additional excavation and replacement material will be paid for according to Contract provisions for changes in the Work.
2. Bulk Excavation: Excavations more than 10 feet (3m) in width and pits more than 30 feet (9m) in either length or width.
3. Unauthorized Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions without direction of the CO. Unauthorized excavation, as well as remedial work directed by CO, shall be without additional compensation.
4. Unclassified Excavation: Excavation to subgrade elevation and to lines and dimensions indicated regardless of the character of surface and subsurface conditions encountered, including rock, soil materials, and obstructions.
5. Rock Excavation: Rock material in beds, ledges, unstratified masses, and conglomerate deposits and boulders of rock material exceeding 2.5 cubic yards that cannot be removed by rock excavating equipment equivalent to the following in size and performance ratings, without systematic drilling, ram hammering, ripping, or blasting when permitted:
    a. Excavation of Footings, Trenches, and Pits: Late-model, track-mounted hydraulic excavator; equipped with a 30-inch- (1065-mm-) wide, short-tip-radius rock bucket; rated at not less than 120-hp (89-kW) flywheel power with bucket-curling force of not less than 25,000 lbf (111 kN) and stick-crowd force of not less than 18,700 lbf (83 kN); measured according to SAE J-1179.

b. Bulk Excavation: Late-model, track-mounted loader; rated at not less than 210-hp (157-kW) flywheel power and developing a minimum of 45,000-lbf (200-kN) breakout force; measured according to SAE J-732.

E. Fill: Soil materials used to raise existing grades.

F. Structures: Buildings, slabs, tanks, vaults, or other man-made stationary features constructed above or below the ground surface.

G. Subgrade: Surface or elevation remaining after completing excavation, or top surface of a fill or backfill immediately below subbase, aggregate base, drainage fill, initial or subsequent backfill materials.

H. Utilities include on-site underground pipes, conduits, ducts, and cables, as well as underground services within building.

1.3 SUBMITTALS

A. Contractor shall submit to the CO for approval source of backfill materials and certified sieve analysis.

B. Proctor test results.

1.4 CLOSEOUT SUBMITTALS

A. Test results.

1.5 PROJECT CONDITIONS

A. Existing Utilities: Do not interrupt utilities serving facilities occupied by Government or others unless permitted in writing by Contracting Officer (CO) and then only after arranging to provide temporary utility services according to requirements indicated:

1. Notify CO not less than two days in advance of proposed utility interruptions.
2. Do not proceed with utility interruptions without CO's written permission.

PART 2 - PRODUCTS

2.1 BACKFILL MATERIALS, GENERAL

A. Excavated material may be processed and used for backfill if the Contractor can show compliance with the material specified herein to the satisfaction of the CO. If excavated material is not sufficient to meet requirements, Contractor shall import needed material.

B. Satisfactory Soils: ASTM D 2487 soil classification groups GW, GP, GM, SW, SP, and SM, or a combination of these group symbols; free of rock or gravel larger than 6 inches in any dimension, debris, waste frozen materials, vegetation, and other deleterious matter.

C. Unsatisfactory Soils: ASTM D 2487 soil classification groups GC, SC, ML, MH, CL, CH, OL, OH, and PT, or a combination of these group symbols.

1. Unsatisfactory soils also include satisfactory soils not maintained within 2 percent of optimum moisture content at time of compaction.

D. Backfill and Fill: Satisfactory soil materials.

2.2 INITIAL BACKFILL

A. Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand; ASTM D 2940; except with 100 percent passing a 1 inch sieve and not more than 8 percent passing a No. 200 sieve. Excavated material may be used if approved by the Contracting Officer.

2.3 SUBSEQUENT BACKFILL

A. Subsequent backfill shall be excavated material or borrow material, free from rock larger than 6 inches in the greatest dimension, frozen material, clods, debris, or organic material. Borrow material used as subsequent backfill shall have a gradation and character suitable to achieve the compaction specified in this Section.

2.4 AGGREGATE BASE

A. As specified in Section 321123.

2.5 WARNING TAPE

A. As specified in Section 312317.

2.6 SHEETING AND SHORING

A. Timber or metal members used by the Contractor shall provide adequate support of the soils during his work. The Contractor shall be responsible for selection of proper material to maintain adequate support.

PART 3 - EXECUTION

3.1 LOCATION, ALIGNMENT AND GRADE

A. The location of all pipelines and structures shall be staked out and grades established by the Contractor. Locations shall be approved by the CO before excavation is started.

3.2 PREPARATION

A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout and other hazards created by earthwork operations.

B. Protect subgrades and foundation soils against freezing temperatures or frost. Provide protective insulating materials as necessary.

C. Provide erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties, walkways and drainages.

3.3 DEWATERING

A. Prevent surface water and ground water from entering excavations, from ponding on prepared subgrades, and from flooding Project site and surrounding area.

B. Protect subgrades from softening, undermining, washout, and damage by rain or water accumulation.

1. Reroute surface water runoff away from excavated areas. Do not allow water to accumulate in excavations. Do not use excavated trenches as temporary drainage ditches.
2. Install a dewatering system to keep subgrades dry and convey ground water away from excavations. Maintain until dewatering is no longer required.

3.4 EXPLOSIVES

A. Do not use explosives.

3.5 EXCAVATION, GENERAL

A. If excavated materials intended for fill and backfill include unsatisfactory soil materials and rock, replace with satisfactory soil materials.

B. Maintain the excavations to guard against and prevent injury to employees and the public. Provide adequate shoring and bracing as required by OSHA and other local governing regulations.

C. Excavations left open at the end of the working day shall be fenced to protect the public.

3.6 EXCAVATION FOR STRUCTURES

A. Excavate to indicated elevations and dimensions within tolerance of plus or minus 1 inch. Extend excavations a sufficient distance from structures for placing and removing concrete formwork, for installing services and other construction, and for inspections.

1. Excavations for Footings and Foundations: Do not disturb bottom of excavation. Excavate by hand to final grade just before placing concrete reinforcement. Trim bottoms to required lines and grades to leave solid base to receive other work.
2. Excavation for Underground Tanks and Mechanical or Electrical Utility Structures: Excavate to elevations and dimensions indicated within a tolerance of plus or minus 1 inch. Do not disturb bottom of excavations intended for bearing surface.

3.7 EXCAVATION FOR UTILITY TRENCHES

A. As specified in Section 312317.

3.8 APPROVAL OF SUBGRADE

A. Notify CO when excavations have reached required subgrade.

B. If CO determines that unsatisfactory soil is present, continue excavation and replace with initial backfill material as directed.

C. Reconstruct subgrades damaged by freezing temperatures, frost, rain, accumulated water, or construction activities, as directed by CO.

3.9 UNAUTHORIZED EXCAVATION

A. Fill unauthorized excavation under foundations or wall footings by extending bottom elevation of concrete foundation or footing to excavation bottom, without altering top elevation. Lean concrete fill may be used when approved by CO.

1. Fill unauthorized excavations under other construction or utility pipe as directed by CO.

3.10 STORAGE OF SOIL MATERIALS

A. Stockpile, borrow materials and satisfactory excavated soil materials. Stockpile soil materials without intermixing. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.

1. Stockpile soil materials away from edge of excavations. Do not store within drip line of remaining trees.

3.11 STRUCTURE BACKFILL

A. Place and compact backfill in excavations promptly, but not before completing the following:

1. Construction below finish grade including, where applicable, damp proofing, waterproofing, and perimeter insulation.

2. Surveying locations of underground utilities for record documents.
3. Inspecting and testing underground utilities.
4. Removing concrete formwork.
5. Removing trash and debris.
6. Removing temporary shoring and bracing, and sheeting.

B. Place and compact subsequent backfill adjacent to structures in such a manner as to prevent wedging action or eccentric lodging upon or against the structures.

C. Do not place backfill against any concrete footings or structure without prior permission of the CO and in no case less than 7 days after placement of concrete.

D. Heavy equipment shall not be operated within four feet of any structure.

E. Provide for anticipated settlement and shrinkage of the backfill and for the finished grades required.

3.12 UTILITY TRENCH BACKFILL

A. As specified in Section 312317.

3.13 FILL

A. Preparation: Remove vegetation, topsoil, debris, unsatisfactory soil materials, obstructions, and deleterious materials from ground surface before placing fills.

B. Plow, scarify, bench, or break up sloped surfaces steeper than 1 vertical to 4 horizontal so fill material will bond with existing material.

C. Place and compact subsequent backfill material in layers to required elevations.

3.14 MOISTURE CONTROL

A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill layer before compaction to within 2 percent of optimum moisture content.

1. Do not place backfill or fill material on surfaces that are muddy, frozen, or contain frost or ice.
2. Remove and replace, or scarify and air-dry, otherwise satisfactory soil material that exceeds optimum moisture content by 2 percent and is too wet to compact to specified dry unit weight.

3.15 COMPACTION OF BACKFILLS AND FILLS

A. The minimum degree of compaction required shall be a percent of the maximum laboratory density obtained by the standard proctor test AASHTO T-99 or ASTM D698. The in-place field density shall be determined by AASHTTO T238 or ASTM D2922. The minimum compaction requirements are:

1. Under structures, building slabs, steps, and pavements, scarify and recompact top 12 inches of existing subgrade and each layer of backfill or fill material at 95 percent.
2. Under walkways, scarify and recompact top 6 inches below subgrade and compact each layer of backfill or fill material at 92 percent.
3. Under unpaved areas, scarify and recompact top 6 inches below subgrade and compact each layer of backfill or fill material at 85 percent.
4. In utility trenches, including each layer of backfill up to 1 ft. above the pipe, compact each layer of backfill material at 90 percent. Backfill material shall at a minimum reach the same compaction as the adjacent soil as determined by the CO.

3.16 GRADING

A. General: Uniformly grade areas to a smooth surface, free from irregular surface changes. Comply with compaction requirements and grade to cross sections, lines, and elevations indicated.

   1. Provide a smooth transition between adjacent existing grades and new grades.

B. Site Grading: Slope grades to direct water away from buildings and to prevent ponding. Finish subgrades to required elevations within the following tolerances:

   1. Lawn or Unpaved Areas: Plus or minus 1 inch.
   2. Walks: Plus or minus 1 inch.
   3. Pavements: Plus or minus 1/2 inch.

C. Finishing Slopes: Finished slopes shall conform reasonably to the lines staked on the ground or shown on the drawings. The finished slope shall be left in a roughened condition to facilitate the establishment of vegetative growth. The finish associated with template and stringline or hand-raking methods will not be allowed.

3.17 AGGREGATE BASE COURSE

A. As specified in Section 321123.

3.18 FIELD QUALITY CONTROL

A. Testing Agency: Contractor shall engage a qualified independent geotechnical engineering testing agency to perform field quality control testing.

B. Allow testing agency to inspect and test subgrades and each fill or backfill layer. Proceed with subsequent earthwork only after test results for previously completed work complies with requirements.

C. Testing agency will test compaction of soils in place according to ASTM D 2922. Tests will be performed at the following locations and frequencies:

1. Toilet Buildings: One test for each Toilet Building foundation.
2. Trench Backfill: At each compacted layer (initial and subsequent backfill), at least one test for every 500 feet of trench length.
3. Backfill around structures and holding tanks: At subgrade and at each compacted backfill layer, at least one test per each 100 cubic yards or 2 feet of depth, but in no case fewer than two tests.

D. When testing agency reports that subgrades, fills or backfills have not achieved degree of compaction specified, scarify and moisten or aerate, or remove and replace soil to depth required; recompact and retest until specified compaction is obtained.

E. Excessive settlement or other evidence of improper backfill shall be corrected by reopening the trench or excavation to the depth required for proper compaction and then shall be refilled and satisfactorily compacted.

F. The correction and retesting of unacceptable work shall be paid by the Contractor at no expense to the Government.

G. The CO may reduce the testing frequency contingent upon the uniformity of material at the project site, compactive effort put forward by the Contractor, and previous test results for similar materials and compactive effort.

3.19 PROTECTION

A. Protecting Graded Areas: Protect newly graded areas from traffic, freezing and erosion. Keep free of trash and debris.

B. Repair and reestablish grades to specify tolerances where completed or partially completed surfaces become eroded, rutted, settled, or where they lose compaction due to subsequent construction operations or weather conditions.

1. Scarify or remove and replace soil material to depth as directed by CO, reshape and recompact.

C. Where settling occurs before Project correction period elapses, remove finished surfacing, backfill with additional soil material, compact, and reconstruct surfacing.

1. Restore appearance, quality, and condition of finished surfacing to match adjacent work, and eliminate evidence of restoration to the greatest extent possible.

3.20 SURFACE FINISH

A. In unpaved areas, surface finish shall be subsequent backfill to elevations shown on drawings.

B. In paved areas, apply surface treatment as specified and shown on the drawings.

3.21 DISPOSAL OF SURPLUS AND WASTE MATERIALS

A. Disposal: Remove surplus satisfactory soil and waste material, including unsatisfactory soil, trash, and debris, and legally dispose of it off Government’s property.

END OF SECTION

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PART 1 - GENERAL

1.1 SUMMARY

A. The work in this Section includes excavation, dewatering, preparation of the trench bottom, supply and installation of bedding and backfill materials; and disposal of waste material for the installation of utilities and related appurtenances.

1.2 QUALITY ASSURANCE

A. Conduct compaction tests as necessary to monitor the installation procedure and assure the quality of the work.

B. Perform all trench excavation in accordance with minimum requirements of OSHA Regulations. Provide shoring, bracing and tight sheeting to prevent caving and to protect workers in accordance with OSHA requirements and to protect adjacent property and structures.

C. Direct dewatering operations to natural drainages or storm sewers as appropriate. Perform discharge from dewatering operations in accordance with rules and regulations established by the Colorado Department of Environmental Quality.

1.3 SUBMITTALS

A. As specified in Division 1 Section General Requirements.

1. Submit the source of bedding and backfill materials and certified sieve analysis.
2. Product Data: Provide manufacturer's literature for the following:

    a. Warning tape.
    b. Soil Moisture Test Results.

1.4 CLOSEOUT SUBMITTALS

A. Test results.

1.5 PROJECT CONDITIONS

A. The type, size and location of known underground utilities have been shown on the drawings; however, no guarantee is made as to the true location of such utilities.

B. Verify the existence and location of all underground utilities at the project site. Notify 'One-Call' at 811 at least three business days prior to any excavation work so local utility companies can locate their lines.

1.6 DEFINITIONS

A. Bedding: Layer placed over the excavated subgrade in a trench before laying pipe.

B. Backfill: Soil materials used to fill an excavation.

1. Initial Backfill: Backfill placed beside and over pipe in a trench, including haunches to support sides of pipe.
2. Subsequent Backfill: Backfill placed over initial backfill to fill a trench.

C. Borrow: Satisfactory soil imported off-site for fill or backfill.

D. Unclassified excavation: Excavation and disposal of all material encountered in the work, including excavation obtained from borrow source.

E. Rock excavation: Removal and disposal of igneous, metamorphic, and sedimentary rock which cannot be excavated without blasting or using rippers, and all boulders or other detached stones each having a volume of one-half cubic yard or more. Note: Boulders exceeding 2 feet in diameter measured along the shortest cross section shall be left on site for landscaping purposes.

F. Muck excavation: Removal and disposal of saturated organic mixtures of soils or organic matter which requires additional work or equipment which would not normally be required for unclassified excavation.

PART 2 - PRODUCTS

2.1 BEDDING

A. Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand, ASTM D 2940, except with 100 percent passing a ½-inch sieve and not more than 8 percent passing a No. 200 sieve. On site materials may be used.

2.2 BACKFILL AND FILL

A. Excavated material may be processed and used for backfill if compliance with the material specified herein can be documented. Provide documentation to the Contracting Officer for review. Import material if excavated material is not sufficient to meet requirements.

B. Satisfactory Soils: ASTM D 2487 soil classification groups GW, GP, GM, SW, SP, and SM, or a combination of these group symbols, free of rock or gravel larger than 3 inches in any dimension, debris, waste frozen materials, vegetation, and other deleterious matter.

C. Initial Backfill: Naturally or artificially graded mixture of natural or crushed gravel, crushed stone, and natural or crushed sand, ASTM D 2940, except with 100 percent passing a ½- inch sieve and not more than 8 percent passing a No. 200 sieve. On site materials may be used.

D. Subsequent Backfill: Excavated material or borrow material, free from rock larger than 6 inches in the greatest dimension, frozen material, clods, debris, or organic material. Borrow material used as subsequent backfill shall have a gradation and character suitable to achieve the compaction specified in this Section.

2.3 WARNING TAPE

A. Detectable Warning Tape: Acid- and alkali-resistant polyethylene film warning tape manufactured for marking and identifying underground utilities, minimum 3 inches wide and 4 mils (0.1 mm) thick, continuously inscribed with a description of utility, with metallic core encased in a protective jacket for corrosion protection, detectable by metal detector when tape is buried up to 30 inches deep, colored as follows:

1. Red: Electric.
2. Blue: Water systems.
3. Green: Wastewater systems.

PART 3 - EXECUTION

3.1 ALIGNMENT AND GRADE

A. Stake out and establish grades for the location of pipelines and utility structures. Obtain Contracting Officer approval before starting excavation.

3.2 PREPARATION

A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout and other hazards created by earthwork operations.

B. Protect subgrades and foundation soils against freezing temperatures or frost. Provide protective insulating materials as necessary.

C. Provide erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.

3.3 SHEETING AND SHORING

A. Provide adequate shoring and bracing as required by OSHA and other local governing regulations.

B. Support excavations and sides of trenches with adequate materials as necessary for the safety of personnel and protection of the utility before backfilling.

C. Install and remove all sheeting, shoring, piling, and bracing to maintain the required trench or excavated section, and to maintain the undisturbed state of the soils at and below the bottom of the excavation.

D. Withdraw sheet piling and timbers in trench excavations in such a manner as to prevent subsequent settlement of the pipe or additional backfill loadings which might overload the pipe.

E. Remove all sheeting and shoring when safe to do so.

3.4 DEWATERING

A. Prevent surface water and ground water from entering excavations, from ponding on prepared subgrades, and from flooding Project site and surrounding area.

B. Protect subgrades from softening, undermining, washout, and damage by rain or water accumulation.

1. Reroute surface water runoff away from excavated areas. Do not allow water to accumulate in excavations. Do not use excavated trenches as temporary drainage ditches.
2. Install a dewatering system to keep subgrades dry and convey ground water away from excavations. Maintain until dewatering is no longer required.

3.5 EXPLOSIVES

A. Do not use explosives.

3.6 ASPHALT CROSSING

A. Wherever new utility trenches cross an asphalt surface, saw cut the surface prior to removal. Once the surface has been removed and the area has been excavated, place all utilities that lie directly under asphalt in schedule 80 PVC sleeves, backfill and compact and replace with new asphalt.

3.7 EXCAVATION, GENERAL

A. If excavated materials intended for fill and backfill include unsatisfactory soil materials and rock, replace with satisfactory soil materials.

B. Remove topsoil from the area to be excavated and from the area where excavated material will be piled prior to excavation. Store topsoil as specified below.

C. Maintain the excavations to guard against and prevent injury to employees and the public. Provide adequate shoring and bracing as required by OSHA and other local governing regulations.

D. Fence excavations left open at the end of the working day to protect the public.

E. Notify the Contracting Officer when excavation is completed. Do not place pipe, bedding or backfill until the excavation is approved by the Contracting Officer. Do not cover pipe until placement has been approved by the CO.

3.8 EXCAVATION FOR UTILITY TRENCHES

A. Excavate trenches to indicated gradients, lines, depths, and elevations.

B. Neatly saw cut pavement to the limits indicated where trenches are cut across existing pavement

C. Excavate trenches to uniform widths to provide a working clearance on each side of pipe or conduit. Excavate trench walls vertically from trench bottom to 12 inches higher than top of pipe or conduit, unless otherwise indicated.

1. Clearance (General): 6 inches on each side of pipe or conduit.
2. Clearance from other utilities: 12 inches on each side of pipe or conduit.

D. Trench Bottoms: Excavate and shape trench bottoms to provide uniform bearing and support of pipes and conduit. Shape subgrade to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits. Remove projecting stones and sharp objects along trench subgrade.

1. Pipes and conduit less than 6 inches in nominal diameter: Hand-excavate trench bottoms and support pipe and conduit on an undisturbed subgrade.
2. Pipes and conduit 6 inches or larger in nominal diameter: Shape bottom of trench to support bottom 90 degrees of pipe circumference. Fill depressions with tamped initial bedding course.
3. Excavate trenches 6 inches deeper than elevation required in rock or other unyielding bearing material to allow for initial bedding course.

3.9 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified independent geotechnical engineering testing agency to perform field quality control testing including inspection and testing subgrades and each fill or backfill layer. Proceed with subsequent earthwork only after test results for previously completed work complies with requirements.

B. Test compaction of soils in place according to ASTM D 2922. Perform tests at the following locations and frequencies:

1. Trench Backfill: At each compacted layer (bedding and backfill), at least one test for every 75 feet or less of trench length, but in no case fewer than two tests.

C. When testing agency reports that fills or backfills have not achieved degree of compaction specified, scarify and moisten or aerate, or remove and replace soil to depth required; recompact and retest until specified compaction is obtained.

D. Correct excessive settlement or other evidence of improper backfill by reopening the trench or excavation to the depth required for proper compaction. Refill, compact, and retest.

E. Pay for the correction and retesting of unacceptable work at no expense to the Government.

3.10 STORAGE OF SOIL MATERIALS

A. Stockpile borrow materials and satisfactory excavated soil materials. Stockpile soil materials without intermixing. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.

B. Keep topsoil separate from trench-excavated material by stockpiling. Reuse topsoil after backfilling.

3.11 UTILITY TRENCH BACKFILL

A. Place bedding material on trench bottoms by hand and tamp to the density specified. Install bedding to the full width of the trench bottom to a minimum depth of 6 inches. Shape initial bedding course to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits. Carefully compact bedding under pipe haunches

B. After individual pipe is jointed, place initial backfill material by hand along each side of the pipe to offset conditions that might tend to move the pipe off line and grade. Place backfill to prevent pipe movement under pressure tests. Tamp backfill to the density specified. Compact evenly up on both sides and along the full length of utility piping or conduit to avoid damage or displacement of piping or conduit. Keep all joints, valves, etc., uncovered for inspection purposes during testing.

C. Do not complete backfill until pressure and leakage tests have been performed and the pipeline installation is approved by the Contracting Officer. Place and compact initial backfill to a height of 12 inches over the utility pipe or conduit. Place initial backfill and bedding in layers not to exceed 4 inches in loose depth.

D. Place and compact subsequent backfill to final subgrade in layers not to exceed twelve inches in loose depth. Compact each layer as specified in this section. Do not place large rock remaining in the backfill material and approved by the Contracting Officer within three feet of the pipe.

E. Fill voids with backfill materials while shoring and bracing, and as sheeting is removed.

F. Install warning tape directly above utilities, between 4 and 8 inches below finished grade.

G. Place 6 inches of topsoil as final backfill in trench areas where this material had been previously stripped and stockpiled. Compact topsoil layer to remain in place but remain loose enough to allow plant growth.

3.12 MOISTURE CONTROL

A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill layer before compaction to within 2 percent of optimum moisture content.

1. Do not place backfill or fill material on surfaces that are muddy, frozen, or contain frost or ice.
2. Remove and replace, or scarify and air-dry, otherwise satisfactory soil material that exceeds optimum moisture content by 2 percent and is too wet to compact to specified dry unit weight.

3.13 COMPACTION OF BACKFILLS AND FILLS

A. See Section 312250.

3.14 DISPOSAL OF SURPLUS AND WASTE MATERIALS

A. Disposal: Remove surplus satisfactory soil and waste material, including unsatisfactory soil, trash, and debris, and legally dispose of it off Government property.

END OF SECTION

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PART 1 - GENERAL

1.1 SUMMARY

A. Work Included. Furnish, install, maintain, and remove temporary erosion and sedimentation controls specified herein, or as required to complete the work.

B. Related Sections include the following:
   1. Division 31 Section "Earthwork" for soil materials, site excavating, filling and grading.

B. Permits and Fees: Obtain and pay for all permits and fees required for the work of this section, including erosion and sediment control and water quality permits required by the authority having jurisdiction and the South Dakota Department of Public Health and Environment, Water Quality Control Division.

C. Erosion Control: Provide erosion control on all exposed surfaces created by construction activities.

1.2 SUBMITTALS

A. Storm Water Management Plan:

   1. The Contractor is to provide a Storm Water Management Plan (SWMP) and report addressing erosion and sediment control measures for all sites with over one acre of disturbed ground. The Engineer may assist in preparation of the General Permit application.
   2. The Contractor is responsible for obtaining all required permits including a General Permit application for Storm Water Discharges associated with construction activities at least ten (10) days prior to start of construction. Permits are to be filed with the South Dakota Department of Public Health and Environment, Water Quality Control Division.
   3. Contractor shall have the Storm Water Management Plan (SWMP) and report available on-site at all times.

1.3 QUALITY ASSURANCE:

A. Regulatory Requirements: Comply with applicable local, State and Federal ordinances, rules and regulations concerning sedimentation control and storm water runoff.

B. In case of conflict between the above codes, regulations, references and standards and these specifications, the more stringent requirements shall govern.

C. Preconstruction Conference: Conduct conference at Project site as directed by Contracting Officer prior to start of construction.

1.4 PROJECT/SITE CONDITIONS

A. Existing Conditions: Verify all existing conditions affecting the work of this section prior to submitting bids or proposals. Additional compensation will not be allowed for revisions or modification of work resulting from failure to verify existing conditions.

1.5 WARRANTY

A. Temporary Erosion and Sediment Control measures shall be maintained until permanent measures are in place. All damaged, disturbed or devices filled with sediment, which may occur within the specified project warranty period, shall be corrected at no cost to the Government. Any devices damaged by erosion or sediment shall be restored to their original condition by the Contractor, at no cost to the Government.

PART 2 - PRODUCTS

MATERIALS

A. Erosion and Sedimentation Control Materials: Provide one or more of the following materials, as shown on the plans or as applicable for site conditions:

1. Sand bags.
2. Clean, seed-free, certified, cereal hay or grain straw bales.
3. Silt fences.
4. Rock riprap.
5. Temporary seeding.
6. Biodegradable wood excelsior, straw, or coconut-fiber mat enclosed in a photodegradable plastic mesh.
7. Biodegradable twisted jute or spun-coir mesh, 0.92 lb/sy minimum, with 50 to 65 percent open area.
8. Drainage geotextile.
9. Impervious fill.
10. Other materials proposed for use on-site.

PART 3 - EXECUTION

3.1 PREPARATION

A. General:

1. Determine the existing ground elevations, drainage patterns, and changes to such patterns during excavation in order to satisfactorily plan and provide materials for adequate erosion and sediment control devices.

TEMPORARY EROSION AND SEDIMENTATION CONTROL

A. Provide temporary erosion and sedimentation control measures to prevent soil erosion and discharge of soil-bearing water runoff or airborne dust to adjacent properties and rights-of-way according to requirements of authorities having jurisdiction.

B. Inspect, repair, and maintain erosion and sedimentation control measures during construction until permanent vegetation has been established.

C. Remove erosion and sedimentation controls, and restore and stabilize areas disturbed during removal.

D. Secure grading permit from agency have jurisdiction prior to commencing grading operations.

EXAMINATION

A. Verification of Conditions: Examine areas and conditions under which the work of this section will be performed. Do not proceed with the work until unsatisfactory conditions have been corrected. Commencement of work implies acceptance of all areas and conditions.

INSTALLATION

A. Erosion and Sedimentation Control Devices. Erosion and sedimentation control measures to be taken during construction include, but are not necessarily limited to the following:

1. Apply soil stabilization within 14 days to all disturbed areas that are to be dormant for a period longer than 30 calendar days after reaching grade. Stabilize soil with mulch anchored per criteria of authorities having jurisdiction. Temporarily revegetate areas that will remain in an interim condition for more than three months.
2. Roads and parking areas indicated to be paved may be covered with an appropriate aggregate base course in lieu of mulch. Temporary mulching or aggregate base course is not required if final pavement construction will take place within 30 days after grading to final contours.
3. Soils that will be stockpiled for more than 30 days must be mulched and seeded within 14 days after stockpile construction.
4. Prevent sediment from leaving the project site by installing a silt fence or other BMPs as indicated on the plans. Protect existing storm inlets adjacent to the site by an approved gravel filter.

5. Haybales may be used at specific locations to provide temporary filtration of sediment from runoff.
6. Provide temporary erosion controls consisting of berms at the top of slopes and interceptor ditches at ends of berms and at those locations which will eliminate or minimize erosion during construction, along with temporary seeding, temporary diversion, chutes, and down pipes and lining of water courses.
7. Temporary sedimentation controls shall consist of silt dams, traps, silt fence, barriers, and appurtenances at the top of spoil and borrow area slopes and where runoff water exits the site.
8. Maintain the available silt holding capacity of silt dams, fence traps and barriers until no longer needed. The sediment capacity of sediment retainage areas shall be at a minimum, the capacity shown on the plans in conformance with Urban Drainage Criteria Manual, Volume 3. Prior to removal, obtain concurrence of the Contracting Officer and Engineer.
9. Remove accumulated sediment and debris from a BMP when the sediment level reaches one-half the height of the BMP, or at any time the sediment or debris adversely impacts the functioning of the BMP.
10. Remove hay bales which have deteriorated and filter stone or cloth which has become dislodged. Place new hay bales and new filter and fence.
11. The erosion/sediment control plan shows the minimum required for the project. If it becomes apparent that additional controls are necessary, the Contractor shall install additional controls.

B. Chemicals and Pollutants:

1. Store construction materials and chemicals that could contribute pollutants to the runoff within an enclosure, container, or dike located around the perimeter of the storage area, to prevent discharge of these materials into runoff from the construction site.
2. Locate areas used for collection and temporary storage of solid and liquid waste away from the storm drainage system. Provide covering or fencing as required to prevent windblown materials; construct perimeter dike to contain liquid runoff. These measures may not be necessary if materials are immediately placed in covered waste containers.
3. Perform equipment maintenance in designated areas using measures such as drip pans to control petroleum products spillage.
4. Immediately clean up and properly dispose of spills of construction related materials such as paints, solvents, or other chemicals.

C. Final Stabilization and Long-Term Management:

1. Final stabilization shall be achieved through permanent vegetation and landscaping after construction of all buildings and paved surfaces.
2. With approval of Contracting Officer temporary erosion and sediment control measures may be removed within 30 days after final site stabilization is achieved or after temporary measures are no longer needed.

A. Inspection and Maintenance: Inspect erosion and sediment control measures weekly during construction. In addition, inspect all facilities immediately after any significant runoff or snowmelt which results in runoff. Repair or otherwise mitigate any damage to the erosion and sediment control facilities at no additional cost to the Government.

CLEANING

A. Removal of Controls: Remove controls upon completion of that portion of the work for which controls were furnished. Leave the site and work area in a clean condition.

END OF SECTION

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PART 1 – GENERAL

1.1 SUMMARY

A. This work consists of the installation of barrier boulders and includes furnishing all labor, materials, equipment, and tools, necessary for completion of this work.

1.2 RELATED SECTIONS

A. The following items of associated work are included in other sections of these specifications:
1. Section 311000 Clearing and Grubbing
2. Section 312250 Earthwork

PART 2 – PRODUCTS

2.1 BOULDERS

A. Barrier boulders shall be at least 30 inches long by 30 inches high by 30 inches thick.

B. Boulders meeting the minimum dimension requirements that are found on site at locations approved by the Contracting Officer may be used for barrier boulders.

C. If boulders are not available onsite meeting the minimum dimension requirements, the Contractor shall furnish and install barrier boulders as shown on drawings.

D. Boulders shall be approved by CO prior to placement.

2.2 BACKFILL MATERIAL

A. Native soil from excavations shall be used as backfill material around the installed barrier boulders.

PART 3 – EXECUTION

3.1 LOCATION

A. Asphalt shall be sawcut prior to placement of boulders.

B. The Contractor shall place barrier boulders as shown on drawings and as approved by COR. Locations of all boulders shall be staked in the field and approved by COR prior to installation.

3.2 INSTALLATION

A. The boulders shall be placed as shown on the drawings.

B. The boulder shall be placed in the excavation in a stable condition. Fractured, jagged or pointed sides of the boulder should be buried. The weathered side of the boulder should be placed upward where possible. Boulders shall be positioned so that the long axis of each is parallel with the adjacent edge of road. Backfill shall be native material moistened and compacted to the surrounding undisturbed soil density.

C. Asphalt patch shall be installed around boulders as shown on drawings. Depth of asphalt shall match depth of existing asphalt surface.

D. All damage that results to vegetation from excavating, loading, transporting, and placing barrier boulders shall be repaired and the borrow and work areas cleaned and graded by the Contractor in accordance with Section 311100 at no additional cost to the Government.

3.3 SURPLUS EXCAVATION MATERIAL

A. Surplus excavation material shall be spread on the ground adjacent to or near the boulders not to exceed two inches in depth and raked to provide a neat appearance.

END OF SECTION

PART 1 - GENERAL

1.1 SUMMARY

A. The work of this section consists of furnishing and placing aggregate, crusher fines, and filler if required, on a prepared subgrade.

1.2 SUBMITTALS

A. As specified in Division 1 Section “Submittals”.

B. Two copies of certified weight tickets for each load of aggregate and crusher fines delivered to project site.

C. If materials are obtained from a commercial source, submit certification from the supplier certifying that aggregate base course and crusher fines meet the requirements of this section.

D. Submit all Material and Compaction Test results directly to Contracting Officer (CO) from testing services, with copy to Contractor.

1.3 QUALITY ASSURANCE

A. Material Test and Compaction Test to be the responsibility of the Contractor. Compaction Test, Paragraph 3.03 and Material Test, listed in 2.01 shall be performed at the rate of one test for each 1,000 tons of aggregate and crusher fines delivered or fraction thereof with a minimum of one test per day of operation.

PART 2 - PRODUCTS

2.1 AGGREGATE

A. Clean, hard, durable fragments or particles of crushed stone, crushed slag, or crushed or natural gravel. Materials that break up due to freeze-thaw or wet-dry cycling shall not be used.

B. Coarse Aggregate: AASHTO T96-77, percentage of wear of not more than 50.

C. Fraction passing No. 40 sieve shall have a liquid limit not to exceed 25 and a plasticity index of not more than three, as determined by AASHTO T89-81 and T90-81, respectively.

D. Material, inclusive of filler, shall meet the requirements shown in the table below:

SIEVE DESIGNATION	PERCENT PASSING
1 inch	100
¾ inch	80 - 100
½ inch	68-90
No. 4	42-70
No. 8	29-53
No. 40	10 - 28
No. 200	3 - 12

E. Mineral filler or binder shall be added, to meet quality and/or gradation requirements. Mineral filler or binder shall be uniformly blended during crushing when a crusher operation is used.

2.2 CRUSHER FINES

A. Clean, hard, durable fragments or particles of crushed stone, crushed slag, or crushed or natural gravel. Materials that break up due to freeze-thaw or wet-dry cycling shall not be used.

B. Material, inclusive of filler, shall meet the requirements shown in the table below:

SIEVE DESIGNATION	PERCENT PASSING
¼ inch	100
No. 8	53-83
No. 16	33-53
No. 40	20-40

C. Mineral filler or binder shall be added, to meet quality and/or gradation requirements. Mineral filler or binder shall be uniformly blended during crushing when a crusher operation is used.

D. Installed crusher fines surfacing shall be ABA compliant.

PART 3 – EXECUTION

3.1 PLACING

A. If the required compacted depth of the aggregate base course or crusher fines exceeds six inches, place course in two or more layers of approximately equal thickness. The maximum compacted thickness of any one layer will not exceed six inches.

3.2 MIXING

A. Mix the aggregate or crusher fines by any one of the three following methods:

1. Stationary Plant Method: Mix aggregate base course or crusher fines and appropriate amount of water for compaction in an approved mixer. After mixing, transport aggregate to the job site while it contains the proper moisture content and place on the roadbed with an approved aggregate or crusher fines spreader. Before compaction, remove excess moisture.
2. Travel Plant Method: After the material for each layer has been placed through an aggregate or crusher fines spreader or windrow-sizing device, it shall be uniformly mixed by a traveling mixing plant.
3. Road Mix Method: After placing each layer, mix materials at optimum moisture content using motor graders or other approved equipment until the moisture is uniform throughout.

3.3 COMPACTION

A. Compact each layer to a density of not less than 95 percent of the maximum density, as determined by AASHTO T180-74, Method D. Test density in-place, in accordance with AASHTO T191-61, T205-64, or other recognized method. Density tests to be performed at the rate of one test per lift per 2,500 square feet.

3.4 SURFACE FINISHING

A. Use a smooth steel wheel roller for the final rolling of top surface base course or final surface. Water surface and evenly spread loose stones before final rolling. Make minimum of two complete passes over area to embed stones. Correct soft spots developed during rolling.

B. Compacted base course surface shall be smooth and free from waves and other irregularities. Unsatisfactory portions of base course shall be torn up, reworked, relaid, and rerolled, at no additional expense to Government.

3.5 MATERIAL ACCEPTANCE REQUIREMENTS

A. Acceptance will be based on periodic samples and tests taken following mixing and before laying.

3.6 TOLERANCES

A. Surface: Contractor will test finished surface of the base course with a 10-foot straightedge or other device. The variation between any two contacts with the surface shall not exceed one-half inch. Any areas not complying with these tolerances shall be reworked to obtain conformity.

B. Width: Plan dimension, plus or minus two inches.

C. Thickness: Plan dimension, plus or minus one-half inch.

3.7 MAINTENANCE

A. Maintain base course in a satisfactory condition until surfaced or until final acceptance.

END OF SECTION

SECTION 321217
HOT-MIX ASPHALT PAVING

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes hot-mix asphalt **paving** and the following:

1. Aggregates
2. Asphalt binder
3. Tack coat
4. Herbicide
5. Cold milling

1.2 SUBMITTALS

A. Product Data: For each type of product indicated. Include technical data and tested physical and performance properties.

B. Job-Mix Designs: Certification, by authorities having jurisdiction, of approval of each job mix proposed for the Work.

C. Material certificates.

1.3 QUALITY ASSURANCE

A. Manufacturer Qualifications: Manufacturer shall be registered with and approved by authorities having jurisdiction or South Dakota DOT.

B. Regulatory Requirements: Comply with South Dakota DOT for asphalt paving work.

C. Asphalt-Paving Publication: Comply with AI MS-22, "Construction of Hot Mix Asphalt Pavements," unless more stringent requirements are indicated.

1.4 PROJECT CONDITIONS

A. Environmental Limitations: Do not apply asphalt materials if subgrade is wet or excessively damp or if the following conditions are not met:

1. Tack Coat: Minimum surface temperature of 60 deg F (15.5 deg C).
2. Asphalt Base Course: Minimum surface temperature of 40 deg F (4 deg C) and rising at time of placement.
3. Asphalt Surface Course: Minimum surface temperature of 60 deg F (15.5 deg C) at time of placement.

B. Pavement-Marking Paint: Proceed with pavement marking only on clean, dry surfaces and at a minimum ambient or surface temperature of 40 deg F (4 deg C) for oil-based materials, 50 deg F (10 deg C) for water-based materials, and not exceeding 95 deg F (35 deg C).

PART 2 - PRODUCTS

2.1 AGGREGATES

A. Coarse Aggregate: ASTM D 692, sound; angular crushed stone, crushed gravel, or properly cured, crushed blast-furnace slag.

B. Fine Aggregate: **ASTM D 1073**, sharp-edged natural sand or sand prepared from stone, gravel, properly cured blast-furnace slag, or combinations thereof.

C. Mineral Filler: **ASTM D 242**, rock or slag dust, hydraulic cement, or other inert material.

2.2 ASPHALT MATERIALS

A. Asphalt Binder: AASHTO MP 1, PG 58-28.

B. Tack Coat: **ASTM D 977**, emulsified asphalt or **ASTM D 2397**, cationic emulsified asphalt, slow setting, diluted in water, of suitable grade and consistency for application.

2.3 AUXILIARY MATERIALS

A. Herbicide: Commercial chemical for weed control, registered by the EPA. Provide in granular, liquid, or wettable powder form.

B. Pavement-Marking Paint: Alkyd-resin type, lead and chromate free, ready mixed, complying with FS TT-P-115, Type [**I**] [**II**] or AASHTO M 248, Type [**N**] [**F**].

1. Color: [White]

C. Pavement-Marking Paint: Latex, waterborne emulsion, lead and chromate free, ready mixed, complying with FS TT-P-1952, with drying time of less than [**45**] minutes.

1. Color: [White]

2.4 MIXES

A. Hot-Mix Asphalt: Dense, hot-laid, hot-mix asphalt plant mixes approved by authorities having jurisdiction; [**designed according to procedures in AI MS-2, "Mix Design Methods for Asphalt Concrete and Other Hot-Mix Types";**] and complying with the following requirements:

1. Provide mixes with a history of satisfactory performance in geographical area where Project is located.
2. Base Course: Aggregate Base per SDDOT 882.2.
3. Surface Course: SDDOT Class E, Type 1.

PART 3 - EXECUTION

3.1 PATCHING

A. Hot-Mix Asphalt Pavement: Saw cut perimeter of patch and excavate existing pavement section to sound base. Excavate rectangular or trapezoidal patches, extending 12 inches (300 mm) into adjacent sound pavement, unless otherwise indicated. Cut excavation faces vertically. Remove excavated material. Recompact existing unbound-aggregate base course to form new subgrade.

B. Tack Coat: Apply uniformly to vertical surfaces abutting or projecting into new, hot-mix asphalt paving at a rate of 0.05 to 0.15 gal. /sq. yd. (0.2 to 0.7 L/sq. m).

C. Patching: Fill excavated pavements with hot-mix asphalt base mix and, while still hot, compact flush with adjacent surface.

3.2 SURFACE PREPARATION

A. Proof-roll subbases using heavy, pneumatic-tired rollers to locate areas that are unstable or that require further compaction.

B. Immediately before placing asphalt materials, remove loose and deleterious material from substrate surfaces. Ensure that prepared subgrade is ready to receive paving.

   1. Sweep loose granular particles from surface of unbound-aggregate base course. Do not dislodge or disturb aggregate embedded in compacted surface of base course.

C. Tack Coat: Apply uniformly to surfaces of existing pavement at a rate of 0.05 to 0.15 gal. /sq. yd. (0.2 to 0.7 L/sq. m).

   1. Allow tack coat to cure undisturbed before applying hot-mix asphalt paving.
   2. Avoid smearing or staining adjoining surfaces, appurtenances, and surroundings. Remove spillages and clean affected surfaces.

3.3 HOT-MIX ASPHALT PLACING

A. Machine place hot-mix asphalt on prepared surface, spread uniformly, and strike off. Place asphalt mix by hand to areas inaccessible to equipment in a manner that prevents segregation of mix. Place each course to required grade, cross section, and thickness when compacted.

1. Spread mix at minimum temperature of 250 deg F (121 deg C).
2. Regulate paver machine speed to obtain smooth, continuous surface free of pulls and tears in asphalt-paving mat.

B. Place paving in consecutive strips not less than 10 feet (3 m) wide unless infill edge strips of a lesser width are required.

C. Promptly correct surface irregularities in paving course behind paver. Use suitable hand tools to remove excess material forming high spots. Fill depressions with hot-mix asphalt to prevent segregation of mix; use suitable hand tools to smooth surface.

D. Place asphalt in 2-inch lifts to match depth of existing surrounding asphalt.

3.4 COMPACTION

A. General: Begin compaction as soon as placed hot-mix paving will bear roller weight without excessive displacement. Compact hot-mix paving with hot, hand tampers or vibratory-plate compactors in areas inaccessible to rollers.

1. Complete compaction before mix temperature cools to 185 deg F (85 deg C).

B. Breakdown Rolling: Complete breakdown or initial rolling immediately after rolling joints and outside edge. Examine surface immediately after breakdown rolling for indicated crown, grade, and smoothness. Correct laydown and rolling operations to comply with requirements.

C. Intermediate Rolling: Begin intermediate rolling immediately after breakdown rolling while hot-mix asphalt is still hot enough to achieve specified density. Continue rolling until hot-mix asphalt course has been uniformly compacted to the following density:

1. Average Density: 92 percent of reference maximum theoretical density according to ASTM D 2041, but not less than 90 percent nor greater than 96 percent.

D. Finish Rolling: Finish roll paved surfaces to remove roller marks while hot-mix asphalt is still warm.

E. Protection: After final rolling, do not permit vehicular traffic on pavement until it has cooled and hardened.

F. Erect barricades to protect paving from traffic until mixture has cooled enough not to become marked.

3.5 INSTALLATION TOLERANCES

A. Thickness: Compact each course to produce the thickness indicated within the following tolerances:

1. Base Course: Plus or minus 1/2 inch (13 mm).
2. Surface Course: Plus 1/4 inch (6 mm), no minus.

B. Surface Smoothness: Compact each course to produce a surface smoothness within the following tolerances as determined by using a 10-foot (3-m) straightedge applied transversely or longitudinally to paved areas:

1. Base Course: [**1/4 inch (6 mm)**].

2. Surface Course: [**1/8 inch (3 mm)**].

3. Crowned Surfaces: Test with crowned template centered and at right angle to crown. Maximum allowable variance from template is 1/4 inch (6 mm).

3.6 PAVEMENT MARKING

A. Do not apply pavement-marking paint until layout, colors, and placement have been verified with Contracting Officer.

B. Allow paving to age for 30 days before starting pavement marking.

C. Sweep and clean surface to eliminate loose material and dust.

D. Apply paint with mechanical equipment to produce pavement markings, of dimensions indicated, with uniform, straight edges. Apply at manufacturer's recommended rates to provide a minimum wet film thickness of 15 mils (0.4 mm).

1. Broadcast glass spheres uniformly into wet pavement markings at a rate of 6 lb/gal. (0.72 kg/L).

3.7 FIELD QUALITY CONTROL

A. Testing Agency: Contractor will engage a qualified independent testing and inspecting agency to perform field tests and inspections and to prepare test reports.

B. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.

C. Remove and replace or install additional hot-mix asphalt where test results or measurements indicate that it does not comply with specified requirements.

3.8 DISPOSAL

A. Except for material indicated to be recycled, remove excavated materials from Project site and legally dispose of them in an EPA-approved landfill.

END OF SECTION 321217

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SECTION 321313
CONCRETE FOR MINOR STRUCTURES

PART 1 - GENERAL

1.1 SUMMARY

A. This specification is intended for minor structures such as thrust blocks, anchor blocks, utility pads, and wheel stops.

1.2 SUBMITTALS

A. Product Data: For each manufactured material and product indicated.

B. Design Mixes: For each concrete mix indicated.

C. Shop Drawings: Include details of steel reinforcement placement including material, grade, bar schedules, stirrup spacing, bent bar diagrams, arrangement, and supports.

PART 2 – PRODUCTS

2.1 PREPACKAGED CONCRETE MIX

A. Prepackaged concrete mix shall not be used where the hardened concrete will be exposed to freeze-thaw conditions unless surface deterioration can be tolerated, as determined by the (CO). It shall not be used where any single placement will be more than one cubic yard.

B. Material shall be identified as conforming to ASTM C387, Normal Strength Concrete; Normal Weight Concrete.

2.2 COMMERCIAL PLANT CONCRETE

A. 4,000 psig minimum compressive strength, meeting as a minimum the requirements of JOB-BATCHED CONCRETE.

2.3 JOB-BATCHED CONCRETE

A. Aggregates shall be clean, washed materials supplied by a firm regularly engaged in the production of concrete aggregates. Separate stockpiles shall be maintained of sand passing a 3/8-inch sieve and coarse aggregate of 3/4 inch to No. 4 nominal size.

B. Cement:
   1. ASTM C150, type II or I.

C. Proportion dry ingredients by volume in the following ratio:
   1. Cement: 3 parts.
   2. Sand: 5 parts.
   3. Coarse aggregate: 7 parts.

D. Add water to obtain a slump of 3-6 inches. Based on the ratio of dry ingredients of 3-5-7, the water will be approximately two parts.

E. The relative proportions of sand and coarse aggregate may be adjusted, upon approval of the CO, if necessary to obtain better workability.

2.4 FORMS

A. Steel, Wood, or other suitable material of size and strength to resist movement during concrete placement and to retain horizontal and vertical alignment until removal. Use straight forms, free of distortion and defects.

B. Coat forms with a non-staining form release agent that will not discolor or deface surface of concrete.

2.5 REINFORCING BARS

A. Deformed steel bars: ASTM A 615, Grade 40.

PART 3 – EXECUTION

3.1 SURFACE PREPARATION

A. Remove loose material from compacted subbase surface immediately before placing concrete.

3.2 FORM CONSTRUCTION

A. Set forms to require grades and lines, rigidly braced and secured.

B. Install sufficient quantity of forms to allow continuous progress of work so that forms can remain in place at least 24 hours after concrete placement.

C. Clean forms after each use, and coat with form release agent as often as required to ensure separation from concrete without damage.

3.3 PLACING REINFORCEMENT

A. Clean reinforcement of loose rust and mill scale, earth, ice, and other materials, which reduce or destroy bond with concrete.

B. Place reinforcement to obtain at least minimum coverage's for concrete protection.
  1. Arrange, space and securely tie bars and bar supports to hold reinforcement in position during concrete placement operations.
  2. Set wire ties so ends are directed into concrete, not toward exposed concrete surfaces.

C. Install welded wire fabric in as long lengths as practicable.
  1. Lap adjoining pieces at least one full mesh and lace splices with wire. Offset end laps in adjacent widths to prevent continuous laps in either direction.

3.4 MIXING OF JOB-BATCHED CONCRETE

A. Thoroughly mix ingredients.

B. Use mechanical mixer if total quantity to be placed is over 8 cubic feet.

3.5 CONCRETE PLACEMENT

A. Thoroughly dampen compacted subgrade before placement.

B. Forms shall be free of debris, and surfaces shall be clean, free of frost, ice, mud, and water.

C. Place in layers to avoid segregation and begin placing at corners of structure.

D. Consolidate with mechanical vibrators or hand tampers.

E. Thoroughly work the external surface to bring mortar against the forms to produce a smooth finish, substantially free of air or water pockets or honeycomb.

3.6 COLD WEATHER

A. Heat or protect concrete, if necessary, to maintain a temperature of more than 35 degrees F for at least 24 hours after placing.

3.7 CURING

A. Keep surfaces moist a minimum of 2 days. Acceptable methods include ponding, wet burlap, wet straw or hay, curing paper, plastic sheets, and membrane curing compound. The CO must approve other methods.

3.8 CONCRETE FINISHING

A. After striking-off and consolidating concrete, smooth surface with screeding and floating. Use hand methods only where mechanical floating is not possible.

B. After completion of floating and troweling when excess moisture or surface sheen has disappeared, complete surface finishing, as follows:
   1. Broom finish, by drawing a fine-hair broom across concrete surface, perpendicular to line of traffic. Repeat operation if required to provide a fine line texture acceptable to the CO.

C. Do not remove forms for 24 hours after concrete has been placed.

D. After form removal, clean ends of joints and point-up any minor honeycombed areas.

E. Remove and replace areas or sections with major defects.

3.9 REPAIRS AND PROTECTIONS

A. Repair or replace broken or defective concrete.

B. Protect concrete from damage until acceptance of work.
   1. Exclude traffic from pavement for at least 14 calendar days after placement.
   2. When construction traffic is permitted, maintain pavement as clean as possible by removing surface stains and spillage of materials as they occur.
   3. Sweep concrete pavement and wash free of stains, discolorations, dirt and other foreign material just prior to final inspection.

END OF SECTION 321313

PART 1 - GENERAL

1.1 DESCRIPTION

A. This item shall consist of concrete curbing wheel stops as shown on the drawings in conformity with the typical layout and dimensions to the grade established.

PART 2 - PRODUCTS

2.1 REQUIREMENTS

A. The precast concrete wheelstops shall be the size and shape as shown on the drawings. The concrete shall be reinforced as shown on the drawings and shall be 4,000 psi (28 day), 6.5 bag mix, with 6 percent entrained air.

B. Before any of the precast sections are placed in the work, the Contractor shall furnish to the Contracting Officer a "certificate confirming compliance for reinforcement and strength" issued by the Precast Plant.

PART 3 - EXECUTION

3.1 PERFORMANCE

A. Wheelstops shall be placed at the location shown on the drawings. All devices shall be constructed to the dimensions shown on the drawings.

B. The Forest Service shall have the right to inspect the precast curb sections either at the Precast Plant or after delivery and reject any that do not meet the specifications. All wheelstops that are damaged through improper handling or placing shall be rejected.

3.2 INSTALLATION

A. Placing and Anchoring - Wheelstops shall be placed or set as illustrated on the drawings or as directed by the CO. They shall, when set in final location, set solidly on the bedding or surfacing.

B. When lifting wires have been cast in the bottom of the sections, they shall be removed. The wires may be removed by cutting reasonably flush so that they do not interfere with the section setting solidly.

C. The wheelstops shall be anchored in place by driving #6 rebar flush with the top through the two holes provided. These pins shall be 3/4 inch round or #6 deformed reinforcing steel, 30 inches long.

END OF SECTION

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SECTION 323000
SIGNS

PART 1 - GENERAL

1.1 DESCRIPTION

A. This work shall consist of furnishing and installing delineators, signs, hardware, sign supports, panels, and posts or removing and disposing of existing signs, posts, and hardware.

1.2 TRAFFIC CONTROL SIGN DETAILS

A. Traffic control sign details not SHOWN ON THE DRAWINGS shall meet the requirements of the MUTCD. The Contractor shall be responsible for furnishing and installing all necessary Traffic Control signs during the construction operations.

PART 2 - PRODUCTS

2.1 REQUIREMENTS

A. All concrete and reinforcing steel shall meet the requirements of Section 321313.

2.2 SIGN PANELS

A. Plywood Panels. The panels shall be exterior Type B-B, high-density overlay, 60/60 with black overlay on both sides, 3/4-inch 7 ply or 1/2-inch 5 ply thick, Douglas fir plywood or better, meeting the requirements of National Bureau of Standards PS 1, current edition or AS SHOWN ON THE DRAWINGS. Other overlay colors may be used provided the back of the panel is printed with two heavy coats of black paint.

B. Aluminum Panels. All sheets and plates shall meet the requirements of ASTM B 209, alloy 6061-T6, or 5052-H38 and shall be of the thickness shown below unless otherwise SHOWN ON THE DRAWINGS.

Sign Width (Inches)	**Sheet Aluminum Thickness (Inches)**
Less than 8	0.022
8-12	0.040
13-19	0.063
20-30	0.080
31-48	0.100
Over 48	0.125

2.3 SITE IDENTIFICATION PANELS FOR CAMPGROUND ENTRANCE SIGN

A. All signs shall be furnished and installed by the Contractor. Contractor shall be responsible for all timber material and attachment hardware.

B. Product Data:
   1. ¾” thick routed recycled HDPE Plastic brown and cream-yellow colors to match U.S. Forest Service sign standards and guidelines.
   2. Top of sign shall be Brown legend/text on yellow-cream background.
   3. Bottom half of sign shall be yellow-cream legend/text on brown background.
      a. Brown color = Federal Color #20059
      b. Yellow-cream = Federal Color #23695
   4. Digital graphic inlayed routed text to meet Forest Service Signs and Poster Handbook - EM-7100-15 & MUTCD for sign and letter size requirements.
   5. Recycled plastic shall be Ultra-Violet radiation stabilized for durability and to inhibit color change over time. Signs shall not be painted.

2.4 OTHER CAMPGROUND SIGNS

A. All signs shall be furnished and installed by the Contractor. Contractor shall be responsible for all timber material and attachment hardware.

B. Product Data:
   1. Aluminum backing thickness shall be determined from table above (Section 2.2B).
   2. Text and boarder edging shall be white with brown background.
      a. Brown color = Federal Color #20059
   3. Both white and brown elements of signs shall have high intensity prismatic (HIP) retro reflectivity.
   4. Text and symbology shall meet Forest Service Signs and Poster Handbook - EM-7100-15 & MUTCD sign and letter size requirements.
   5. Sticker shall be Ultra-Violet radiation stabilized for durability and to inhibit color change over time.
   6. Back of sign shall have brown color treatment or painted brown to reduce shiny glare.

2.5 REGULATORY SIGNS

A. All signs shall be furnished and installed by the Contractor. Contractor shall be responsible for all timber material and attachment hardware.

B. Product Data:
   1. Aluminum backing thickness shall be determined from table above (Section 2.2B).
   2. Signs shall have high intensity prismatic (HIP) retro reflectivity.
   3. Shall meet Forest Service Signs and Poster Handbook - EM-7100-15 and MUTCD sign requirements.

4. Sticker shall be Ultra-Violet radiation stabilized for durability and to inhibit color change over time.
5. Back of sign shall have brown color treatment or painted brown to reduce shiny glare.
6. Shall have fabrication details provided, in sticker form, on the back of the sign. Details shall include date of fabrication and manufacturer's name, according to Sect. 2A.04 Desing of Signs in MUTCD.

2.6 POSTS

A. Posts shall be wood, aluminum, steel, or other material as specified.

B. Wood Posts. Wood posts shall meet the dimensional requirements SHOWN ON THE DRAWINGS and meet the grading, species, and treatment requirements of Section 323000 Section 2.4 and Section 2.5.

C. Steel Posts. Steel posts shall meet the requirements of ASTM A 499, galvanized in accordance with AASHTO M 111.

D. The posts shall have 7/16-inch holes drilled or punched, before galvanizing, along the centerline of the web. The punching or drilling should begin 1 inch from the top of the post, at 2-inch centers for the upper 5 feet of the post.

E. Aluminum Posts. Aluminum posts shall be standard shapes as SHOWN ON THE DRAWINGS and shall be aluminum alloy 6061-T6 or 6351-T5 meeting the requirements of ASTM B 221.

2.7 STRUCTURAL TIMBER & LUMBER

A. Treated timber shall be rough sawn Lodgepole Pine #2 Grade or Ponderosa Pine #2 Grade as graded by WWPA, SIZE AS SHOWN ON THE DRAWINGS.

2.8 PRESERVATIVE TREATMENTS

A. All timber for timber structures shall be pressure treated with CCA with a retention of 0.4 pounds per cubic foot or to refusal in accordance with AWPA C2. Lumber shall be kiln dried after treatment to moisture content of 19 percent or less. All treated timber shall be completely and accurately fabricated before treatment. All surfaces greater than 2 inches in width shall be incised before treatment.

B. All lumber shall also be treated with 'Sunwood' as manufactured by Osmose or approved equal. No 'green colored' treated lumber shall be used on this project.

2.9 FITTINGS

A. Lag screws, washers, clip angles, wood screws, shear plates, U-bolts, clamps, bolts, nuts, and other fasteners shall be galvanized steel, cadmium-plated steel, aluminum alloy, or as SHOWN ON THE DRAWINGS.

B. Galvanizing of steel hardware shall be in accordance with AASHTO M 232. High-strength steel bolts, nuts, and washers shall meet the requirements of ASTM A 325, except as SHOWN ON THE DRAWINGS.

PART 3 - EXECUTION

3.1 SIGN INSTALLATION

A. All signs shall be installed in accordance with the Forest Service Sign Installation Guide (EM-7100).

B. Sign supports shall be erected plumb and in accordance with the details SHOWN ON THE DRAWINGS.

C. The sign panels shall be securely fastened to the posts as SHOWN ON THE DRAWINGS.

D. To reduce specular glare, sign panel face shall be erected in accordance with MUTCD.

3.2 SIGN REMOVAL

A. Sign assemblies to be removed shall be SHOWN ON THE DRAWINGS. Where signs are to be replaced, signs shall be removed just before the installation of replacement signs. All sign material removed shall become the property of the Government. Posts shall be removed to a minimum of 3 inches below natural ground line. Postholes remaining shall be backfilled with suitable material and compacted.

END OF SECTION

PART 1 - GENERAL

1.1 SUMMARY

A. This item shall consist of constructing trailer pads. It includes excavation, backfilling, and placement, treated timbers when required by site conditions as shown on the drawings, and placement of surface material in accordance with these specifications and details shown on the drawings.

PART 2 - PRODUCTS

2.1 SURFACING

A. See Specification Section 321123.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Refer to Section 312250 Earthwork.

B. The trailer pads shall be constructed as shown on the drawings.

C. Uniformly prepare the subgrade, (taking care not to damage adjacent vegetation or trees to be saved), to a level, smooth, compacted surface, free from irregular surface changes. Adjustments shall be made in the shape of the trailer pad to accommodate large rocks or trees at the periphery of the trailer pad. Subgrade material shall be moistened or dried to a uniform moisture content suitable for compaction and compact with a minimum of two passes with a mechanical tamper or until visible deformation ceases.

D. After the subgrade is complete, place compacted surface material as shown on the drawings. Moisten or dry surface material to a uniform moisture content suitable for compaction and compact with a minimum of two passes with a mechanical tamper or until visible deformation ceases.

E. After the surface material construction is complete, the area around each trailer pad shall be backfilled with subsequent backfill or borrow material and graded to blend with the surrounding topography, taking care not to damage existing vegetation.

F. Upon final grading and completion of the trailer pad, the area shall be cleaned by removing and disposing of all debris and material not utilized.

END OF SECTION

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PART 1 - GENERAL

1.1 SUMMARY

A. This item shall consist of constructing use pads. It includes excavation, backfilling, and placement, treated timbers when required by site conditions as shown on the drawings, and placement of surface material in accordance with these specifications and details shown on the drawings.

PART 2 - PRODUCTS

2.1 SURFACING

A. See Section 321123.

2.2 TREATED TIMBERS:

A. Treated timbers for walls, steps, border, and edger shall be rough sawn #2 grade or better Ponderosa Pine or equal, landscape timbers shall be brown and not green in color, uniform in size, straight, and free from sweep and crook. The timbers shall be treated with ACQ (Alkaline Copper Quaternary) or CBA (Copper Boron Azol) to a net retention of 0.4 lbs. per cu. ft. or approved equal. Sizes as shown on drawings.

B. All field cuts or abrasions and holes made in fabricated timbers after treatment shall be given 3 brush coats 2% copper Napthanate.

C. Handling Precautions: Wear protection such as dust masks, goggles, and gloves when sawing treated lumber.

PART 3 - EXECUTION

3.1 INSTALLATION

A. The use pads shall be constructed as shown on the drawings.

B. Uniformly prepare the use pad subgrade, (taking care not to damage adjacent vegetation or trees to be saved), to a level, smooth, compacted surface, free from irregular surface changes. Adjustments shall be made in the shape of the use pad to accommodate large rocks or trees at the periphery of the use pad. Subgrade material shall be moistened or dried to a uniform moisture content suitable for compaction and compact with a minimum of two passes with a mechanical tamper or until visible deformation ceases.

C. After the subgrade is complete, place surface material as shown on the drawings. Moisten or dry the surface material to a uniform moisture content suitable for compaction and compact with a minimum of two passes with a mechanical tamper or until visible deformation ceases.

D. After the surface material construction is complete, the area around each use pad shall be backfilled with subsequent backfill or borrow material and graded to blend with the surrounding topography, taking care not to damage existing vegetation.

E. Upon final grading and completion of the use pad, the area shall be cleaned by removing and disposing of all debris and material not utilized.

END OF SECTION

PART 1 - GENERAL

1.1 SUMMARY

A. This work shall consist of supplying and spreading commercial topsoil or stockpiled on site topsoil, seed and mulch at all disturbed areas. The placement of the topsoil, seed and straw mulch shall be on all disturbed areas including staging sites as directed by the Contracting Officer. All abandoned, scarified roads or spurs and all disturbed areas shall be seeded.

PART 2 - PRODUCTS

2.1 ON-SITE AND COMMERCIAL MATERIAL

A. Topsoil stripped from all on-site disturbed areas shall be stockpiled and replaced on areas to be restored.

B. Topsoil shall consist of loose, friable, free draining, vegetative matter free of clods, stumps, logs, brush, weeds, non-native plants or other material that would be detrimental to the development of native vegetative growth. No rocks greater than 4" shall be in the topsoil and rocks must compose less than 5% of the topsoil material.

2.2 SEED

A. The kinds of grass, legume, and cover crop seed furnished shall be those stipulated in the section 3.6. Seed shall be "certified" or "blue tag".

B. Seed shall be furnished separately or in mixture in standard containers with (1) seed name; (2) lot number; (3) net weight; (4) percentages of purity and of germination ; and (5) percentage of maximum weed seed content clearly marked for each kind of seed. The contractor shall furnish the Contracting Officer duplicate signed copies of a statement by the vendor, certifying that each lot of seed has been tested by a recognized laboratory for seed testing within 6 months of date of delivery. This statement shall include (1) name and address of laboratory, (2) date of test, (3) lot number for each kind of seed, and (4) results of tests as to name, percentages of purity and of germination, and percentage of weed content for each kind of seed furnish, and, in case of a mixture, the proportions of each kind of seed.

C. All seed shall be certified noxious weed free and labels submitted to the Contracting Officer for a record of what products were supplied and placed on the project by the contractor.

2.3 STRAW MULCH

A. Wheat or barley straw to be hand crimped in areas where seed is broadcast.

B. Free of mold or other evidence of decomposition.

C. Certified free from weed seed by South Dakota Department of Agriculture.

PART 3 – EXECUTION

3.1 TOPSOIL LOCATIONS

A. All disturbed areas shall receive topsoil and then be seeded.

3.2 STRIPPING

A. Complete clearing and grubbing as applicable.

B. Strip topsoil to 4-inch depth in a manner to prevent intermingling with underlying subsoil or other objectionable material.

C. Remove large roots, stones (over 4” in diameter) and other debris.

3.3 TOPSOIL STOCKPILING

A. The Contractor may stockpile provided the location and method of stockpiling are as shown on the drawings approved by the Contracting Officer. Stockpiling shall employ such protective measures as the Contracting Officer deems feasible to hold segregation, intermixing and waste to a practical minimum. Stockpiles shall be constructed to freely drain surface water. Upon completion of topsoil placement, the stockpile area shall be cleaned and restored to original condition. Excess topsoil shall be wasted upon site in locations approved by the Contracting Officer.

3.4 PLACEMENT OF TOPSOIL

A. Surfaces where topsoil is to be placed shall be at the elevation that will accommodate specified topsoil depth so that the finished surface is at the correct elevation.

B. Disturbed surfaces to receive topsoil shall be ripped and scarified to depth of 6”to 12”. Fill surfaces shall not be compacted on the top 6”.

C. Topsoil shall be deposited and spread to the top 3’-4” of the existing surface and raked smooth. Topsoil depth shall be 3’-4”.

D. Topsoil shall not be compacted and shall be graded to blend with the adjacent ground. Rocks larger than 4” in diameter or other organic material brought to the surface shall be buried.

E. Topsoil shall not be spread when the ground or topsoil is frozen, excessively wet or in a condition detrimental to the work.

F. The roadbed surfacing shall be kept clean during hauling operations. Topsoil or other soil deposited upon the surfacing shall be removed before it becomes compacted by traffic.

3.5 SEEDING SEASONS

A. Seeding season shall consist of all the months of the year (fall preferred) except June, July and August, whenever the ground is free of snow, not frozen or muddy as determined by the Contracting Officer. Seeding materials shall not be applied if wind velocity exceeds 5 MPH or when the ground is excessively wet or frozen. Work shall be performed during each specified seeding season on all completed and previously untreated sections.

3.6 APPLICATION METHODS FOR SEED

A. Material may be placed by the following methods:

1. Drilled Method. Mechanical seed drills or other approved mechanical seeding equipment shall be used to apply the seed evenly over entire area by sowing equal quantity in 2 directions at right angles to each other.

B. Seed shall be applied at the rate of 20 pounds per acre. The seed mix, per 50 pound bag, shall be applied according to the percentages as follows: (CONTRACTOR SHALL CONFIRM SEED MIX WITH C.O. PRIOR TO SEEDING).

TYPE	% OF MIX
Annual Rye (Lolium multiflorum)	35
Slender Wheatgrass (Elymus trachycaulus)	25
Green Needlegrass (Nassella viridula)	15
American Vetch (Vicia Americana) or Purple Prairie Clover (Dalea purpurea)	5
Warm season combination – any of the following: Blue Grama (Bouteloua gracilis) Switchgrass (Panicum virgatum) Indiangrass (Sorghastrum nutans) Sideoats Grama (Bouteloua curtipendula)	20

3.7 STRAW MULCH

A. Straw will be hand crimped.
1. Rate: 2 tons per acre uniformly spread
2. Anchor with threader (i.e., tracked vehicle).
3. Operate crosswise to slope.

4. Depth: 3 to 4 inches.
5. Interval: 6 to 12 inches across slope.

3.8 CARE DURING CONSTRUCTION

A. The Contractor shall be responsible for protecting and caring for seeded areas until final acceptance of the work. The Contractor shall repair all damage to seeded areas caused by his construction operations without additional compensation.

END OF SECTION

SECTION 331116
WATER DISTRIBUTION

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes piping and appurtenances for potable-water service outside the buildings. This Section also includes flushing, testing, and disinfecting waterlines.

1.2 SYSTEM PERFORMANCE REQUIREMENTS

A. Minimum Working Pressures: The following are minimum pressure requirements for piping and specialties, unless otherwise indicated:

1. Potable-Water Service: 25 psig.

1.3 SUBMITTALS

A. Product data which shows compliance with the reference standards for the following:

1. Pipe and fittings
2. Compression couplings for similar and dissimilar pipes
3. Suction Strainer
4. Insect Screen
5. Gradation of Rip-Rap
6. Curb Valve and Box
7. Operating Tools For Curb Valves
8. Air/Vacuum Valve and Valve Pit
9. Method for disposing water used in testing and disinfection of waterlines.

1.4 QUALITY ASSURANCE

A. Regulatory Requirements:

1. Comply with standards of authorities having jurisdiction for potable water-service piping, including materials, installation, testing, and disinfections.

B. Piping materials shall bear label, stamp, or other markings of specified testing agency.

C. NSF Compliance:

1. Comply with NSF 61 for materials for water-service piping and specialties for domestic water.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver piping with factory-applied end-caps. Maintain end-caps through shipping, storage, and handling to prevent pipe-end damage and to prevent entrance of dirt, debris, and moisture.

B. Protect flanges, fittings, and specialties from moisture and dirt.

C. Store plastic piping protected from direct sunlight. Support to prevent sagging and bending.

1.6 SEQUENCING AND SCHEDULING

A. Coordinate piping materials, sizes, entry locations, and pressure requirements with building water distribution piping.

B. Coordinate with other utility work.

PART 2 – PRODUCTS

2.1 GENERAL

A. All waterlines, pipes, and fittings shall be new and unused, of the type, pressure rating or class, and size specified and as shown on the drawings.

2.2 HIGH DENSITY POLYETHYLENE (HDPE) PIPE

A. Pipe

1. PE plastic pipe shall be PE-3408 as defined in ASTM D1248, with a minimum cell classification of 345444C in accordance with ASTM D3350. The pipe shall be SDR11, 160 psig minimum pressure rating and comply dimensionally with ASTM D3035.
2. Pipe shall be manufactured in accordance with AWWA C901 for sizes ½” through 3”.
3. PE pipe larger that 2” diameter shall be provided in 40-foot lengths rather than rolls.
4. The joining method shall be butt fusion method and shall be performed in strict accordance with the pipe manufacturer’s recommendations. Butt fusion equipment used in the joining procedures shall be capable of meeting all conditions recommended by the pipe manufacturer, including but not limited to, temperature requirements, alignment, and fusion pressures. Ribbed steel inserts with stainless steel clamps are not approved for PE pipe.
5. The HDPE pipe shall be Geo-Flo pipe as manufactured by PW Pipe, P.O. Box 10049, Eugene, OR, (541) 343-0200 or Cen Fuse pipe as manufactured by Centennial Plastics LLC, Hastings, NE (866) 851-2227 or equal and supplied by Grand Junction Pipe and Supply, Grand Junction, CO (970) 243-4604.

B. Fittings

1. PE fittings shall be made from material meeting the same requirements as the pipe.
2. Molded fittings shall be manufactured in accordance with either ASTM D2683 for socket fused or ASTM D3261 for butt fused and shall be so marked.
3. All PE fittings shall have the same or higher pressure rating and compositions,
4. HDPE elbows, tees and reducers shall be as manufactured by Central Plastics Co., Shawnee, OK, (800) 654-3872 or equal, and as supplied by Grand Junction Pipe and Supply, Grand Junction, CO (970) 243-4604.

2.3 GALVANIZED STEEL PIPE (GSP)

A. Pipe

1. ASTM A53 for pipe 4-inch diameter and smaller. Schedule 40.
2. Plain ends for joining with standard manufactured couplings.
3. Threaded ends shall meet requirements of ASME B1.20.1.

B. Fittings

1. ASTM A53. Schedule 40.
2. For threaded ends use Teflon thread sealing tape.

2.4 COUPLINGS FOR SIMILAR OR DISSIMILAR PIPE

A. Compression type transition fittings for making connections between like and dissimilar pipe materials shall be designed to enable a variety of connections between polyethylene pipe to various pipes, sizes 2” and smaller, using the same central body and different inserts and collets. Fittings shall be Cepex performance series compression fittings as manufactured by Cepex Holding, S.A. and distributed by The Lateral Connection Corp., Sparks, NV 89434 (775) 626-6777 or equal, and supplied by Crow Company, Denver, CO, (303) 571-4444.

2.5 PIPE CORROSION PROTECTION FOR GSP

A. Cold applied mastic type 1200, as manufactured by Protecto Wrap Co., Denver, CO (303) 777-3001 or equal.

2.6 FLAT STEEL SUCTION STRAINER

A. Zinc plated, 16-gauge, cold-rolled steel strainer with threaded end to attach to GSP pipe, part SSS6, 1 ½”, as manufactured by Campbell Manufacturing, Inc., Bechtelsville, PA, (800) 523-0224 or equal.

2.7 INSECT SCREEN

A. 24-mesh woven wire cloth, type 304 stainless steel standard grade, non-corrodible.

2.8 RIPRAP AT OUTLET DRAIN

A. Stone for riprap shall be hard, dense, durable, angular in shape and resistant to weathering. Rounded stone or boulders will not be accepted as riprap material. Individual stones shall have dimensions of 6” to 12”. Source for the stone shall be from rocks uncovered on site during construction operations or from an approved off-site source.

2.9 CURB VALVE

A. One-piece, closed-bottom body with tee head and plug integrally cast, TFE coated brass ball that is supported by two Buna-N-seats, and 300 psig maximum working pressure.

B. Minneapolis pattern.

C. Female iron pipe inlet by female iron pipe outlet.

D. Hole for attaching curb box rod or handle is provided in tee head.

E. Heavy cast bronze body construction conforming to AWWA C800.

F. Curb Valves shall be 300 psig Ball Curb Stops, sized for pipe diameter and material, both ends threaded with Minneapolis pattern top, model number 6105, as manufactured by A.Y. McDonald, Mfg. Co., P.O. Box 508, Dubuque, Iowa, (800) 292-2737, or equal, and supplied by Hughes Supply Inc., Denver, CO (303) 394-0004.

2.10 CURB BOX

A. Extension type curb box with threaded Minneapolis pattern base, cast iron base, steel pipe telescoping upper section.

B. Curb boxes shall be manufactured by the same company as the curb stop and sized to fit curb stop. Box length, when installed, shall permit adjustment between 3 inches above and below final ground surface.

C. Lids shall be two-piece, cast iron with brass pentagon plug and have lettering “WATER” permanently imprinted on them.

D. Curb box, complete with lid and stationary rod, shall be Minneapolis style, series number 5610 for 1” valves, series number 5611 for 2” valves, length as needed, as manufactured by A.Y. McDonald Mfg., P.O. Box 508, Dubuque, Iowa, (800) 292-2737, or equal, and supplied by Hughes Supply, Inc., Denver, CO (303) 394-0004.

E. Six curb box keys shall be supplied to the COR. Keys shall be as supplied by the curb valve and box manufacturer, and shall be model number 304B, 3 foot length, as manufactured by A.Y. McDonald Mfg., P.O. Box 508, Dubuque, Iowa, (800) 292-2737, or equal, and supplied by Hughes Supply Inc., Denver, CO (303) 394-0004.

2.11 AIR/VACUUM VALVE AND VALVE PIT

A. Air/Vacuum Valve

1. Valve shall operate by sealing the BUNA-N rubber outlet seat with a peripheral float as the liquid enters the valve chamber to raise the float. The valve shall satisfactorily withstand hydrostatic pressures of 300 psig. Cast iron body, top flange with brass trim, ½” NPT screwed inlet and outlet. Crispen A5 valve, beige color, standard pressure 20 to 150 psi, as manufactured by Multiplex Manufacturing Co., Berwick, PA (800) 247-8258 or equal as supplied by Flo Controls, Englewood, CO (303) 762-8215.

B. Valve Pit

1. Underground valve pit, molded out of modified polyethylene plastic with the following ASTM specs, 2300 psi tensile strength at yield, 765% ultimate elongation and have UV stabilization. Meter pit shall be 24” diameter, 30” depth, with T top and crush resistant ribbing as manufactured by DFW Plastics, Inc., Bedford, TX, (800) 806-3251 and supplied by Grand Junction Pipe and Supply, Gypsum, CO (800) 558-4440, or equal.
2. Aluminum valve pit cover with 24” diameter riser and 12” diameter cast iron lid, with “WATER METER” imprinted on the lid. Lid shall fit valve pit. Model number M-70-24 as manufactured by Castings, Inc., 860 4th Avenue, Grand Junction, CO 81506, (970) 243-2032, or equal.

C. Post

1. 4” x 4” pressure treated as specified in Section 331219.
2. Smooth four sides, as shown on the drawings.

D. Vent

1. Vent shall be ½” diameter galvanized pipe. Attach bronze insect screen with stainless steel clamp to down turned 90-degree bend as shown on the drawings.

2.12 WASHED ROCK

A. One-inch minus washed rock.

2.13 UNDERGROUND DETECTION TAPE AND TRACER WIRE

A. As specified in Section 312317.

2.14 CONCRETE

A. As specified in Section 033000.

2.15 SODIUM HYPOCHLORITE CHLORINE SOLUTION

A. As specified in Section 332600.

2.16 PVC SLEEVE

A. Sch 40 4-inch PVC sleeve to protect all waterlines installed under asphalt surfacing.

PART 3 - EXECUTION

3.1 PIPE INSTALLATION, GENERAL

A. Construct the water system to the lines and grades shown or established in the field.

B. Stake locations of waterlines as shown on the drawings. Waterline location shall be approved by COR before installation of waterline.

C. All water line installation shall be graded to drain.

D. All pipe, fittings, and appurtenances shall be handled and laid in strict conformance to the manufacturer's recommendations.

E. Daylight drains, valves, and other regulating and controlling devices shall be installed in the line where called for on the drawings.

F. The interior of all pipe, fittings, and other accessories shall be kept as free as possible from dirt and foreign matter at all times. Each piece of line as it is laid shall be cleaned of all debris. When the pipeline has become dirty on the inside during shipment or storage, it shall be swabbed out by drawing a damp swab through the line before lying. Care shall be exercised to keep all joining surfaces clean. Any pipe with contaminated or damaged joining surfaces, which cannot be satisfactorily cleaned or repaired, shall be discarded. Under no circumstances shall pipe be laid in water, and no pipe shall be laid when trench conditions or the weather is unsuitable for such work. At all times when work is not in progress, all open ends of the pipe and fittings shall be securely closed to the satisfaction of the COR so that no trench water, rodents, earth, or other substance will enter the pipe or fittings.

G. Any section of pipe already laid and found to be defective shall be taken up and replaced with new pipe at the Contractor's expense.

H. Where water line crosses a road culvert, adjust grade of water line to provide a minimum of 12 inches of separation between the water line and culvert, maintaining slope of water line to daylight drain.

I. A minimum of one foot separation shall be maintained between electric and water lines in the same trench. Unless the electric conductors are in conduit the water line shall be a minimum of one foot below the electric conductors.

J. Bury piping with at least 24 inches of cover over top of pipe.

3.2 EXCAVATION, TRENCHING AND BACKFILLING

A. Trenching and backfilling shall meet the requirements of Section 312317.

3.3 CUTTING AND HANDLING OF PIPE

A. Cutting of pipe for closure pieces or for other reasons shall be done in a neat manner by methods, which will not damage the pipe. The pipe and fittings shall be handled in such a manner as to insure delivery and final placement in good, undamaged condition. Particular care shall be exercised not to injure the pipe surfaces or coatings. Damaged pipe shall not be used in the work.

3.4 LAYING PIPE

A. No pipe shall be placed in the trench until the COR has approved the trench and bedding.

B. Each section of the pipe shall rest upon the pipe bed for the full length of its barrel.

C. Backfill the pipe within 24 hours after installing the pipe in the trench, leaving the joints and one foot of pipe on each side of each joint exposed until testing is complete.

D. When work is not in progress, securely close open ends of pipe and fittings.

E. When high density polyethylene pipe is used, it shall be weaved in the trench to provide horizontal slack in the line for expansion and contraction.

F. High density polyethylene pipe shall be allowed to cool to within 20°F of the shaded trench bottom temperature before backfilling.

3.5 PIPE ALIGNMENT AND GRADE

A. Complete water system shall be constructed to drain by gravity to daylight drains for winter campground shutdown. Waterlines shall be constructed to drain at a minimum grade of 0.5%. Waterlines not meeting this requirement shall be removed by the Contactor and replaced at no cost to the Government.

B. Pipelines may be relocated slightly, if necessary, to avoid existing obstacles. However, all lines shall be installed to slope towards daylight drains as shown on the drawings.

C. Deflections from a straight line or grade, as required by vertical curves, horizontal curves, or offsets, shall not exceed the manufacturer's recommended maximum joint deflection or curvature for their pipe. If the alignment requires curvature or deflections in excess of these limitations, the Contractor shall provide special bends, standard fittings, or a sufficient number of shorter lengths of pipe to provide angular deflections at the joints within the limit set forth by the manufacturer.

D. Bends resulting in tension on joints will not be allowed.

3.6 HDPE PIPE INSTALLATION

A. Install HDPE pipe according to ASTM D 2774 and ASTM F 645.

3.7 JOINT CONSTRUCTION

A. Joints shall be made using jointing materials and applied with the proper accessories, in accordance with the manufacturer's instructions. The joints shall be adjusted and the work performed in a manner so as to obtain the degree of water tightness required. If pipe and fittings are assembled with lubricant, it must be nontoxic. Where connections are made between new work and existing lines, the connections shall be made by using special fittings to suit the actual conditions.

B. Threaded Joints:

1. Thread pipes with tapered pipe threads according to ASME B1.20.1, apply tape and apply wrench to fitting and valve ends into which pipes are being threaded.
2. Exposed threaded pipe joints on galvanized steel pipe shall be painted with mastic following the manufacturer's application procedures.
3. Field cut ends shall be squared off by grinding or filing; ends shall be reamed with all chips and flakes removed.
4. Field cut threads or machined joints shall be made with sharp tools to sharp finished ends.

C. PE Heat-Fusion Joints

1. According to ASTM D 2657 and piping manufacturer's written instructions.
2. The joining method shall be butt fusion method and shall be performed in accordance with the pipe manufacturer's recommendations. Butt fusion equipment used in the joining procedures shall be capable of meeting all conditions recommended by the pipe manufacturer, including but not limited to, temperature requirements, alignment, and fusion pressures.
3. Personnel trained in butt fusion techniques by the manufacturer of the pipe, shall supervise or perform butt fusion operations.
4. Butt fusion shall be performed between pipe ends or pipe ends and fitting outlets, of like outside diameter and wall thickness, SDR or DR. Butt fusion joining between same diameters but unlike wall thickness shall not be permitted. Transitions between unlike wall thickness shall be made with a transition nipple (a short length of the heavier wall pipe with one end machined to the lighter wall) or by mechanical means.
5. Branch connections to the main shall be made with polyethylene fittings, and butt fused.
6. On every day butt fusions are to be made, the first fusion of the day shall be a trial fusion. The trial fusion shall be allowed to cool completely, and then fusion test straps shall be cut out. The test strap shall be 300 mm (12") (min) or 30 times the wall thickness in length with the fusion in the center, and 25 mm (1") (min) or 1.5 times the wall thickness in width. Bend the test strap until the ends

of the strap touch. If the fusion fails at the joint, a new trial fusion shall be made, cooled completely and tested. Butt fusion of pipe to be installed shall not commence until a trial fusion has passed the bent strap test.

D. Dissimilar Materials Piping Joints

1. Polyethylene pipe and fittings may be joined to other materials by means of couplings designed for joining polyethylene pipe to another material. Couplings shall be fully pressure rated and fully thrust restrained such that when installed in accordance with manufacturer's recommendations, a longitudinal load applied to the coupling will cause the pipe to yield before the coupling disjoints.
2. Use adapters compatible with both piping materials, OD, and system working pressure.

3.8 BRANCH CONNECTIONS

A. Branch connections to the main shall be made with tees.

B. Horizontal tees for waterline laterals shall not be permitted. The branch side of tees shall be up when drainage is toward the tee, and down when drainage is away from the tee.

C. Waterline tees shall be the same diameter as the largest pipe connected to the tee.

3.9 INSTALLATION OF CURB VALVE AND BOX

A. Each curb valve shall be housed in a curb box as specified.

B. Comply with manufacturers written instructions. Install underground curb valves with head pointed up.

C. Install concrete pad around top of box as shown on the drawings.

3.10 INSTALLATION OF AIR/VACUUM VALVES

A. Install air relief valves at high points in the water lines as shown on the drawings to release air in the line as the line is filling and to allow air to enter the system when draining.

B. Air/Vacuum valves shall be housed in valve pits as specified. Comply with manufacturers written instructions.

C. Place washed rock under the valve pit as shown on the drawings. Place the housings over the valves and rest on compacted gravel with the entire assembly being plumb.

D. Notch out the valve pit for water distribution lines and for the vent pipe at all Air/Vacuum valve locations. Install galvanized steel vent pipe to atmosphere as shown on the drawings.

E. Attach the insect screen to the down turned 90 degree bend and clamp pipe to the 4”x 4” pressure treated post.

3.11 DAYLIGHT DRAIN

A. Transition from water system piping to galvanized steel pipe 10-feet from atmospheric outlet of daylight drain.

B. Install suction strainer and insect screens on the atmospheric outlet of daylight drains. Attach insect screens to pipe with stainless steel hose clamps.

C. Minimum grade of slope of drain shall be 1%.

D. Place riprap securely around outlet with a minimum 6” air gap.

3.12 SERVICE ENTRANCE PIPING

A. Extend water-service piping and connect to water-supply source and building water piping systems at outside face of building wall in locations and pipe sizes indicated.

1. Terminate water-service piping at building wall until building water piping systems are installed. Terminate piping with caps, plugs, or flanges as required for piping material. Make connections to building water piping systems when those systems are installed.

3.13 BACKFILLING AND WATERLINE MARKING

A. Backfilling and Waterline Marking shall meet the requirements of Section 312317.

3.14 FINAL PIPE CLEANING AND FLUSHING

A. Before testing, clean new water distribution piping systems and parts of existing systems that have been altered, extended, or repaired by high pressure water jet or mechanical means.

B. If the new waterlines are extensions or modifications of an existing water system, the new system(s) and all of the existing system immediately downstream of the control valve shall be flushed to remove sediment. The control valve referred to here is the valve in the existing system that must be closed to allow construction of new facilities.

C. During the flushing operation, operate all valves, hydrants, hose bibs and other outlets.

D. Flush system with clean potable water until dirty water does not appear at points of outlet.

3.15 PRESSURE TESTING WATERLINES

A. The Contractor shall conduct pressure and leakage tests on all newly installed or laid pipeline, or any valved section thereof. Pneumatic testing will not be allowed.

B. Conduct piping tests before joints are covered and after thrust blocks, if required, have hardened sufficiently. Joints shall remain uncovered until testing is completed satisfactorily.

C. The Contractor shall furnish the pump, pipe, gauge, measuring device, connections, and all other necessary apparatus, and shall furnish the necessary personnel to conduct the tests. All equipment, gauges, and attachments shall be subject to approval by the COR.

D. The COR shall be present during the testing period.

E. Test all newly constructed portions of the water system, including water mains, service lines, fittings, valves, and hydrants. New waterlines that are extensions or modifications of an existing water system shall be tested at the test pressure specified here unless the COR specifies another pressure.

F. Conduct piping tests before joints are covered. Joints shall remain uncovered until testing is completed satisfactorily.

G. Use only potable water.

H. Waterlines shall be tested in sections not to exceed 2,000 ft in length or a maximum pressure differential of 30 psi in the test section.

I. Test Procedure

1. Slowly fill each valved or plugged test section of pipe with water. The line shall be filled by means of a pump connected to the pipe. Operate all hydrants and taps to bleed entrapped air from the test section. Operate the pump continuously during bleeding operations. The pump shall take suction from a small auxiliary tank for measuring the water the water pumped into the test section.
2. Raise the internal pressure of the system by pumping in water. Increase pressure in 50-psig increments and inspect each joint between increments. The test pressure between the well and the storage tank shall be 100 psi. The test pressure for the campground distribution system shall be 100 psi.
3. The system should be allowed to stabilize for 3 hours at the test pressure before conducting the hydrostatic test. Make up liquid may be added during this time to return to test pressure.
4. After obtaining the approved test pressure record the water level in the auxiliary tank.
5. The test phase follows immediately and shall last 2 hours.
6. Immediately following the initial expansion phase, reduce the test pressure by 10 psi, and stop adding test liquid. If the test pressure remains steady (within 5% of the target value) for one (1) hour, no leakage is indicated.

J. Leakage shall be verified by the Contractor and the COR.

K. If there is leakage in excess of the above requirements, the Contractor shall, at his own expense, locate and replace any defective pipe, fittings, valves, joints, and hydrants as required, until the leakage is within the specified allowance. All visible leaks are to be repaired regardless of the amount of leakage. Flush the system, and repeat the pressure test until leak is repaired.

L. No paints, asphalts, tars, enamels, or other types of pipe compounds shall be used to eliminate leaks.

M. Backfill for waterlines shall be completed after the COR approves testing.

N. The Contractor shall furnish a written report to the COR describing the results of each test. The report must identify the specific portions of the pipeline tested, the pressure, the duration of the tests, and the amount of leakage.

3.16 DISINFECTION OF WATERLINES

A. Disinfection by chlorination shall be performed after flushing and pressure testing are completed and approved by the COR.

B. Dosage: Provide chlorine solution with approximately 25 mg/l of chlorine for disinfecting the water system unless otherwise directed by the COR.

C. Procedure: The entire system, including the storage tank, shall be filled with the disinfecting solution. Chlorine solution shall be placed in the system, at a location approved by the COR. All taps, hydrants, and other outlets shall be opened and left open until chlorine is noticeable by odor or testing in the water coming from each. Immediately close taps, hydrants, and other outlets. An effort shall be made to limit the amount of chlorinated water discharged from the taps, hydrants and other outlets. Heavily chlorinated water shall not be drained into streams, rivers, or lakes; rather, open first those drains and hydrants which permit the longest travel time for the solution to reach surface waters.

D. Test Period:

1. Chlorinated water shall remain in the system a minimum of 24 hours, after which the system shall be examined by the COR for residual chlorine. Chlorine residual at the end of the 24-hour period shall be a minimum of 10 mg/1.
2. If the minimum residual is not found present, the system shall be drained and the disinfection testing shall be repeated. When disinfection is complete and approved by the COR, the system shall be flushed with potable water and the water leaving the lines shall have a residual chlorine concentration no higher than 1 mg/l.
3. When emptying the lines or tank after the 24-hour period, whether the system has passed or failed, the chlorine solution shall be neutralized before being

discharged. Neutralize system by adding sodium thiosulfate to the storage tank in accordance with the table below. Let the sodium thiosulfate sit in the storage tank for 24-hours and then release the water from the tank and flush the system. There shall be a trace of residual chlorine at the outlets.

Amounts of Chemical Required to
Neutralize Various Residual Chlorine
Concentrations in 1,000 gallons of Water

Residual Chlorine Concentration mg/L	Pounds of Sodium Thiosulfate ($Na_2 S_2 O_3 - 5H_2O$)
1	0.012
2	0.024
10	0.12
50	0.60

E. Bacteriological Examination: After the system has been disinfected as specified, the COR shall test the water for bacteriological contamination from representative points in the system. The coliform organisms shall be tested for presence/absence in accordance with State drinking water regulations. If coliform bacteria are present, testing shall be repeated in accordance with State drinking water regulations. If the report of this examination is unsatisfactory, the system shall be flushed and the disinfection procedure repeated until the results of bacteriological examinations are satisfactory.

F. The satisfactory quality of water delivered by the waterline should continue for a period of at least two full days, as demonstrated by laboratory examination of samples taken from taps along the system.

G. Bacterial testing of the water system will be the responsibility of the Forest Service.

H. Repeat disinfection of the water system shall be the responsibility of the Contractor with no additional expense to the Government.

END OF SECTION

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SECTION 331219
WATER HYDRANTS

PART 1 – GENERAL

1.1 SUMMARY

A. Provides for furnishing and installing water hydrants in the campground and at the host site. Work includes installation of a support post, concrete pad, rip-rap and hydrant as shown on the drawings. Work also includes grading and shaping the approach apron to the hydrant pad.

1.2 LOCATION

A. General location and orientation of all hydrants are shown on the drawings.

B. Exact location and orientation shall be approved by the Contracting Officer.

1.3 SUBMITTALS

A. Manufacturer's literature for faucets, hose bibb, and vacuum breaker.

B. Certification that riprap meets the gradation requirements of this section.

PART 2 - PRODUCTS

2.1 POST

A. Treated timbers for walls, steps, border, and edger shall be rough sawn #2 grade or better Ponderosa Pine or equal, landscape timbers shall be brown and not green in color, uniform in size, straight, and free from sweep and crook. The timbers shall be treated with ACQ (Alkaline Copper Quaternary) or CBA (Copper Boron Azol), or approved equal, to a net retention of 0.25 lbs. per cu. ft. Sizes as shown on drawings.

B. The top of the post shall be sloped as shown on the drawings. All field cuts or abrasions and holes made in fabricated timbers after treatment shall be given 3 brush coats 2% copper Napthanate.

C. Handling Precautions: Wear protection such as dust masks, goggles, and gloves when sawing treated lumber.

2.2 PIPE AND FITTINGS

A. Galvanized steel pipe, Schedule 40 as specified in Section 331116.

2.3 WATER FAUCET

A. Self-closing bibb faucet with plain end. Plain end means no threads, so a garden hose cannot be attached to the end.

B. Extended "paddle" type handle for accessibility. Operation of handle shall require less than 3.5 lbs pressure for operation.

C. Equipped with integral 2.5 gpm flow control.

D. Finish shall be non-chromed brass. Brass shall be lead free and suitable for dispensing potable water.

E. Model No. 6252-EHLF with a 4-inch elbow lever handle, as manufactured by Haws Corporation, Sparks, NV, (775) 359-4712, or approved equal.

2.4 HOSE BIBB

A. Wall faucet for non-freezing area, model 21CP, ½” MPT outside threads, brass finish, with tee key operating handle, as manufactured by Woodford Manufacturing Company, Colorado Springs, Colorado, (719) 574-1101 or approved equal.

B. Provide 6 tee keys to Contracting Officer.

C. Provide vacuum breakers for each hose bibb.

2.5 VACUUM BREAKER

A. Brass finish, removable, ¾” hose thread female inlet x ¾” hose thread male outlet hose connection type anti-siphon vacuum breaker, with manual drain for freezing conditions, maximum pressure 125 psi, model no. NF8, as manufactured by Watts Regulator, Plumbing and Heating Division, No. Andover, Mass. (978) 688-1811, or approved equal.

2.6 CONCRETE HYDRANT PAD

A. Concrete as specified in Section 033000.

B. Aggregate base as specified in Section 321123.

2.7 CURB VALVE

A. As specified in Section 331000.

2.8 SUPPLY PIPE

A. High Density Polyethylene (HDPE) pipe as specified in Section 331116.

2.9 RIPRAP

A. Stone for riprap shall be hard, dense, durable, round in shape and resistant to weathering. Individual stones shall have dimensions of 4”-6”. Source for stone shall be rocks uncovered on site during construction operations or from an approved off-site source.

PART 3 - EXECUTION

3.1 WATER HYDRANTS

A. Install hydrants as shown on the drawings.

B. Wood posts shall be cut and drilled to the lengths and chamfered, as shown on the drawings. All posts shall be pressure treated with wood preservative. All field cuts, abrasions, and holes made in the timbers after treatment shall be given three (3) brush coats of field cut treatment specified above.

C. The post and riser pipe shall be placed in the trench, plumb and to the elevations shown on the drawings, backfilled and compacted with subsequent backfill as specified in Section 312250.

D. Elevation of edge of concrete pad or riprap shall be the same as adjacent finished surface of edge of road, spur, or trail. Pads not meeting this requirement shall be removed by the Contractor and replaced at no cost to the Government. Backfill and fine grade around the outside of the pad. Slope pad to drain toward rip rap as shown on the drawings.

E. Place riprap at each hydrant location as shown on the drawings. Rock shall be secure and flush with the top of the concrete slab.

END OF SECTION

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SECTION 332000
UNDERGROUND STORAGE TANK

PART 1 - GENERAL

1.1 SUMMARY

A. This item shall consist of furnishing materials and appurtenances and performing all work necessary or required for the installation of a horizontal underground fiberglass potable water storage tank.

B. The installation shall include the ladder, manway, cover, nozzles, all the pipe, fittings, valves, including floats, and connections necessary to connect the storage tank to the water supply and distribution system. The installation shall also include the valves and boxes, the screened vent pipe, and the entire lengths and screened outlets of the overflow, perimeter drain and tank drain lines.

C. Includes furnishing and installing the float switch and control cable from the treatment building to the tank, complete and operational as shown on the drawings.

1.2 SUBMITTALS

A. Tank and Manway

1. The Contractor shall submit to the Contracting Officer Representative (COR), for approval, two copies each of shop drawings of the storage tank and manufacturer’s literature. The drawings shall specifically show:
    a. The manway opening location and dimensions;
    b. Manway cover dimensions, method of connection to the tank, hinge and locking method.
    c. Bracket locations, dimensions and method of connection to the tank.
    d. Nozzle fabrication, locations, sizes and methods of connection;
    e. Ladder location, dimensions and method of connection to tank.
2. Fabrication of the tank shall begin after COR approval of the shop drawings.
3. The National Sanitation Foundation (NSF) shall list the tank as suitable for potable water use. The Contractor shall submit proof of this listing.
4. Certification from the tank manufacturer that the tank was tested by the manufacturer at the factory and met the design criteria specified.

B. Perforated perimeter drain piping, literature and drawing showing perforation pattern.

C. Contractor shall submit to the COR certification that the backfill material meets the gradation specified in this section.

PART 2 - PRODUCTS

2.1 UNDERGROUND WATER STORAGE TANK

A. Tank

1. Tank shall be 3,000 gallon, fiberglass reinforced plastic (FRP), single-wall potable water storage tank with dimensions as shown on the drawings. The interior of the tank shall be smooth, without ribs.
2. Tank shall be manufactured of 100% resin and glass-fiber reinforcement, with no sand fillers. The laminate materials used in the internal coating system of the tank shall conform to the requirements of NFS Standard 61.
3. Tank shall be vented to atmospheric pressure.
4. Internal Load: Tank shall withstand a 5-psig air-pressure test with 5:1 safety factor.
5. Vacuum Test: Tank shall be vacuum tested by the manufacturer at the factory to 11.5 inches of mercury.
6. Surface Loads: Tank shall withstand surface H-20 axle loads when properly installed as recommended by the manufacturer.
7. External Hydrostatic Pressure and Burial Depth: Tank shall be capable of being buried in ground with 7 feet of overburden, the hole fully flooded and a safety factor of 5:1 against general buckling.
8. Tank shall support accessory equipment as shown on the drawings and when installed as recommended by the manufacturer.
9. Tanks shall be manufactured by Xerxes Corporation, Minneapolis, MN, (952) 887-1890 and distributed by Sunbelt Industries, Englewood, CO, (303) 662-8344 or equal.

B. Fittings

1. The tank shall be furnished with all nozzles, flanges and other appurtenances as shown on the drawings. Flange for manway shall have a bolt pattern as shown on the drawings. All nozzles and appurtenances shall be installed by the tank manufacturer.
2. All threaded fittings shall be constructed of carbon steel or FRP.
3. All threaded fittings shall be half-couplings.
4. All FRP nozzles shall be flat-faced and flanged, and shall conform to ANSI B16.5 150 lb. bolting pattern.
5. All flanged connections shall be gasketed.

C. Lifting Lugs

1. Lifting lugs shall be provided as recommended by the manufacturer. They shall be an integral part of the tank.

2.2 LADDER

A. Ladder shall be manufactured with materials conforming to the requirements of NSF 61.

B. Fiberglass ladder shall be provided by tank manufacturer with connections as shown on the drawings.

C. Ladder shall be installed by the manufacturer prior to shipping.

2.3 MANWAY AND COVER

A. Manway shall be FRP, flanged and 30-inch interior diameter, complete with gaskets, bolts, hinged, lockable, with fiberglass handle to open cover. The cover shall have a minimum 2" lip.

B. Manway shall have an exterior tan gel finish coat. Finish coat shall contain between 0.25 to 0.5% UV stabilizers.

C. Bolt pattern in flange to match bolt pattern in fiberglass tank.

D. Manway shall be manufactured by Xerxes Corporation, Minneapolis, MN, (952) 887-1890 and distributed by Sunbelt Industries, Englewood, CO, (303) 662-8344 or equal.

2.4 CONFINED SPACE SIGN

A. A sign reading "DANGER—CONFINED SPACE, ENTER BY PERMIT ONLY" or other similar language as approved by the COR must be permanently mounted in a clearly visible location on the manway cover.

2.5 VENT

A. Mushroom Style tank vent with a 30-mesh screen, 2-inch diameter female iron pipe thread.

B. Tank vent shall be part number 36635K44 as supplied by McMaster-Carr Supply Company, Chicago, IL, (630) 833-0300 or equal.

2.6 TANK BEDDING

A. Tank bedding shall be 3/8" pea gravel. Bedding shall be washed and free flowing.

B. Pea gravel shall meet the gradation requirements of CDOT Coarse Aggregate No. 8, from AASHTO M43 or equal.

Sieve Size	Percentages by Weight Passing Square Mesh Sieves
½"	100
⅜"	85-100
#4	10-30
#8	0-10
#16	0-5

2.7 FILTER FABRIC

A. Filter fabric shall be a non-woven geotextile, Mirafi 140N as manufactured by TC Mirafi, Pendergrass, GA, (888) 795-0808 or equal.

2.8 PIPE AND FITTINGS

A. As shown on the drawings

B. Polyethylene as specified in Section 331116.

C. Galvanized Steel Pipe as specified in Section 331116.

D. PVC as specified in Section 220500.

2.9 CURB STOPS AND BOXES

A. As specified in Section 331116.

2.10 SCREENS

A. Insect screens on the tank drain, vent pipe, and perimeter drain shall be as shown on the drawings and specified in Section 331116.

2.11 PERIMETER DRAIN

A. PE perforated pipe shall be PE-3408 as defined in ASTM D1248. The pipe shall be SDR 11, 160 psig minimum pressure rating and comply dimensionally with ASTM D3035. Perforation shall meet the requirements of ASTM F810 with ½" standard perforations, 4 holes per ring pattern at 90º around pipe, and 5" between ring patterns or equal.

2.12 EXPANSION JOINT

A. Use on storage tank nozzles as shown on the drawings. Shall be neoprene, double arch, loose-ring, 150 lb. ANSI flanged ends. Ends rotate for easy bolt-hole alignment and installation. Flanges ends shall be galvanized steel and joints shall be nylon-reinforced. Vacuum rating shall be 26" Hg and maximum pressure rating shall be 225 psi. Expansion joints shall be part no. 6819K65 as supplied by McMaster – Carr Supply Company, Chicago, Illinois, (630) 833-0300, or equal.

2.13 FLOAT SWITCHES

A. Float Switches will be Contractor furnished.

B. The manufacturer shall certify that the float switch meets NSF standard 61 protocol.

C. Each float switch sensor shall be mechanically-activated and shall not contain mercury.

D. Sensors shall be sealed float type, tethered limit travel beyond their respective levels.

E. Units shall be configured with a normally open contact.

F. Each sensor shall be a wide-angle sensor.

G. Float cord shall be flexible 16 gauge, water resistant (CPE), and shall be permanently molded to the sensor to ensure a watertight unit. Float cord shall be continuous from the float switch to the junction box with no splice.

H. Reference Electrical detail drawings for float switch manufacturer and model number(s).

2.14 SUPPORT BRACKETS FOR FLOAT SWITCH

A. The height of each float shall be adjustable from the top of manway by raising or lowering the cord where it attaches to the support bracket with nylon tie cord. Brackets will be furnished with manway for water storage tank.

2.15 GOVERNMENT SUPPLIED FLOAT SWITCH TRAY CABLES

A. Cable shall be two #16 conductors, in a direct bury jacket, UV resistant.

B. Cable shall be installed in a Contractor furnished 1” PVC conduit with a minimum of 18-inches of cover.

2.16 JUNCTION BOX

A. NEMA Type 4

B. Box and fittings to be made of plastic, to resist corrosion from tank moisture.

2.17 CORD SEAL

A. Cord seal shall provide strain-relief and a liquid-tight seal.

B. Cord seal shall seal 1-3 cables in 2” terminal adapters and include 2 PVC plugs for unused cable positions.

C. Cord seal shall have a neoprene seal, stainless steel washers, nut and bolt and shall accommodate the wire size of the float cord.

D. Cord seal shall be manufactured by SJE-Rhombus Controls, Detroit Lakes, MN (888) 342-5753 or equal

PART 3 - EXECUTION

3.1 EXCAVATION

A. As specified in Section 312250.

3.2 DEWATERING

A. As specified in Section 312250.

3.3 DRAINAGE

A. Underdrains shall be provided under the storage tank as shown on the drawings. Drainage shall be provided by the use of tank bedding under the storage tank, with 3” perimeter drainage tubing around the tank extending to daylight.

3.4 PLACING TANK BEDDING AND SETTING TANK

A. The foundation bed for the water storage tank shall be sloped 1% so the tank drains, as shown on the drawings, and shaped to cradle the tank for about 1/3 of its depth. The cradle shall be checked by the Contractor with a template sweep, which matches the outside radius of the tank. The Contractor must obtain the approval of the COR prior to placing the storage tank. The tank shall be lifted, place, or otherwise handled using a spreader bar to insure vertical support cables at each lift ring.

3.5 MANWAY

A. Cover of manway shall extend 2 to 3 feet above finished grade.

3.6 INSPECTION AND TESTING

A. Leak Test

1. The tank shall be subjected to a leak test, not the manway. The Contractor shall furnish all facilities and personnel for conducting the test and shall conduct the test.
2. All nozzles shall be sealed to prevent water from escaping during the test. Water shall be introduced into the tank until the level reaches the top of the tank not the manway. At least 24 hours shall be allowed for the tank to stabilize. The tank will be termed "Acceptable" if the level remains the same after 8 hours.
3. If the tank fails to meet the acceptable test requirements, the Contractor shall determine the source of leakage and make the necessary repairs or adjustments, as approved by the COR, until the acceptable criteria is met.

3.7 PRODUCT STORAGE REQUIREMENTS

A. Tank shall be vented to atmospheric pressure, as the tank is not designed as a pressure vessel.

3.8 BACKFILLING

A. After the tank is in place and the necessary connections completed, the area around the tank shall be carefully backfilled. Tank bedding shall be manually shoveled and probed into place under the tank belly, below the end caps, and around all bottom piping or appurtenances. Do not strike tank with shovels and probing tools. No voids shall be permitted under the end caps or load bearing bottom of the tank. Continue to hand probe the tank between the 4 o'clock and 8 o'clock position. Once the tank is properly supported up to the 4 o'clock and 8 o'clock level, backfill of tank bedding can be by machine. Care should be exercised during backfilling so as not to damage the tank.

B. Place filter fabric between the tank bedding and subsequent backfill interface as shown on the drawings.

C. Continue backfill with subsequent backfill to the grades shown on the drawings. Subsequent backfill shall be compacted by mechanical means to 90% of maximum dry density in accordance with AASHTO T-99.

3.9 TANK DEFLECTION

A. Maximum allowable tank deflection (diameter out of round) shall be as follows:

1. 4' diameter tank – 1/2" allowable deflection
2. 6' diameter tank – 5/8" allowable deflection
3. 8' diameter tank – 1" allowable deflection

3.10 FINISH GRADING

A. Upon completion of the tank installation and after backfilling operations are complete, the area shall be finish graded with the stockpiled topsoil to the specified grade and lines shown on the drawings to present a smooth appearance.

3.11 LIQUID LEVEL SENSORS (FLOAT SWITCHES)

A. Install float switch per manufacturer's recommendation and as shown on the drawings.

B. Connect sensor cables to the junction box.

C. Do not cut sensor cables, but leave as slack to allow for adjustment. Slack shall be between the electrical junction box and the point of suspension.

D. Float switch cords shall be installed in junction box with cord seal as shown on the drawings.

3.12 CONTROL CABLE INSTALLATION

A. Trenching as specified in Section 312317.

B. Placing cable in trench

1. Do not lay cable when the outside temperature is below the minimum temperatures specified by the cable manufacturer.
2. Whenever possible, cable shall be played out from the reel mounted on a moving vehicle or trailer. Support the reel on a stand. Place cable in the trench by hand.
3. Inspect the cable as it is removed from the reel in laying operations to be certain that it is free from visible defects.
4. Where cable must be pulled through conduit, duct, or other locations which prevent placing by hand, the cable may be lubricated with a suitable cable lubricant prior to pulling into conduit or duct.
5. Leave a minimum of 24-inch slack on each cable at all risers, splices, transformer pads, pedestals, and terminal points.
6. Before any backfilling operations are begun, the Contractor and COR shall jointly inspect all trenches, cable placement, and other construction not accessible after backfilling.

C. Cable shall be grounded at pump controller in the water treatment building. Use bond clamp assembly to clamp single jacket to bonding strap. Bond strap to ground rod.

D. Backfill as specified in Section 312250.

E. Conduit

1. All exposed ends of conduit shall be plugged during construction.
2. Remove burrs or sharp projections which might damage the cable.

F. Terminations of control cable shall be made in accordance with the manufacturer's recommendations. Clean all pairs of control cable filling compound at each splice and termination with solvent, not harming insulation.

END OF SECTION

SECTION 332600
DISINFECTANT FEED EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Includes furnishing, installing and adjusting electrically operated metering system and appurtenances. Also furnish pump repair kit and chlorine test kits. Chemical metering pump shall be salvaged pump from existing water system.

1.2 SUBMITTALS

A. Manufacturer's instructions for installation, operation and maintenance of the chemical metering pump and fittings, solution tank, auto prime valve, injection check valve and tubing, and chlorine residual test kit. Submit a detailed parts list and diagram for the chemical metering pump and fittings.

B. Submit name, address, and telephone number of the sales and/or service office closest to the project area for the metering equipment submitted.

C. The Contracting Officer's Representative (COR) shall approve the chemical metering equipment before installation.

PART 2 – PRODUCTS

2.1 CHEMICAL METERING PUMP

A. The Chemical Metering Pump shall be suitable for use with corrosive liquids.

B. The Chemical Metering Pump is an AA141-150SHB manufactured by Milton Roy, Liquid Metronics, Inc., Acton, MA, (978) 263-9800, and supplied by Tritec, Inc., Lakewood, CO, (303) 233-2004 or equal.

2.2 AUTO-PRIME VALVE

A. The auto-prime valve is designed for constant removal of vapors and gasses to discharge to atmosphere. The body of the valve shall be PVDF. The valve shall include auto-priming, degassing, and anti-siphon functions. Valve shall be manufactured by Milton Roy, Liquid Metronics, Inc., Acton, MA, (978) 263-9800, and supplied by Tritec, Inc., Lakewood, CO, (303) 233-2004 or equal.

2.3 CHECK VALVES AND TUBING

A. A total of 16 feet of polyethylene tubing shall be provided complete with compression connections. Tubing shall be suitable for use with potable water.

2.4 FOOT VALVE

A. A foot valve with integral one piece strainer shall be provided for the suction line. Valve shall have end for connection to tubing.

2.5 INJECTION VALVE

A. An injection check/back pressure valve with 1/2" NPT male connection for the injection point. The injection check valve shall incorporate a dilating orifice, which prohibits scale formation and accumulation of crystalline deposits and shall have end for connection to the tubing.

2.6 PUMP REPAIR KIT

A. Contractor shall furnish and give to the CO one pump repair kit.

B. Model RPM-352, as manufactured by Milton Roy, Liquid Metronics Inc., Acton, MA, (978) 263-9800, and supplied by Tritec, Inc., Lakewood, CO, (303) 233-2004 or equal.

2.7 SOLUTION TANK

A. Chlorine solution tank shall be 10 gallon capacity, ultraviolet resistant, polyethylene tank, and slightly translucent.

B. Tank shall have a mounting surface on the top of the tank for the chemical metering pump.

C. Tank top shall be fitted with a fill hole, and include fill hole cap.

D. Tank shall be suitable for corrosive solutions

E. Model No. 27421 as manufactured by Milton Roy, Liquid Metronics, Inc., Acton, MA, (978) 263-9800, and supplied by Tritec, Inc., Lakewood, CO, (303) 233-2004 or equal.

2.8 CHLORINE RESIDUAL TEST KITS

A. One portable of each test kit specified below shall be furnished by the Contractor and given to the CO. Each test kit shall have its own individual polypropylene or styrene plastic case. The case shall contain the necessary chemicals, related testing apparatus, and instructions to make the required tests.

B. Chlorine test kit capable of determining free and total chlorine. Operating range shall be 0-0.7 mg/L for free chlorine and 0-3.5 mg/L for total chlorine.

1. Model CN-70, as manufactured by the Hach Company, P.O. Box 608, Loveland, Colorado 80539, (800) 227-4224, or equal.

C. Chlorine test kit capable of determining total chlorine. Operating range shall be 10-200 mg/L for total chlorine.

1. Model CN-21P, as manufactured by the Hach Company, P.O. Box 608, Loveland, Colorado 80539, (800) 227-4224, or equal.

2.9 SODIUM HYPOCHLORITE CHLORINE SOLUTION

A. Chlorine or other disinfectants, approved by the COR shall be used. The disinfectant shall be delivered to the jobsite in original closed containers bearing the original label and indicating the percentage of available chlorine.

1. 5.25% Purex Chlorox (Liquid)

2.10 EMERGENCY EYE WASH STATION

A. Mount an emergency eye wash station in the treatment building.

B. 32 oz. Eyewash Station for immediate flushing of eyes or body. Eye wash station shall include isotonic saline wash.

C. Eyewash station shall be North model no. 12-60-32 as supplied by Grainger, stock no. 5T062, Denver, CO, 303-733-8777 or equal.

2.11 SAFETY GOGGLES

A. Clear, anti-fog, splash safety goggles with indirect venting and a goggle case shall be provided in the treatment building. Goggle case shall be wall mounted.

B. Goggles shall fit over most eyeglasses and meet the requirements of ANSI Z87.1 impact and chemical splash standards.

C. Goggles shall be AOSafety model 40305 as supplied by Grainger, stock no. 4YZ57, Denver, CO, 303-733-8777 or equal.

D. Goggle case shall be wall mountable, sized for goggles.

E. Goggle case shall be No. 3ZL25 as supplied by Grainger, Denver, CO, 303-733-8777 or equal.

PART 3 - EXECUTION

3.1 CHEMICAL METERING PUMP INSTALLATION

A. The chemical metering pump, piping, injection check valve, and solution tank shall be installed as shown on the drawings and according to manufacturer's instructions.

B. The chemical metering pump shall be connected to the electrical system such that it operates only when the well pump is supplying water.

C. The injection check valve shall be install in the vertical position into the bottom of the pipe as recommended in the manufacturer's instructions.

D. The water distribution and storage facilities shall be disinfected in accordance with Section 331116 before completing the adjustments specified in this Section.

E. The Contractor shall prepare 5 parts water to 1 part 5.25% (Chlorox) chemical solution and set the initial metering pump application rate.

F. Do not mix more than 10 gallons of chemical solution at any one time.

G. The Contractor shall adjust the stroke frequency and length to provide a solution metering rate to maintain a trace of residual chlorine in the distribution line to the most distant hydrant and 0.2 mg/l residual chlorine at the outlet of the storage tank at all flow rates.

H. The COR will test the water for free residual chlorine using the portable test kit provided.

I. The Contractor, in the presence of the COR, shall conduct all remaining adjustments to the disinfection system.

END OF SECTION

PART 1 - GENERAL

1.1 SUMMARY

A. This work shall consist of installing metal pipe and pipe appurtenances, including all excavation, bedding and backfilling required to complete the work.

PART 2 - PRODUCTS

2.1 CORRUGATED IRON OR STEEL PIPE AND PIPE ARCHES

A. Riveted Pipe and Pipe Arches – these pipes shall meet the requirements of AASHTO M 36.

B. Welded Pipe and Pipe Arches – corrugated metal pipe and pipe arches fabricated by resistance spot welding shall meet the applicable requirements of AASHTO M 36.

B. Helical Pipe – unperforated helically corrugated pipe with continuous lock or welded seams shall meet the applicable requirements of AASHTO M 36.

C. Coupling Bands – coupling bands shall meet the requirements of AASHTO M 36.

D. Special Sections – special sections such as elbows, tees, wyes, etc. shall be the same thickness as the conduit to which they are joined and meet the applicable requirements of AASHTO M 36.

E. Flared End Sections – flared end sections for inlet and outlet ends of pipe and pipe arch culverts shall meet the applicable requirements of AASHTO M 36. End sections shall be fabricated in accordance with the details and dimensions as shown on the drawings, except minor variations may be accepted to permit the use of the manufacturer's standard methods of fabrication.

2.2 STRUCTURAL PLATE FOR PIPE, PIPE ARCHES AND ARCHES

A. The pipes, bolts and nuts used for connecting plates shall meet the requirements of AASHTO M 167.

2.3 ALUMINUM ALLOY STRUCTURAL PLATE FOR PIPE, PIPE ARCHES AND ARCHES

A. Pipes, bolts and nuts for connecting plates shall meet the requirements of AASHTO M 219

PART 3 – EXECUTION

3.1 WATER POLLUTION AND STREAM DEGRADATION SHALL BE CONTROLLED.

A. Pipe which is installed in or which will affect streams that are designated as important fisheries shall be installed only during those periods.

3.2 EXCAVATION

A. Excavation for culverts shall be to the lines and grades or elevations as shown on the drawings or as designated on the ground. Excavations shall be of sufficient size to permit the placing and backfilling of culverts. Boulders, logs and any other unsuitable materials encountered shall be removed and disposed of in areas as shown on the drawings. The width of trenches shall permit satisfactory jointing and thorough tamping of the bedding material under and around the culvert.

B. Unsuitable foundation material shall be excavated below the invert of the culvert to an approximate depth of 2 feet and a width of at least the culvert diameter plus 4 feet. Unsuitable material shall be replaced with selected granular foundation material and compacted in accordance with paragraph 3.8.

C. Where rock, hardpan or other unyielding material is encountered, it shall be removed below the foundation grade for a depth of at least 1 foot. The width of the excavation shall be at least 2 feet greater than outside width of the culvert. This excavated material shall be replaced with selected granular foundation material and compacted in accordance with paragraph 3.8.

3.3 UTILIZATION OF EXCAVATED MATERIALS

A. All suitable excavated material shall be utilized as backfill or embankment. No excavated material shall be placed in live streams. All surplus material shall be disposed of as shown on the drawings. No excavated material shall be deposited in a manner that will endanger the partly finished structure.

3.4 BEDDING

A. Bedding shall consist of bedding the pipe to a depth of not less than 10 percent of its total height. The foundation surface, after excavation in accordance with paragraph 3.2 shall be compacted in accordance with paragraph 3.7 and shaped to fit the pipe.

B. The completed bedding shall have a longitudinal camber when shown on the drawings.

C. The bedding material shall be selected mineral soil meeting the requirements for backfill in paragraph 3.7.

3.5 LAYING PIPE

A. No pipe shall be placed in service until a suitable outlet is provided.

B. Paved or partially lined pipe shall be laid so the longitudinal centerline of the paved segment coincides with the flowline.

C. Elliptical pipe shall be placed with the major axis within 5 degrees of a vertical plane through the longitudinal axis of the pipe.

D. The final installed alignment of all pipe shall have no sag and no point shall vary from a straight line drawn from inlet to outlet by more than 2 percent horizontally and vertically of the culvert length or 1 foot, whichever is less, unless otherwise shown on the drawings.

E. Helically corrugated lock-seam pipe shall be installed with the seam at the inlet end placed below the horizontal centerline.

F. Longitudinal laps on riveted or spot-welded pipe shall be positioned at any location between 45 degrees above or below horizontal.

3.6 JOINING PIPES

A. Pipe shall be firmly joined by form-fitting coupling bands. End sections shall be attached to pipe by connecting bands or other means as recommended by the manufacturer. Rubber gaskets shall be installed at each joint to form a watertight connection when shown on the drawings. Dimpled bands shall not be used when the slope of the pipe is greater than 25 percent.

B. The coupling bands shall meet the strength requirements of field joints for Non-Erodible Soil Condition—Special Joint Type according to Division 2, Section 23 of the "Standard Specification for Highway Bridges" by AASHTO.

3.7 BACKFILLING

A. Pipe meeting any of the following conditions shall not be placed or back-filled until the excavation and foundation has been approved by the Contracting Officer.

1. Embankment height greater than 10 feet at subgrade centerline.
2. Installation in a live stream.

B. After the bedding is prepared and the pipe is placed, selected material shall be placed in layers not exceeding 6 inches loose thickness and compacted under the haunches and alongside the pipe. The material shall have the moisture content needed for effective compacting. Backfill shall be readily compactable material free of frozen lumps, chunks of highly plastic clay or other objectionable material. Rocks larger than 3 inches in diameter shall not be used within 1 foot of the pipe. On each side of the pipe there shall be an area of compacted material at least as wide as 2 diameters of the pipe or 12 feet, whichever is less. Backfill shall be compacted without damaging or displacing the pipe using equipment approved by the CO. Make at least three complete passes. Backfilling and compacting shall be continued until the backfill is 12 inches above the top of the culvert. After being bedded and backfilled, pipe shall be protected by an adequate cover of embankment before heavy equipment is permitted to cross during roadway construction. Pipe distorted more than 5 percent of nominal dimension, ruptured or broken shall be replaced.

END OF SECTION